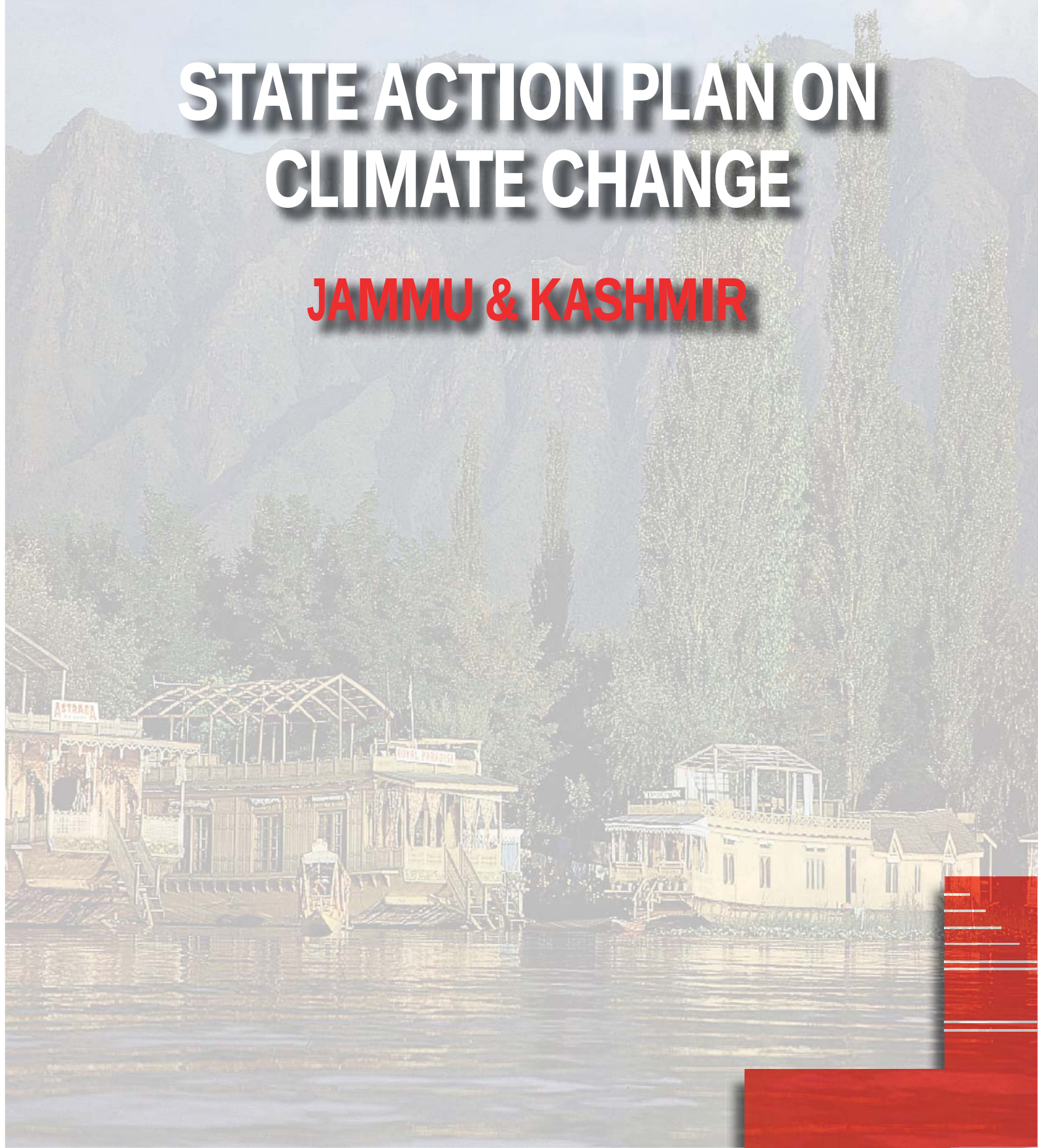


STATE ACTION PLAN ON CLIMATE CHANGE JAMMU & KASHMIR

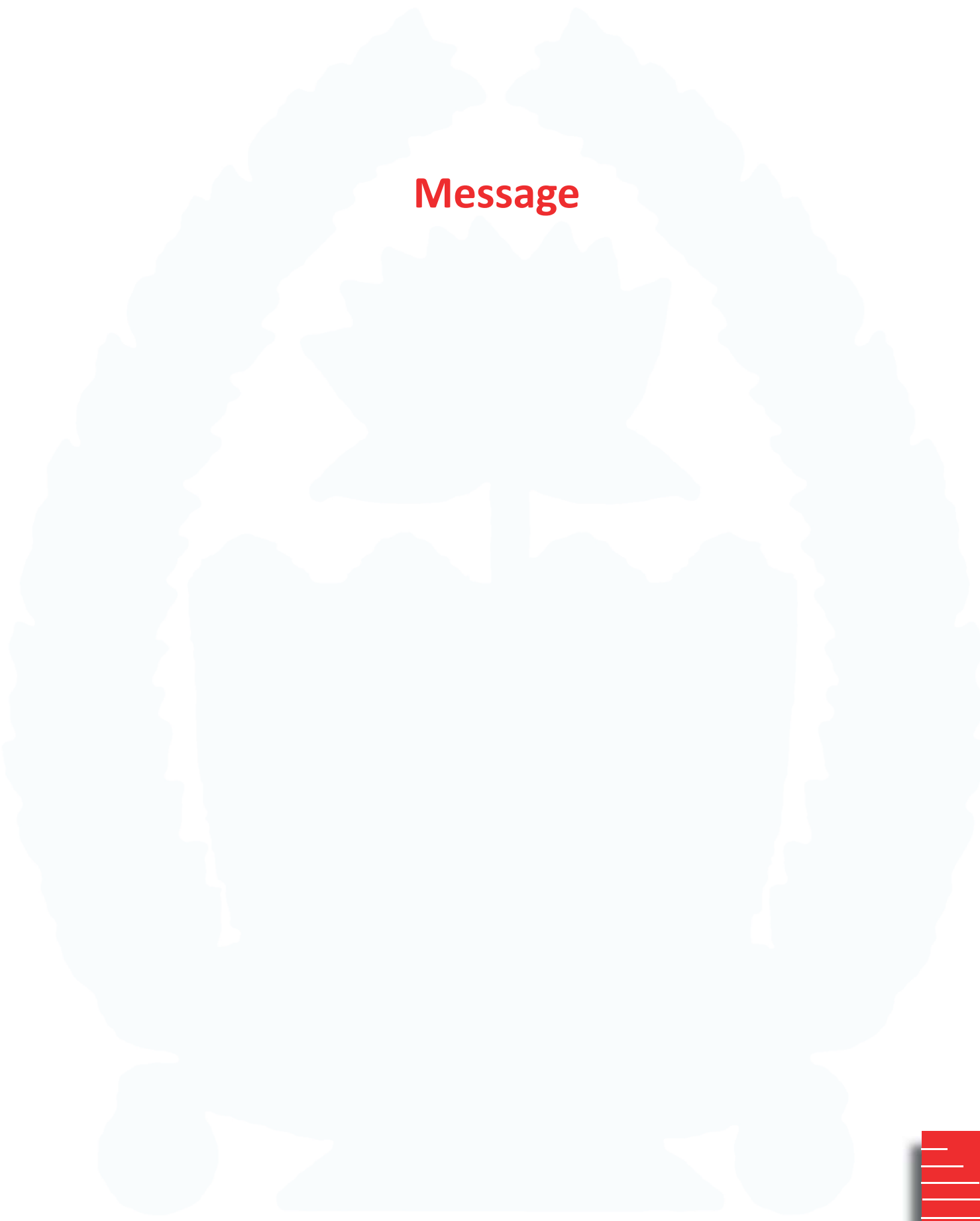


STATE ACTION PLAN ON CLIMATE CHANGE

JAMMU & KASHMIR







Message





Foreword



Acknowledgement





Contents

1	Background	5
1.1	Introduction	5
1.2	Objective	6
1.3	Methodology	6
1.4	Structure of the Report	10
2	National Action Plan on Climate Change	11
2.1	Introduction	11
2.2	Domestic Actions	11
2.3	National Mission	11
2.4	Other Initiatives	14
2.5	National Mission and Objectives	14
2.6	Mapping National Mission in the State	16
2.7	Institutional Mechanism	17
3	Vulnerability Analysis	18
3.1	Introduction	18
3.2	Climate Sensitivity	19
3.3	Vulnerability of agriculture and allied activities due to climate sensitivity	25
3.4	Bio-Physical Factors	27
3.5	Soil Water Regime	29
3.6	Socio-economic factors contributing to the vulnerability	30
4	Sustainable Energy Mission	35
4.1	Introduction	35
4.2	Key Trends in the sector	36
4.3	Vulnerability of the sector	37

4.4	Key issues in the sector	39
4.5	Programme and Policies in the sector	40
4.6	Key Priorities	42
4.7	List of Key Priority Action	51
5	Enhanced Energy Efficiency	55
5.1	Introduction	55
5.2	Key trends in the sector	55
5.3	Key issues in the sector	59
5.4	Programmes and Policies in the sector	63
5.5	Key Priorities	68
5.6	List of Key Priority Action	75
6	Mission on Sustainable Habitat	78
6.1	Introduction	78
6.2	Key trends in the sector	78
6.3	Vulnerability of the sector	80
6.4	Key issues in the sector	81
6.5	Programme and Policies in the sector	82
6.6	Key Priorities	84
6.7	List of Key Priority Action	92
7	Green India Mission	93
7.1	Introduction	93
7.2	Key trends in the sector	96
7.3	Vulnerability of the sector	96
7.4	Key issues in the sector	99
7.5	Programme and Policies in the sector	100
7.6	Key Priority	102
7.7	List of Key Priority Action	106
8	Water Mission	107
8.1	Introduction	107
8.2	Key trends in the sector	107
8.3	Vulnerability of the sector	117
8.4	Key issues in the sector	119
8.5	Programme and Policies in the sector	121
8.6	Key Priorities	123
8.7	List of Key Priority Action	129
9	Annex I: Mission wise Comprehensive List of Actions proposed under	

SAPCC	136
9.1 Sustainable Energy Mission	136
9.2 Mission on Enhanced Energy Efficiency	142
9.3 Mission on Sustainable Habitat	144
9.4 Green India Mission	146
9.5 Water Mission	148
Table 1: National Mission on Climate Change its goal and link with development goal	12
Table 2: Relating State Mission under Climate Change and National Mission	16
Table 3: Temperature variation at Srinagar (°C)	22
Table 4: Major glaciers in the Indus basin of India	23
Table 5: Characteristic of glacial coverage in the state is given below.	24
Table 6: Recent retreat pattern of selected glaciers in Jammu and Kashmir	24
Table 7: Estimates of Poverty of J&K	31
Table 8: population distribution of Jammu & Kashmir	36
Table 9: Electricity Demand in the state	36
Table 10: Possible Climate Change Impacts on the Energy Sector	38
Table 11: List of Mini Hydro Project Considered Under Implementation	41
Table 12: List Of Wind Power Project Under Implementation	42
Table 13: List of Key Priority Action Proposed under Sustainable Energy Mission	51
Table 14: Capacity wise installed hydro power project State Sector	56
Table 15: Capacity wise installed hydro power project Central Sector	56
Table 16: Capacity wise installed hydro power project Independent Power Producers	56
Table 17: List of Key Hydro Power Project and their capacity	57
Table 18: Category wise Electrical Energy Consumption	58
Table 19: Peak and Power Availability Scenario in the State	58
Table 20: Energy Demand and Availability	59
Table 21: T&D and AT&C Losses (from 2006-07 to 2011-12)	60
Table 22: Sector wise Energy Conservation Potential in Jammu & Kashmir (2007-2008)	61
Table 23: Sector Specific Electricity consumption & GHG emission from electricity consumption	62
Table 24: List of Hydro Power project (with capacity) under renovation and modernization	69

Table 25: List of Key Priority Action Mission on Enhanced Energy Efficiency	75
Table 26: Urban Population	78
Table 27: Types of vehicles registered in the State	80
Table 28: Adaptation Pathways with respect to climate change in State	82
Table 29: List of Key Priority Action Mission on Sustainable Habitat	92
Table 30: Assessment of variation in forest cover (Area in km ²)	94
Table 31: Forest Cover Change Matrix (Area in km ²)	94
Table 32: List of Key Priority Action Green India Mission	106
Table 33: District wise number of lake and water bodies	109
Table 34: Availability of Water from precipitation across the administrative divisions of Jammu	110
Table 35: Availability of Water from precipitation across the administrative divisions of Kashmir	110
Table 36: Major glaciers in the Indus basin of India	111
Table 37: The characteristic of glacial coverage in State	111
Table 38: Basin-wise Replenishable Groundwater Resources (in bcm)	112
Table 39: Basin-wise Replenishable Groundwater Resources (in bcm)	112
Table 40: Total area of irrigated land from different sources	113
Table 41: River basin wise irrigated crop area permitted under Indus Water Treaty	116
Table 42: Recent retreat pattern of selected glaciers in Jammu and Kashmir	118
Table 43: Maximum area irrigated	121
Table 44: Aggregate Storage Capacity	122
Table 45: Key Priority Actions State Water Mission	129
Table 46: Comprehensive List of Actions proposed under Sustainable Energy Mission	136
Table 47: Comprehensive List of Actions proposed under Mission on Enhanced Energy Efficiency	142
Table 48: Comprehensive List of Actions proposed under Mission on Sustainable Habitat	144
Table 49: Comprehensive List of Actions proposed under Green India Mission	146
Table 50: Comprehensive List of Actions proposed under Water Mission	148

Figure 1: Sectors of J&K mapped to National Mission	7
Figure 2: The process of formulation of SAPCC in J&K	9
Figure 3: Eight fold mission under NAPCC	11
Figure 4: Vulnerability Profiling	19
Figure 5: Vulnerability map of Indian Subcontinent	20
Figure 6: Agricultural Vulnerability based on NDVI variability	26
Figure 7: All forested grids in India	27
Figure 8: All forested grids in India (2071-2100)	28
Figure 9: Location of Hydro Power Project	55
Figure 10: Source wise hydro power potential and sector wise hydro power resource harnessed	56
Figure 11: Gross Per capita Electricity (kWh) consumption in J & K	59
Figure 12: Sector specific GHG emission from electricity consumption	63
Figure 13: Projects under execution/planned for execution in 12th Plan	67
Figure 14: District Map of J&K	79
Figure 15: Forest cover in the state	93
Figure 16: Different Forest Types in States	95
Figure 17: Forest Grids in India	97
Figure 18: Indus River and its tributaries with in Jammu and Kashmir	108
Figure 19: Indus Basin	115
Figure 20: Predicted change in rainfall (mm) by 2085, (Source: Current Science)	118





Abbreviations and Acronyms

AC	Alternating Current
AT&C	Aggregate Technical and Commercial
BEE	Bureau of Energy Efficiency
BOOT	Build Own Operate and Transfer
BPL	Below Poverty Line
C:N	Carbon: Nitrogen
CCAP	Climate Change Action Plan
CDM	Clean Development Mechanism
CEA	Central Electricity Authority
CFA	Central Financial Assistance
CFL	Compact fluorescent lamp
CGWB	Central Ground Water Board
CIC	Community Information Centre
CM of J&K State	Chief Minister
CO ₂	Carbon - di- oxide
CPP	Captive Power Plant
CSDRM	Climate Smart Disaster Risk Management
CWC	Central Water Commission
DC	Direct Current
DPR	Detailed Project Report
DRDA	District Rural Development Agency
DRDO	Defence Research and Development Organisation
DRM	Disaster Risk Management

DRR	Disaster Risk Reduction
DSM	Demand Side Management
ECBC	Energy Conservation Building Code
EEFP	Energy efficiency financing platform
EFA	External Funding Agency
EMP	Environment Management Plan
ESCCerts	Energy Savings Certificates
ESCO	Energy Service Company
ET	Evapotranspiration
FEEED	Framework for energy efficient economic development
FSI	Forest Survey of India
FY	Financial Year
GDP	Gross Domestic Product
GHG	Green House Gas
GIS	Geographic Information System
GLOF	Glacial Lake Outburst Floods
Goi	Government of India
GSDP	Gross State Domestic Products
H1N1	Swine influenza A H1N1 virus
H7N9	Avian influenza A(H7N9) virus
HCN	Hydrogen cyanide
HVAC	Heating, ventilation, and air conditioning
HVDS	High Voltage Distribution System
IBIS	Integrated biosphere simulator : Vegetation Model
ICDS	Integrated Child Development Services
ICIMOD	The International Centre for Integrated Mountain Development
IGEA	Investment Grade Energy Audit
IMD	Indian Meteorological Department
INCCA	Indian Network on Climate Change Assessment
INM	Integrated nutrient management
IPCC	Intergovernmental Panel on Climate Change
IPP	Independent Power Producer
IREDA	Indian Renewable Energy Development Agency
J & K EC act	J & K Energy Conservation Act

J&K	Jammu and Kashmir
JAKEDA	Jammu and Kashmir Energy Development Agency
JKSERC	J&K State Electricity Regulatory Commission
JKSPDC	Jammu and Kashmir State Power Development Corporation
JNNSM	Jawaharlal Nehru National Solar Mission
KREDA	Kargil Renewable Energy Development Authority
IAY	Indira AwaasYojana
LED	Light-emitting diode
LoC	Line of Control
LREDA	Ladakh Renewable Energy Development Agency
MDF	Moderately Dense Forest
MDG	Millennium Development Goal
MGNREGS	Mahatma Gandhi National Rural Employment Guarantee Scheme
MNRE	Ministry of New and Renewable Energy
MoEF	Ministry of Environment and Forest
MOU	Memorandum of Understanding
MOWR	Ministry of Water Resources
MSME	Micro, Small and Medium Enterprise
MSW	Municipal Solid Waste
MTEE	Market transformation for energy efficiency
NAPCC	National Action Plan on Climate Change
NASA	National Aeronautics and Space Administration
NATCOM	National Communication
NBMMP	National Biogas & Manure Management Programme
NDMA	National Disaster Management Authority
NEWNE	North East West North-East Grid
NF	No Forest
NGO	Non-governmental organizations
NIC	National Informatics Centre
NIH	National Institute of Hydrology
NMEEE	National Mission on Enhanced Energy Efficiency
NSDP	Net State Domestic Product
O&M	Operation and Maintenance
OF	Open Forest

PAT	Perform Achieve and Trade
PDD J&K /JKPDD	Power Development Department J&K
PFC	Power Finance Corporation
PHC	Public Health Centre
PHE	Public Health Engineering
PoA	Programme of Activities
PRECIS	Providing Regional Climates for Impacts Studies
PV	Photovoltaic
PWD	Public Works Department
R-APDRP	Restructured- Accelerated Power Development and Reforms Programme
REC	Rural Electrification Corporation
REDD	Reduce Emission from Deforestation and forest Degradation
RESCO	Renewable Energy Service Company
RKVY	RashtriyaKrishiVikasYojana
RMU	Renovation and Modernization
RS	Remote Sensing
SADP	Special Area Demonstration Programme
SAPCC	State Action Plan on Climate Change
SDMA	State Disaster Management Authorities
SHLS	Solar Homelighting System
SKUAST	Sher-E-Kashmir University of Agricultural Sciences and Technology
SME	Small and medium enterprises
SOC	Soil Organic Carbon
SOER/SOE Report	State of Environment Report
SRES	Special Report on Emission Scenarios - IPCC
STP	Sewage Treatment Plant
SWH	Solar Water Heater
T&D	Transmission and Distribution
TOE	Tons of Oil Equivalent
UDD	Urban Development Department J&K
UNEP	United Nation Environment Programme
UNFCCC	United Nation Framework Convention on Climate Change
VDF	Very Dense forest

WGI	Working Group I of IPCC
WHO	World Health Organisation

Units

0 C	Degree Centigrade - unit of temperature
KVA	Kilo - volt - ampere - unit of electrical power
bcm	Billion cubic meters
GW	Giga watt
ha	Hectare
J	Joule
kg	Kilogram
km	Kilometer
km ²	Square Kilometer
kW	Kilo - watt - unit of electrical energy
LPD	Litre Per Day
m	Meter
Mha	Million Hectare
mm	Millimeter
MT	Metric Tons
MU	Million Unit of Electrical Energy (1 unit = 1 kilo watt hour)
MW	Mega Watt
MWh	Megawatt hour
Wp	Watt peak





Executive Summary

Introduction

Climate projections suggest that impacts are likely to be varied and heterogeneous in India. Some regions will experience more intense rainfall and flood risks, while others will encounter sparser rainfall and droughts including spatial shift in the pattern of rainfall. The Fifth Assessment Report has further emphasized the loss of ice from glacial range and projected that the current glacial extents are out of balance with current climatic condition indicating that glaciers will continue to shrink in the future even without further temperature increase. This has a significant impact not only on the Himalyan ecosystem of state of Jammu and Kashmir but also several other regions in India and some regions in Pakistan Occupied Kashmir and beyond.

A global scientific consensus with regard to the unequivocal nature of the climate change and its association with increasing concentration of greenhouse gas emissions is widely established. Regional climate projection indicates that climate variability

will manifests through alteration in frequency, intensity, spatial extent, or duration of weather and climate extremes, including climate and hydro-meteorological events. This is likely to pose greater risks to human life and endanger the sustainability of the economy of several states in India. Indeed the likely projection of the hydrometeorological and geophysical hazards has the potential to derail the current growth strategy and deepen poverty amongst the vulnerable. The impacts of extreme climate-induced events (such as cloud burst, enhanced glacial flow, untimely dry and wet days) could result in loss of life, livelihoods, assets and infrastructure and affect the state's economic growth and pro poor initiatives. Given its profile, climate change is an important concern for the state as it is presently on a carbon-oriented development path and at the same time, it is vulnerable to climate variations owing to the fragile ecosystem. It is therefore in the context above a comprehensive climate change action plan with institutional structure has been conceptualized by the state of J&K that would address both adaptation and mitigation concerns with an objective of

increasing resilience and climate proofing of the sectors keeping in mind the conditions of the local vulnerabilities.

Process of formulating the CCAP

Immediately after the meeting of Chief Secretaries in February 2010 where the issue of developing the State Specific Climate Change Action Plan was urged by Hon'ble Prime Minister the State Government of J&K institutionalized the J&K State Council on Climate Change on 25th of April 2011. The State Council on Climate Change is a three-tier structure comprising of Governing Council headed by Hon'ble Chief Minister, Executive Council headed by the Chief Secretary and Expert Committee headed by the Director, Environment & Remote Sensing. While framing the council and the working group, the membership was drawn both from the line departments and also from the different associated departments/organizations. A nodal officer was appointed to co-ordinate the meetings, to collect/collate the required information and to develop the plans pertaining to each of the sectors.

Following are the sector identified as part of the Climate Change Action Plan.

1. Energy - Solar Mission and Renewable Energy
2. Enhanced Energy Efficiency
3. Water
4. Sustainable Habitat
5. Sustainable Agriculture

6. Tourism
7. Sustainable Himalayan Ecosystem
8. Health
9. Disaster Management
10. Strategic Knowledge mission
11. Green India Mission

Both top down and bottom up approach was followed in developing the SAPCC. Consultations were carried out with members of working groups as well as with the administrative departments. Administrative heads were consulted recognizing the need for involvement of the senior bureaucracy within the government towards assessing climate change issues and mapping the solutions. Tools and templates were formulated for listing climate change specific issues, prioritizing the activities as high, medium and low and developing an outline of the sub-activities to be taken up under the high priority activities, estimating the budget and sources of funding as well as formulation of the monitoring and verification protocol. GIZ provided support for this action planning facilitation and engaged CTRAN to coordinate and manage the process on behalf of the Government of J&K.

Vulnerability

In the context of India, especially Jammu and Kashmir, which nestles in fragile Himalayan Ecosystem; there are natural fluctuations in climate, human induced changes due to large scale urbanization that drive the warming trend. The biodiversity loss and water stress owing to the climate change are the greatest

challenge for the state over the coming decades. Climate change poses a serious threat to the species diversity, habitats, forests, wildlife, fisheries and the water resources in the region. Many wetlands in J&K that support 20% of the known range of biodiversity in the region are adversely affected. The climatic changes resulting in changing weather movements are relatively unpredictable, but could be important in the context of the incidence of different diseases like avian influenza. As per UNEP report some parts of the state are moderate to high vulnerability. As per INCCA assessment the number of rainy days in the Himalayan region in 2030s may increase by 5-10 days on an average, with an increase by more than 15 days in the eastern part of the Jammu and Kashmir region. The intensity of rain fall is likely to increase by 1-2 mm/ day. This is likely to impact some of the horticultural crops though rangelands and livestock are likely to benefit. The rate of recession of glaciers is reportedly varying which is being attributed to winter precipitation climate warming and anthropogenic elements. Temperature, precipitation and cold wave are most likely to significantly impact the agricultures sector and enhance its vulnerability. Deficit in food production is growing in recent times in Jammu & Kashmir. With the reduction in rainfall, the rain-fed agriculture will suffer the most. Horticultural crops like apple are also showing decline in production and a real coverage particularly due to decline in snowfall. Study by Indian Institute of Science using dynamic vegetation model

has revealed that about 39% and 34% of the forested grids are likely to undergo shifts in vegetation type under A2 and B2 climate scenarios, respectively with a trend towards increased occurrence of the wetter forest types. Approximately 47% and 42% of tropical dry deciduous grids are projected to undergo shifts under A2 and B2 scenarios respectively, as opposed to less than 16% grids comprising of tropical wet evergreen forests. Similarly, the tropical thorny scrub forest is projected to undergo shifts in majority of forested grids under A2 (more than 80%) as well as B2 scenarios (50% of grids). The increase in temperatures may lead to increasing morbidity due to heat stress. Flash floods due to Glacial Lake Outburst Floods (GLOF) may lead to large scale landslides and affect food security and hence nutritional health.

Sustainable Energy Mission

The action plan proposed under sustainable energy mission is strategized in line with the National Action Plan on Climate Change with objectives of meeting the country's development goals while simultaneously yielding co-benefits to address climate change effects. The state is currently focused on generation through big hydro power projects. Potential of micro hydel power and Solar are increasingly being tapped. The challenges in the sector are quite a few. Grid electricity penetration in remote hilly areas of J&K is techno-economically unviable by virtue of geographical disadvantages and scattered household pattern. Moreover, the power sector of the state is already facing difficulties like slow rate of capacity addition, poor power evacuation facility, high AT&C

losses and mismatch in load profile and power utilization mix is not commensurate with the state climatic conditions. Promotion and utilization of renewable energy is the most feasible solution which would also promote low carbon growth and can meet the decentralized energy requirement of the remote locations. The power available in 2011-12 from all sources was able to satisfy only 64% of total power demand of the state. Due to the shortfall of 6,231.74 million units during 2011-12, the state faced problems of frequent power cuts for 8- 10 hours a day which has a detrimental effect on the growth

potential of the state. To meet its power demand of 17,323.00 million units during 2011-12, the state has procured 66.76% of its total power supplied (i.e. 11,091.26 million units) from the NEWNE grid which is highly fossil fuel dominated. Promotion of renewable energy in this context will help in bridging up of the demand supply gap but will also promote low carbon societal development and meeting up of the electrical and thermal energy requirement without contributing to the greenhouse gas emission to the atmosphere. The key priorities proposed under the missions are as follows:

Key Priorities: Sustainable Energy Mission

1. Promotion and implementation of solar city project
2. Promotion and facilitation of Off-grid and decentralized renewable energy generation for electrification, cooking and other thermal energy requirement
3. State Govt. by amendment of building bye-laws suitable to state condition would promote and mandate use of solar water heating and/or lighting systems
4. Institutional development and strengthening of JAKEDA, LREDA and KREDA for promotion of Renewable Energy technology applications.
5. Harnessing renewable energy potential scenario of the state for power generation
6. Strengthening of technical competency of various stakeholders of renewable energy which includes O&M persons, technicians, installers, manufacturers and other relevant service providers as per international standard
7. Enhancing state's own power generation capacity through hydro power plant set up and provides support to private/public investors in projects implementation and undertakes micro/mini hydro projects for remote area to meet up local demand
8. Promotion of grid connected Rooftop and Small Solar Power projects in the state of J&K
9. Promotion of Green buildings in the state

Mission on Enhanced Energy Efficiency

The power generation in the state is mainly governed by hydro power with a little share of fossil fuel based grid interactive power plant. Total estimated potential for hydro power generation in the state is 20,000 MW out of which 16,480 MW (excluding 251 Projects up-to 25 MW generation potential identified totaling to 986 MW Capacity) has been identified as the actual generation potential. Out of the identified potential only 2457.96 MW or 15% has been harnessed so far. The power sector of the state is subjected to high AT&C losses of the order of 61%. Technical losses pertain to outdated and old electrical device in the network. To minimize such losses, the system needs up-gradation and improvements. Commercial losses include power theft, unaccounted and uncontrolled consumption beyond agreement load, unregistered consumers, lesser contract demand, etc.

Domestic consumers as a whole are the major electricity consuming segment in the state with cumulative consumption in tune of 34% followed by industrial 20.21%, government institution 17.6%, Public water works 14.50%, commercial 8.6%, agriculture 4.9% and public lighting in tune of 0.7% respectively. A substantial potential of energy saving are there across all the energy consuming segments. The energy saving potential across agricultural sector is highest in tune of 25-35%, followed by domestic sector 15-20%, building sector in tune of 20-30%, industrial sector 7-10% and municipalities around 20%. The key priorities

proposed under the missions are as follows:

Key Priorities: Sustainable Energy Mission

1. Establishment and incorporation of Bureau of Energy Efficiency, J & K EC act 2011
2. Promoting Energy Efficiency in Small and Medium Enterprises (SMEs)
3. Promoting Energy Conservation Building Code (ECBC)
4. Renovation and Modernization of Hydro-power Projects
5. Development of comprehensive scheme for promotion of energy efficient pumps in agriculture sector
6. Reducing Transmission and Distribution (T & D) Losses
7. Promoting Demand Side Management (DSM) and energy efficiency
8. Decentralized power generation
9. Harnessing the biomass potential
10. Development of policy for mandatory use of efficient light particularly for commercial organizations in the state
11. Expansion Conversion of conventional street lights to solar LED/CFL street lights
12. Maximizing solar power usage and Encouraging use of Solar Gadgets (especially in Industries)
13. State energy audit policy and Implementation of pilot Energy efficiency project and IGEA
14. Promoting Energy Efficiency Practices in the State
15. Awareness and implementation for use of CFL and replacing incandescent lamp under Bachat Lamp Yojna, and Umbrella program of BEE
16. Standards And Labelling Programme

Mission on Sustainable Habitat

Urban sector faces significant impacts of climate change. It impacts human health, livelihood and productive assets, especially for the urban poor, informal settlements and other vulnerable groups. Climate change impacts range from increase in extreme weather events, flooding to high temperatures leading to public health concerns, resource congestions, etc. As the urban regions of the state are dynamic systems that face unique climate impacts, their adaptation must be location specific and tailored to local circumstances.

The importance of infrastructure and improvement of living standards of population is crucial to sustain economic development. A detailed analysis of the development process over last four decades shows that one of the major reasons for slow economic and social development is the unplanned population growth. The capital cities and other district towns of the state are challenged with problems of rapid urbanization, expansion of informal settlements, substantial poverty, inadequate infrastructure and environmental degradation. These and other concerns plague cities' ability to grow and prosper. Many of these conditions also limit resilience to address current climate variability.

Solid waste management is the most emerging and pressing issues in the urban areas of the state. Agglomeration of solid and liquid waste not only creates health impact, water contamination and pollution but also its unplanned decomposition results

into release of methane to the atmosphere. Sanitation possesses major stress in the urban areas as absence of adequate sewerage treatment facilities in urban areas results in release of effluent from domestic and hospitality industry into nearby rivers, lakes and streams leading to contamination of water sources. Indiscriminate developmental activities also add to the problem by jamming drains and encroaching on rainwater flow paths. The urban transport sector has been largely neglected in the State, characterized by heavy traffic congestion due to narrow roads, rapid growth in number of vehicles along with highly topographic and concentric development. Public transport is limited due to inadequate road network, poor infrastructure and scattered demand.

The state government is thus planning to mainstream climate change adaptation in urban infrastructure development. Creating this new possibility of adaptive resilience would support a new future in the face of climate change. The key priorities proposed under the missions are as follows:

Key Priorities: Sustainable Habitat

1. Promotion of battery operated (charged from solar power) transport systems in tourist places, battery driven motor boats instead of diesel engine boats (battery charged from solar module mounted on the boat) and use of alternative fuel instead of diesel to drive Barges and other diesel driven equipment used for maintenance purpose in the lake

2. Capacity building of all levels of stakeholders to make them understand their role on their regards and maintain proper environment management system
3. GHG accounting study on emission by transport vehicles
4. Establishment of intelligent transport system
5. Policy framework in transport sector
6. Promotion of public transport and mass transport & Promotion of water transport
7. Establishment of Rail based transport system
8. Reallocation of the hotels or application of STP near Dal lake & DPR preparation on STP
9. Establishment of end-to-end solid waste management system
10. Base line study on solid waste and industrial waste generation and DPR preparation & four pilot project implementation
11. Capacity Building/ Awareness Generation Program on segregation of solid waste at household level
12. Establishment of segregation mechanism of bio-medical waste in urban hospitals and in rural PHCs
13. Application of methane capture technology and preparation of fertilizer from weeds generated in Dal lake

Green India Mission

According to the interpretation of satellite data of October-December 2008, the forest cover of Jammu and Kashmir is spread over 22,539 km² (including area outside the LoC) accounting for 10.14% of the State's geographical area. The State of Jammu and Kashmir has rich diverse forest resources and plays an important role to preserve the fragile ecosystem of the state and it also serves as catchments for Himalayan Rivers. In Jammu and Kashmir, the perennial water supply including groundwater recharge, and health of soils, is primarily depending upon the area and quality of forest cover. A healthy forest cover is essential for long term operation of hydroelectric projects. In the State forest also provide some environmental services like carbon sequestration, pollution abatement, amelioration of climate, in-situ conservation of biodiversity and maintenance of ecological balance. Since there is unique aesthetic value of forest landscape and huge potential for tourism in the State, the forests plays a pivotal role to make the State an ideal tourist destination. Forests and climate change are inherently linked. When kept healthy, forests play a key role in our national climate change strategy. Over geologic time, changes in disturbance regimes are a natural part of all ecosystems. However, loss of forest's ability to mitigate the effects of climate change will mean losing many important ecosystem good and services. Unplanned development, climate change, invasive species, unsustainable forestry practices, diseases and insects, all of these factors and more, endanger the forests. Number of

pressures is responsible for deforestation and degradation in Jammu and Kashmir. Due to high dependency on forest for timber and fuel wood deforestation is an ongoing process. On the other hand the livestock population in the state is 9.90 million resulting in huge demand for fodder. In addition most of these live-stocks graze in forests. Grazing and trampling of saplings by livestock is the biggest threat to regeneration of vegetation in all forested areas of the state. Low tree density, low bio-diversity status as well as

higher levels of fragmentation contribute to the vulnerability of these forests. The current biotic and abiotic stresses are likely to exacerbate the projected impact of climate change that indicates towards shift of vegetation of the forest grid including tropical dry deciduous grids, tropical wet evergreen forests and tropical thorny scrub forest. The key priorities proposed under the missions to reduce the impact of climate change are as follows:

Key Priorities: Green India Mission

1. Implementation of J&K State Forest Policy
2. Capacity building and awareness for all levels of stakeholders
3. Gene bank development for climate adaptable species
4. Eco-restoration through afforestation and Climate oriented eco-restoration plan
5. Study on identification of tree phenology
6. Flora and Fauna vulnerability study
7. Carbon sequestration study for Jammu and Kashmir forest
8. Study on per capita fuel wood consumption and alternative livelihood
9. Study on soil organic carbon of forest area
10. Climate impact study in undisturbed/ protected forest areas.
11. E-green portal with geo-reference
12. Nursery development for climate adaptable species
13. REDD+ feasibility study for carbon sequestration.
14. Study of climate change impact on wild life species
15. Climate change impacts on undisturbed forest area like national parks

Water Mission

Climate change is projected to influence the hydrological cycle of water bodies, water supply systems and exacerbate requirement of water in different regions. The disturbance of the hydrological cycle as projected due to the variability in the climatic conditions are likely to enhance the current stress on water resources from surging population, growing economic development and land use change including urbanization. Increase in ambient temperature and changes in rainfall pattern might influence the hydrological cycle to a large extent. Other consequences like increase in rates of evapotranspiration decrease in fresh water availability, mass losses from glaciers and reductions in snow cover might severely impact the hydrological cycle. Change in the hydrological cycle may affect the spatial and temporal distribution of runoff, soil moisture, ground-water reserves and may increase the frequency of droughts and floods. Event of increased frequency of rainfall and variation in rainfall pattern including extreme events like flood risk poses challenge to the society, physical infrastructure and quality.

Economy of the State relies heavily on water intensive sectors like agriculture, horticulture, energy generation, tourism, etc. Since most of the water-bodies in the state are supplied by melt-water from mountain range therefore loss of glacier might reduce the water availability impacting hydro power potential and lower availability of water for irrigation impacting the agrarian economy in the state. Regions in the state that are already subjected to water stress are projected to

be adversely affected with the potential of impact on multiple sectors e.g. agriculture, domestic water supply and health. The key priorities proposed under the missions to undermine the impact of climate change and also to enhance the sustainability of the sectors are as follows:

Key Priorities: Water Mission

1. Generate baseline data with terrain analysis (contour/slope/ gradient/ aspect, etc.), with comprehensive land classification with catchment demarcation, river networks, lakes and water bodies, forest coverage, etc
2. Promoting and implementing water use water use Efficiency measures
3. Develop hydrological models for different seasons and calibration with the existing data
4. Comprehensive water budgeting in lieu of the current and future demand from domestic supply, irrigation, industry, tourism and for other sectors
5. Monitoring river flows and mapping of flood and drought zones
6. Formulate comprehensive rejuvenation plan for Dal Lake, Mansar Lake, Manasbal Lake, Pangong Lake and all other major lakes
7. Assessment and Implementation of rain water harvesting and artificial recharging of ground water
8. Emphasize the awareness and capacity building for conservation, concern and co-management of water
9. Dealing with flood in the changing climate scenarios.
10. Reducing impact of increased erosion and sedimentation
11. Creation of Storage facilities on Distributaries of River Chenab and Jhelum As per Indus Water Treaty

Sustainable Mission

Agriculture

Agriculture and its allied sector is the mainstay of state economy providing livelihood opportunity to 70% of the state's population and contributing to around 27% to the state's income. Horticulture plays a significant role in State's economy as the sector contributes 7% to GSDP and provides greater opportunities for development of agro-industries in the state. Weather variability or climate extremes resulting from climate change could impart direct and indirect effects on the crops, soils, livestock and pests. While the magnitude of impact might vary over the region but it is expected that variability of the climatic condition will impact agricultural productivity and shifting crop pattern. The vulnerability of the State agriculture sector to climatic variability would be accentuated at multiple levels including the crops or livestock's, farm or cropping system and the food system. The implication of climatic variability or extremes over agricultural sector could be wide-reaching and it could affect food security, livelihood activities, trade policies, water conservation issues impacting large portion of population. Adverse impacts of climate change on agricultural production would be severe in the absence of appropriate adaptation and mitigation measures with far reaching consequences in terms of food shortages and consequent inflation which could endanger the food and livelihood security of the State. The key priorities proposed under the missions to undermine the impact of climate change both through

adaptation and mitigation measures and also to enhance the sustainability of the sectors are as follows:

Key Priorities: Sustainable Agriculture Mission

1. Planning of cropping system and crop varieties through Crop diversification, drought tolerant crops and water saving crops:
2. Capacity building of Planners farmers and extension workers and dissemination of new and appropriate technology:
3. Integrated nutrient management (INM) in 50000 ha area:
4. Zero Tillage:
5. Trash mulching in agriculture and Horticulture
6. Combating climate related risk through Micro Irrigation programme
7. Management of climate change risk for sustainable productivity
8. Weather based Crop insurance
9. Enhancing capacity for livestock disease management and forecasting monitoring and management
10. Conservation of water bodies and fish stock augmentation in natural water bodies and promotion of fish farming
11. Resource mapping (Agriculture) by using GIS and RS technology and Strengthening of agro advisory services by using Android Technology

12. Screening of crops for moisture/heat/disease/pest tolerance and nutrient use efficiency
13. Crop yield and weather modeling for future projections
14. Carbon pool assessment/sequestration in forest and agricultural ecosystems
15. Institutionalizing the role of research universities in developing sustainable agricultural practice and facilitate pilot implementation
16. Promoting Sustainable Floriculture practice
17. Promoting sustainable horticulture practice

Disaster Management

Adverse impacts of climate change on society often increases disaster risk, disasters themselves erode environmental and social resilience, and thus increase vulnerability to climate change. Moreover climate variability is projected to result in more frequent and intensive disasters – with most severe consequences on the infrastructure, food security and livelihoods of natural resource dependent vulnerable communities. Since both disaster risk (including climate associated disaster risk) and climate related vulnerabilities are likely to undermine the economic sustainability and development it is therefore planned that disaster risk management strategies and climate change adaptation planning be integrated in the state. J&K is a multi-hazard prone region with natural disasters like earthquakes, floods,

landslides, avalanches, high velocity winds, snow storms, besides manmade disasters including road accidents and fires etc. Such incident frequently occurs thereby disturbing the ecological balance and in most of the case directly results to disastrous event. The key priorities proposed under the missions to undermine the impact of climate extremes are as follows:

Key Priorities: Sustainable Agriculture Mission

1. Hazard risk mapping using GIS and Remote sensing
2. Risk reduction through implementable planning and policy development
3. Strengthening Communication Networks and Disaster Management Facilities
4. Hazard specific multi-parameter vulnerability assessments and societal impacts assessments at the regional and local levels for preparing contingency plan
5. Developing and promoting concept of Climate Smart Disaster Risk Management (CSDRM)
6. Preparation of disaster rescue and rehabilitation plan.
7. Awareness/ Sensitization/ Capacity Building in form of :
8. Community based disaster management in the Disaster Management Plan at all levels
9. Disaster response training at the community level to build infrastructure and human resources for medical

preparedness and emergency medical response to manage mass casualties during extreme events

10. Institutional development and operationalization of the regulatory framework for effective management of disaster under State Disaster Management Act 2005 for preparation of Disaster response and management plan

Tourism

Other than the destinations of tourism value, climate is another principal attraction for tourists from all over the world. Preference of locations during a particular period of time and the profitability of enterprise is largely determined by the climatic conditions. Thus any regional manifestation of climate variability and extremes are likely to influence the popularity of any tourist destination. So the projected climate variability and change can severely affect the tourism industry in the state through increased infrastructure damage, additional emergency preparedness requirements, higher operation expenses and business interruptions. Barring the vulnerability of the sector the tourism sector also contributes to climate change through GHG emissions from transportation and accommodation facilities for tourists.

The unplanned and unprecedented growth of the tourism industry in the state has resulted in severe and detrimental impact on local environment. The inflow of tourist beyond the carrying capacity of the location has put tremendous pressure on existing resources

and infrastructure. The pressing issues are:

1. Increased Pollution load, enhanced health/safety concern
2. Increased issues of sanitation and garbage management
3. Weather related risks e.g. changes in temperature may discourage tourists to visit existing tourist destinations. Extreme climate events may lead to disruption of services.
4. Pressure on drinking water resources
5. Siltation and pollution of water bodies
6. Lack of collection and disposal mechanism of solid waste
7. Increased urban congestion and lack of proper transportation facilities
8. Lack of sewage disposal facility
9. Poor Maintenance and Use of Existing Tourist Infrastructure
10. Inadequate medical and DRM plan at climate sensitive tourist spots
11. Deforestation and land degradation
12. Unplanned infrastructure in climate sensitive zones
13. Lack of transport infrastructure
14. Lack of sufficient connectivity with rest of the country and adjacent countries
15. Un-regulated development of tourists destination including problem of accommodation
16. Poor waste disposal system
17. Restricted & protected access to important tourist destinations in State.
18. Requirement of Herringbone drainage system

The key priorities proposed under the missions to undermine the impact of climate change and also to enhance the sustainability of the sectors are as follows:

Key Priorities: Tourism

1. Undertaking carrying capacity studies and developing EMP for ten tourist locations
2. Construction of Bio-toilet
3. Implementation of solid and liquid waste management facility at six locations
4. Improve climate change related knowledge base of tour operators
5. Undertake study on tourist-related transport emissions
6. Strategizing Restoration of Dal Lake
7. Undertake study to identify new environmental friendly natural tourist destinations for their full-fledged operation as tourist spots

Health

Weather and climate variability has a profound influence on human health. The impact of climate change over human health is likely to be multifaceted involving increased incidence of vector, water and food borne diseases, malnutrition and undernourishment, injuries and death caused by extreme hydrogeological events and thermal stress. Temperature, precipitation and humidity have a strong influence on the reproduction, survival and biting rates of the mosquitoes that determine the malaria and dengue fever, and temperature effects on the life-cycle of the infectious agents themselves. The same meteorological factors also influence the transmission of water and food-borne diseases such as cholera and other forms of diarrhoeal diseases. Climate and weather related hazards on the contrary results in destruction of social and economic infrastructure and degradation of fragile ecosystems. The vulnerability due to the incidence of diseases or hazards will however depend upon the level of exposure, sensitivity and the coping capacity of the populace. Short term impacts of climate change are likely to magnify the existing socio-economic threats due to rapid urbanization, population growth, poverty, health infrastructure, contamination of air and water, unplanned urbanization, issues of solid and liquid waste management resulting in increasing the risks of diseases in terms of morbidity and mortality. Long term climate change impacts will exacerbate the existing stress while undermining growth and development. The key priorities proposed under the missions to undermine the impact

of climate change are as follows:

Key Priorities: Health Tourism

1. District wise inventory on water born and vector borne diseases with respect to weather variability
2. Capacity building of all stakeholders on preparedness and rapid action on health context in situations of flash flood and earthquake
3. Research and study on impact of climate change on human health
4. Construction of new hospitals with modern infrastructure and equipment's to cope up with climatic impacts on human population
5. Modernization of existing hospitals and PHCs
6. Increase of surveillance –
 - Systematic collection and analysis of data about vector borne diseases
 - Systematic collection and analysis of data about water borne diseases
 - Systematic collection and analysis of data about zoonotic diseases
7. Integration of GIS mapping for collection, analyse and share data and identification of spatial areas of high concern
8. Monitoring system -
 - To understand changes in pattern of diseases
 - To deal with new diseases from emigrants (H1N1, H7N9 etc.)

Sustainable Mission

Himalayan

In J&K, the major causes of increased vulnerabilities to the Himalayan ecology are unplanned construction, changing socio economic profile, over and unplanned exploitation of natural resource (e.g. Dal Lake or deforestation), unplanned urban growth, conflicts, insufficient institutional capacities and many getting aggravated due to climate variability and change. Climate change may adversely impact the Himalayan ecosystem through increased temperature, altered precipitation patterns and episodes of drought. Concerns have also been rising over the Himalayan glaciers along with other entities in the global cryosphere. The Himalayan glaciers may lose significant ice-mass thereby endangering the river flow especially in the lean season when the North Indian Rivers depend on melting snow and ice. Studies by several scientific institutions in India have been inconclusive about the extent of change in glacier mass and whether climate change is a significant causative factor.

As per IPCC, glacial meltdown is projected to increase flooding, snow avalanche fall from established slopes and diminish water supplies to rivers as glaciers recede. The biodiversity which resides in the higher altitudes will have lesser and lesser place to occupy and will be at increased risk of extinction. There will be a gradual shift of plant species towards higher altitudes. This will force local communities to shift to newer agricultural and fodder species. This coupled with the fact that that the area lays in a highly sensitive

seismic zone, there is bound to be a heavy stress on housing sector. Flash floods due to Glacial Lake Outburst Floods (GLOF) may lead to large scale landslides and subsequent loss of life and property. Changes in the rainfall pattern and relative humidity add to weather inconsistencies which also impacts plantation and harvesting. The Himalayan ecology is largely impacted by key human intervention like encroachment, pollution, siltation, increased resource extraction coupled with inadequate and inappropriate plantation. Sustainable development practice that aims at a balanced growth can arrest this regressive trend. The State Action plan for sustaining the Himalayan ecosystem aims to strike a balance between addressing the issues detrimental to the ecosystem while enhancing the activities for economic prosperity like tourism, infrastructure building etc. The key priorities proposed under the missions to undermine the impact of climate change are as follows:

Key Priorities: Sustainable Himalayan Mission

1. Baseline study of climate change impact on flora, fauna species and on glacial ecology
2. Study on vulnerability of mountain ecology
3. Community participation for protection and conservation of mountain ecology, terrestrial ecology and aqua-ecology
4. Capacity building and awareness for all levels of stakeholders
5. Institutional development on

biodiversity conservation and protection

6. Identification of climate grids and effect on forest and ecological resources
7. Study on terrestrial and aquatic ecosystem
8. Study on anthropogenic intervention on ecological hotspot
9. Study of climate impact on migrated birds and their timings
10. Identification of current status of Himalayan ecology and Climatic trend analysis for Himalayan ecology

Strategic Mission

Knowledge

While there has been significant improvement in scientific understanding of climate change in the past few decades, there remains considerable uncertainty about the nature, timing, spatial distribution and severity of the particular impacts. In this context, the state mission on Strategic Knowledge on Climate Change intends to create a knowledge system that would help implement adaptive actions to reduce vulnerabilities as well as take advantage of the mitigation opportunities. It is imperative to build a vibrant and dynamic knowledge system that would inform and support national as well as regional actions for responding effectively to the objective of ecologically sustainable development. A strong and strategic knowledge system is essential for identifying, formulating, planning and implementing policy driven actions while maintaining the necessary economic growth

rate. Such a strategic knowledge system for informing and supporting climate sensitive actions will need to address a number of objectives. It is equally important to address climate science with region specific modeling; assessment of various technology scenarios and alternatives for complying with national objectives; leveraging international cooperation and strengthening initiatives for selection and development of new technologies for adaptation and mitigation; and ensuring that knowledge gaps are bridged. The key priorities proposed under the missions to enhance the coping capacity of the population are as follows:

Key Priorities: Strategic Knowledge Mission

1. Map & Analyse knowledge sharing process, flow, protocol and priority
2. Compile currently available information for assessment frameworks, methods and tools for climate impacts, decision support tools, good adaptation practices
3. Establishment of weather and climate data and easy access of data to climate researchers and end users of vulnerable community as well as livelihood sector
4. Capacity building including developing of Training module and imparting through regional workshop
5. Develop & maintain online portal website
6. Organize annual forum, learning seminar, workshop and dissemination of knowledge based product

Gender Issues and Climate Change

Women are affected disproportionately and differently, due to climate change and associated natural disasters such as floods, droughts, cyclones and storms. This is largely because men and women are bound by distinct socio-economic roles and responsibilities that give rise to differences in vulnerability and ability to cope with these climate change consequence.

Women usually, require firewood, or even modern fuel and water for cooking and any constraint posed by lack of access or contamination, leaves them more vulnerable. The work participation rate of male and female vary significantly. In post disaster recovery period, the females stay in hostile terrain whereas males migrate.

Therefore it is important that issues relating to gender safety, violence against women during climate stressed scenarios and adaptation options which are gender segregated need to be worked upon and friendly policies for women need to be incorporated.

Actions and Budget

Both adaptation and mitigation actions are proposed under the SAPCC. The actions proposed under the action plan are prioritized as high medium and low based upon the degree of importance and barrier envisioned during its implementation. The proposed budgetary estimations for implementation of high priority action under the Climate Change Action Plan in different sectors are only a rough estimate. The total budget has

been estimated at INR 384641 million for a 5-year period. Parts of the Action Plan can be planned to be implemented based on the availability of resources of the departments and additional funds may have to be provided for certain activities. As the implementation

activities make progress, the exact situation will be more easily visualized. The synopsis of the number of actions being proposed under each sector and the budget for the priority actions are as follows:

Sl No	Sector	No of high Priority Actions	Total No of Action Proposed	Indicative Budget (in Rs Million)	
				Existing	Additional
1	Sustainable Energy mission	9	15	1470.1	3970.4
2	Enhanced Energy Efficiency	16	16	1572	5548.4
3	Sustainable Habitat	13	29	0	321788.5
4	Green India Mission	15	22	0	1173.5
5	Water Mission	11	24	0.00	45676.00
6	Sustainable Agriculture	17	41	100.0	1044.2
7	Disaster Management	10	21	0.00	223.5
8	Tourism	7	18	0.00	697.5
9	Health	8	22	0	1047.0
10	Himalayan Mission	10	10	0	290.0
11	Strategic Knowledge	6	11	0	40
	TOTAL	122	229	3142.1	381499





1 Background

1.1. Introduction

The fifth assessment report of the Intergovernmental Panel on Climate Change has clearly established the evidences of variation in global climate parameters such as increase in global average air and ocean temperatures, widespread melting of snow-cover and ice and rise in global average sea level. As per the IPCC WGI Fifth Assessment Report the earth has been in radiative imbalance, with more energy from the sun entering than exiting the top of the atmosphere, since at least circa 1970. It is virtually certain that Earth has gained substantial energy from 1971–2010. The estimated increase in energy inventory between 1971 and 2010 is $274 [196 \text{ to } 351] \times 10^{21} \text{ J}$. The Global Mean Surface Temperature has increased since the late 19th century with the last three decade being comparatively warmer with the decade of 200 being the warmest. The global combined land and ocean temperature data show an increase of about $0.89^{\circ}\text{C} [0.69\text{--}1.08]$ over the period 1901–2012 and about $0.72^{\circ}\text{C} [0.49\text{--} 0.89]$ over the period 1951–2012.

Global mean sea level has risen by $0.19 [0.17 \text{ to } 0.21] \text{ m}$, estimated from a linear trend over the period 1901–2010 with mean rate of sea level rise around $1.7 [1.5 \text{ to } 1.9] \text{ mm yr}^{-1}$ between 1901 and 2010. Between 1993 and 2010, the rate was very likely higher at $3.2 [2.8 \text{ to } 3.6] \text{ mm yr}^{-1}$. Further to AR4, AR5 reveals that glacier change have increased substantially in number. There is very high confidence that, during the last decade, most ice was lost from glaciers in Alaska, the Canadian Arctic and the periphery of the Greenland ice sheet, the Southern Andes and the Asian mountains. Total mass loss from all glaciers in the world, excluding those on the periphery of the ice sheets, was very likely $226 [91 \text{ to } 361] \text{ Gt yr}^{-1}$ (sea-level equivalent $0.62 [0.25 \text{ to } 0.99] \text{ mm yr}^{-1}$)

A global scientific consensus with regard to the unequivocal nature of the climate change and its association with increasing concentration of greenhouse gas emissions is widely established. The change in climatic condition has manifested through alteration

in frequency, intensity and spatial extent of weather and climate extremes. The range of hydro meteorological events include heat waves, heavy precipitation, drought, and tropical cyclones which have emerged as serious threats to both human and natural systems. Such climatic variability is likely to lead to extreme conditions or impacts. The impacts over economy or sustainable development are severe when the intensity of the event goes beyond a critical threshold in a social, ecological, or physical system, or through interplay with any other event.

Climate model projections indicate that global average temperature would increase with disproportionate increase in the tropics and poles and increased frequency and intensity of extreme events. South Asia is likely to be more vulnerable to climate change issues due to high population density, poverty and high dependence on natural resources. Climate projections for India suggest that impacts are likely to be varied and heterogeneous, with some regions experiencing more intense rainfall and flood risks while others might encounter sparse rainfall and prolonged droughts. A projected spatial shift in pattern of rainfall towards the flood-prone areas and drought towards water scarce regions is among the more severe effects. Climate variability and climate change pose huge risks to human life in India and threatens the sustainability of the economic growth. India's geographic diversity adds to the complexity of developing and implementing unified adaptation strategy. The impacts of climate change vary across states, sectors, locations and populations because there can

be no uniform climate change strategy. The approaches need to be tailored to address the state specific issues and local vulnerabilities and conditions.

In June 2008, Government of India's (GoI) National Action Plan on Climate Change (NAPCC) was announced. The objective of NAPCC is to adopt and enhance ecological sustainability of India's development path. In August 2009, the Hon'ble Prime Minister of India urged each state Government to formulate its own state level action plan in line with the strategies of the National plan. This was re-emphasized by the Union Minister of State, Environment & Forests (MoEF), at the meeting of the Chief Secretaries in February 2010. The state Government of J&K has thereafter institutionalized a three tier State Council for formulation of the State Action Plan on Climate Change.

The state of J&K located in the northern part of the Indian sub-continent and sharing international border with Pakistan, Tibet and China has the most strategic location among the states in the country. It falls in the great northwestern complex of the Himalayan Ranges with marked relief variation, snow-capped summits, antecedent drainage, complex geological structure and rich temperate flora and fauna. Irrespective of the ethereal beauty, majesty and grandeur the ecosystem of the state is considered the most fragile on the earth. The lives and livelihood of the state are heavily dependent on the natural resources and climatic conditions. Agriculture and its allied sector including horticulture, floriculture, sericulture remains

the mainstay of the state's economy followed by tourism, mining and MSME sector. Apart from agriculture and allied activities, the economy and livelihood in the state is largely dependent on tourism sector. Owing to the dependency on natural resources and the weather condition, the livelihood of the people needs to be climate independent.

The state economy is characterized by steady growth. However it is impossible to attain a sustainable economic development even with strong economic growth, revenue realization and natural resource base if social issues like poverty, unemployment and gender inequality are not properly addressed. In order to promote sustainable development and enhance the adaptive capacity as well as resilience of the social sector, the state Government has been emphasizing upon improving and strengthening livelihood status, healthcare facilities, disaster management, poverty alleviation, literacy and improving gender inequalities. Moreover the State Government has also recognized that the climate change is detrimental to economic progress. Owing to its importance, Government of Jammu and Kashmir has committed to demonstrate continued leadership in this new and important area.

1.2 Objective

The main objective of State Action Plan on Climate Change (SAPCC) is to strategize adaptation and mitigation initiative towards emission stabilization and enhancement of ecosystem resilience, climate proofing of the livelihood sector and diversification of

dependency on the natural resources. This exercise would serve as a platform to forward the State climate change agenda which in future could be a combination of advocacy, knowledge deepening, policy analysis and operational work. However, there is also a need for encouraging public investment to make the State and community more climate resilient.

1.3 Methodology

Government of India has drafted the 'National Action Plan on Climate Change (NAPCC)' to meet the challenge of sustaining economic growth in an ecologically sustainable manner as well as framing of adaptation strategies to manage the risk associated with extreme weather and climate change related events. In order to achieve the objectives of the NAPCC and link states' initiative for climate proofing life and livelihood of the vulnerable section of the society, Government of Jammu and Kashmir had constituted the J&K State Council on Climate Change on 25th of April 2011. The State Council on Climate Change is a three-tier structure comprising of Governing Council headed by Hon'ble Chief Minister, Executive Council headed by the Chief Secretary and Expert Committee headed by the Director, Environment & Remote Sensing. The first meeting of the Expert Committee held on 9th August 2011 drew the road map of the SAPCC. Considering the issues related to the impacts of climate change on the ecologically sensitive as well as economically important sector. Ten missions were identified specific to the state and corresponding ten working groups were identified and sanctioned by

the Government. While framing the council and the working group, the membership was drawn both from the primary department and also from the different associated departments/ organizations. A nodal officer was appointed to co-ordinate the regular conduct of meetings, to collect/collate the required information and to develop the plans pertaining to each of the sectors. The list of the member representative of the council, the working group and the nodal department is presented in the Annex to the Report.

Sectors in J&K as mapped to National Missions

1. Energy - Solar Mission and Renewable Energy [directly linked to NAPCC mission]
2. Enhanced Energy Efficiency [directly linked to NAPCC mission]
3. Water [directly linked to NAPCC mission]
4. Sustainable Habitat [directly linked to NAPCC mission]
5. Sustainable Agriculture [directly linked to NAPCC Mission]
6. Tourism (indirectly linked to 3, 4 and 7)
7. Sustainable Himalayan Ecosystem [directly linked to NAPCC mission]
8. Health (indirectly linked to 3, 4) and NRHM
9. Disaster Management (indirectly linked to 5, 7 and 3)
10. Strategic Knowledge mission [directly linked to NAPCC mission]

Figure 1: Sectors of J&K mapped to National Mission

A top down and bottom up approach was followed in developing the SAPCC. Consultations were carried out with members of working groups as well as with the administrative departments. Administrative heads were consulted recognizing the need for involvement of the senior bureaucracy within the government towards assessing climate change issues and mapping the solution. Tools and templates were formulated for listing climate change specific issues, prioritizing the activities as high, medium and low and developing an outline of the sub-activities to be done under the high priority activities, estimating the budget and sources of funding as well as formulation of the monitoring and verification protocol.

As part of the above exercise sectoral vulnerability was mapped and climate sensitive issues were identified. These were presented in course of assessment before the working group. A list of possible activities relevant in the climate change context were formulated and deliberated on. Stakeholders consultation workshop were also convened to obtain specific feedback in regard to climate change issues and possible actions to address the concern of anthropogenic emission of GHG into the atmosphere as well as climate proofing of the sector. Recognizing that all activities are equally important a prioritization exercise was undertaken to divide the comprehensive list of activities into high, medium and low priority. Subsequent brainstorming exercise was undertaken for a sharper focus and to finalize 10-12 key priorities for immediate implementation under the five year plan period. Descriptions

of the key priorities were developed, implementing organizations were identified and budgets were estimated. Viability gap funding requirement was assessed through action that has linkage with existing program. The brief of the proposed and the priority actions formulated was shared with external stakeholders from academics, civil societies and government agencies.

The purpose of the consultation was to make the climate change agenda inclusive, transparent and with validated ownership of the different stakeholders. Initially, the process started with involvement of administrative and technical groups within the government, but it was extended to public to validate the proposed actions

from the point of view of the wider interest group. GIZ provided support for this action planning facilitation and engaged CTRAN to coordinate and manage the process on behalf of the Government of J&K. The report tries to best capture the views of the stakeholders and put them together in an unbiased and professional manner. The team with the knowledge partner has tried to use the various frameworks of the National Climate Change Action Plan in the sub-national context encompassing vulnerability, resilience and climate governance to frame policies at the State level. Efforts were put to support climate change adaptation policies and manage the potential social risks associated with climate change mitigation.



The brief methodology adopted is presented as follows

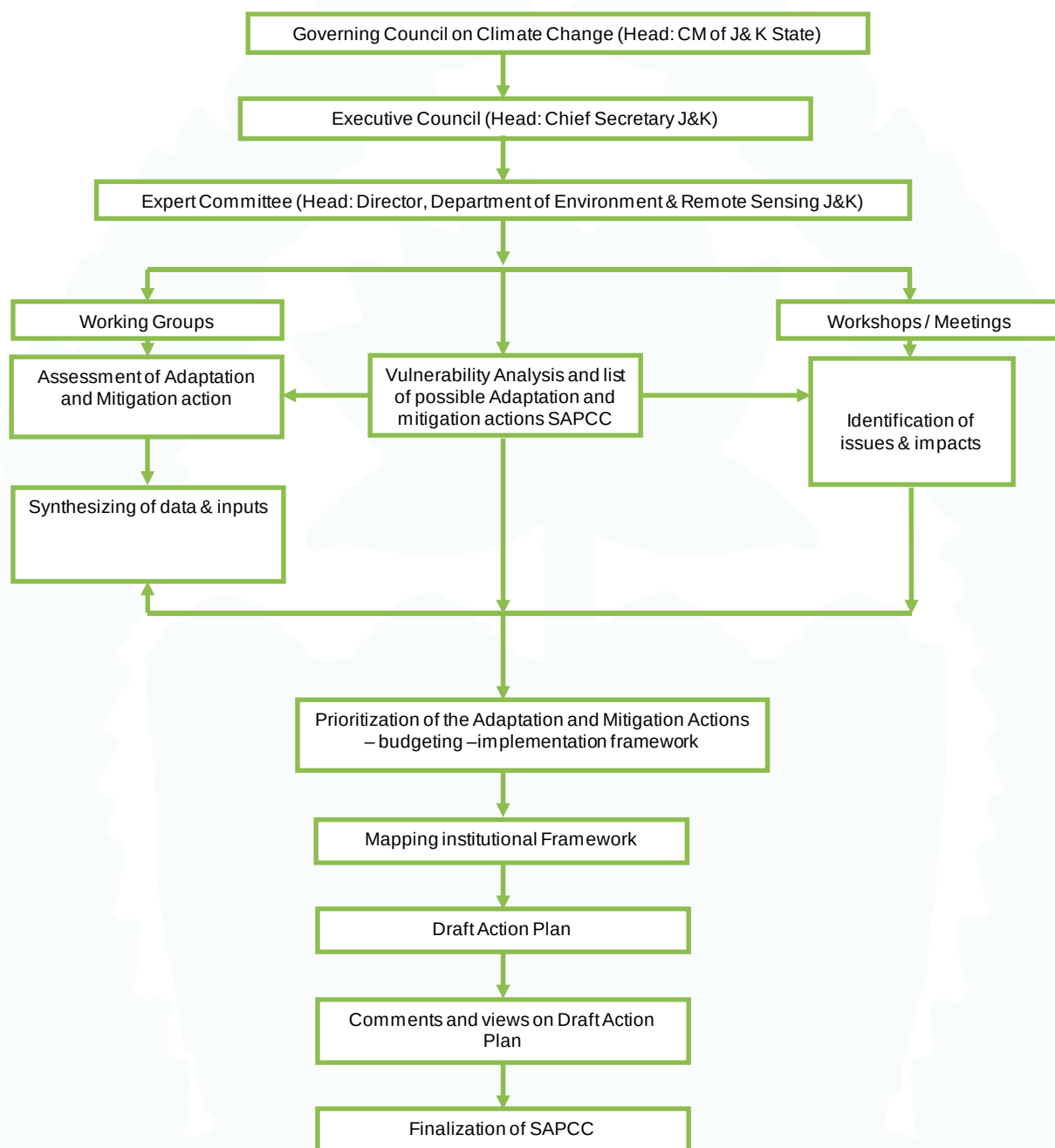


Figure 2: The process of formulation of SAPCC in J&K

1.4 Structure of the Report

- The SAPCC document begins with the background chapter introducing the context. The chapter provides an insight into the methodology and outlines the structure of the document.
- The second chapter gives an overview of the National Action Plan on Climate Change, eight missions and other initiatives.
- The third chapter describes the vulnerability assessment of the State

of Jammu and Kashmir and highlights climate sensitivity from both biophysical and socio economic perspectives.

- The fourth to fourteenth chapter highlights a detailed overview of the sectoral Climate Change Issues and describes actions including prioritized one to address the issues across all the ten sectors.

- The fifteenth chapter outlines over the cross cutting issues.
- The final chapter provides the conclusions and recommendations. The Annexes include the composition of the councils, working groups, nodal department, list of external stakeholder consultations held, sector wise table of key priorities, comprehensive list of activities considered and list of references.







2 National Action Plan on Climate Change

2.1 Introduction

National Action Plan on Climate Change emphasizes on the overriding priority of maintaining high economic growth rates to raise the living standards. The NAPCC aligns the measures to promote development objectives with co-benefits for addressing climate change effectively. It also advocates strategies that promote, firstly, the adaptation to Climate Change and secondly, further enhancement of the ecological sustainability of India's development measures.

2.2 Domestic Actions

India's National Action Plan on Climate Change emphasizes on promoting inclusive and sustainable development strategy sensitive towards climate change so as to protect the poor and vulnerable section of the society. Eight National Missions form the core of the National Action Plan represent multipronged, long term and integrated

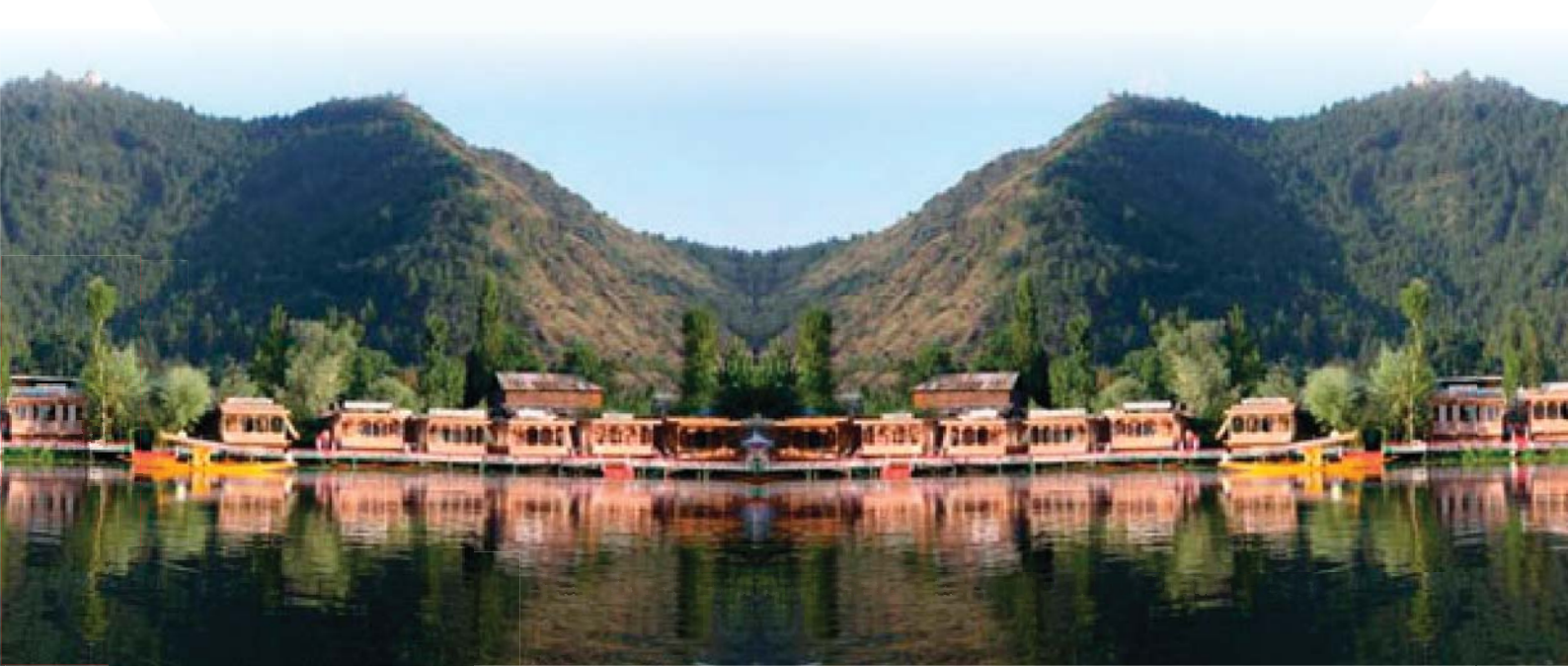
strategies for achieving key goals in the context of climate change. The focus is to promote understanding of Climate Change, framing adaptation and mitigation strategy and promoting energy efficiency and natural resource conservation. While many of these programmes are already a part of the current actions, the Action Plan seeks to enhance them in scope and effectiveness and implement them in an accelerated manner through time bound plans.

2.3 National Mission

On June 30, 2008, Prime Minister Dr. Manmohan Singh released India's first National Action Plan on Climate Change (NAPCC) outlining existing and future policies and programs addressing climate related mitigation and adaptation measures. The plan identifies eight cores 'National Missions' running through 2017. The various missions are presented in the following diagram.



Figure 3: Eight fold mission under NAPCC



The following table gives an overview of the Eight National Missions, the objectives and the responsible entity for the same:

Table 1: National Mission on Climate Change its goal and link with development goal¹

Mission	Goal	Link with Development Goal ¹
National Solar Mission Responsible Entity: MNRE	- Enabling environment to deliver 20 GW of solar power by 2022 - Grid-connected solar power capacity of 1 GW by 2013 - Additional 3 GW through mandatory purchases backed with preferential tariff by 2017 - Favorable conditions for solar manufacturing capabilities - Off-grid applications: 1 GW by 2017, 2 GW by 2022 15 million m² solar thermal collector area by 2017 and 20 million m² by 2022 20 million solar lighting systems for rural areas by 2022	Capacity addition of 16,553 MW hydro, 3380 MW nuclear (out of the total of 78,577 MW capacity addition) Ensuring electricity connection to all villages and below poverty line (BPL) households by 2009 and reliable power by the end of the plan [56% electrification rate (2000–05), 487.2 million population without electricity (2005)] 20% rise in real wage rate of unskilled workers [Youth literacy rate of 76.4% (for ages 15–24)] Reduction in head-count ratio of consumption poverty by 10 percentage Points
National Mission for Enhanced Energy Efficiency Responsible Entity: Ministry of Power	- Specific energy consumption (SEC) reduction targets for energy-intensive units - Incentivizing action through Energy Savings Certificates (ESCerts) – traded and used for compliance - National energy efficiency Clean development Mechanism(CDM) roadmap - National energy efficiency financing platform - Creating markets for energy efficient products and services	Increase energy efficiency by 20% by 2016/17

¹Planning Commission; UNDP (2007); NAPCC (2008)

Mission	Goal	Link with Development Goal ¹
National Mission for Sustainable Habitat Responsible Entity: Ministry of Urban Development	- Increasing energy efficiency in buildings: building bye laws and standards, energy performance monitoring, national standards for construction and recycling of construction waste - Urban transport: norms integrating congestion charges, parking, etc., norms for pedestrian and cycling, integrating transport planning with spatial planning - Water supply: mandatory rainwater harvesting, water and energy audits	Develop minimum standards of education at elementary level, to ensure Quality To attain WHO quality of air standards in all major cities by 2011/12 Provide homestead sites to all by 2012 and step up the pace of house construction for rural poor to cover all the poor by 2016/17
National Water Mission Responsible Entity: Ministry of Water Resources	- Comprehensive water database in public domain and assessment of impact of climate change on water resources - Promotion of citizen and state action for water conservation, augmentation and preservation - Focused attention to vulnerable areas including over-exploited areas - Increasing water use efficiency by 20% - Promotion of basin level integrated water resources management	Ensuring water security and food security (MDG 1: Eradicate extreme poverty and hunger) through efficient use of water and adaptation to climate change Ensure integrated water resources management (MDG 7: Ensure environmental sustainability)
National Mission for Sustaining the Himalayan Ecosystem Responsible Entity: Ministry of Science & Technology	- Continuous monitoring of Himalayan ecosystems - Identification of desirable adaptation and development policies (sustainable urbanization, water security: rejuvenation of springs, infrastructure development: green roads)	Ensure all-weather road connection to all habitations with population 1000 and above (500 and above in hilly and tribal areas) by 2009, and all significant habitations by 2015
National Mission for a Green India Responsible Entity: Ministry of Environment & Forests	2 Mha of moderately dense forests 4 Mha of degraded forests regenerated/afforested 0.10 Mha of mangroves restored, 0.1 Mha wetlands conservation 0.20 Mha urban/peri-urban forests, 1.50 Mha degraded land under agro-forestry	Increase forest and tree cover by 5 percentage points [22.8% of forest cover of the total land area with an annual change of 0.4% between 1990 and 2005]

Mission	Goal	Link with Development Goal ¹
National Mission for Sustainable Agriculture Responsible Entity: Ministry of Agriculture	• Use of genetic engineering to produce carbon responsive crops • Low input sustainable agriculture: enhanced water use efficiency • Micro-irrigation for efficient use of water • Water conservation in rain-fed areas	Average GDP growth rate of 9% [between 1990 and 2005, 34.3% of population earning US \$1/day, while 80.4% earning US \$2/day] Agricultural GDP growth rate on average of 4%
National Mission on Strategic Knowledge for Climate Change Responsible Entity: Ministry of Science & Technology	• Climate change research and fellowship programme • Climate Change Professor Chairs • National Research Chairs Climate Research Institute • Network of climate change research institutes and scientists	

The idea of a sub-national action plan emerged to address the local issues and has high ownership. This would generate better awareness rising from experiences on climate related issues to corrective actions, better preparedness and also to set strategic priorities at the Sub-National level. These priorities would enable the leaders in the states to plan for the resources and also to monitor the savings in terms of long run cost associated with climate change.

2.4 Other Initiatives

Apart from eight fold National Missions, there have been several initiatives that have positive influence on mitigating the adverse impact of climate change. These include

- Establishment of market based instruments in sectors that have maximum influence on climate change viz. Perform Achieve and Trade for energy efficiency, white certificates in renewable in energy sector, air

pollutant trading in industry and mining sector and offset instruments in forestry sector such as compensatory afforestation.

- Encouraging Kyoto market instruments like Clean Development Mechanism (CDM).
- Other initiatives like Bio-Diversity Conservation, Wetland Management.
- Research & development in the area of designing efficient thermal power plants; promoting energy efficiency in transmission and distribution sector including adoption of high-voltage AC and high-voltage DC transmission to reduce technical losses; promoting small and large scale hydro power project; promotion of renewable energy technologies such as bio-mass combustion and gasification-based power generation and geothermal based energy generation.

- e. Enhancements in the regulatory/tariff regimes to help mainstream renewable-based sources in the national power system; and renewable energy technologies for transportation and industrial fuels.

2.5 National Mission and Objectives

2.5.1. National Solar Mission

The NAPCC aims to popularize the use of solar energy for power generation and other purposes over other fossil-based energy options. The plan includes:

- Specific goals for increasing use of solar thermal technologies in urban areas, industry, and commercial establishments;
- A goal of increasing production of photo-voltaic power to 1000 MW/year; and
- A goal of deploying at least 1000 MW of solar thermal power generation.

Other objectives include the establishment of a solar research centre, increased international collaboration on technology development, strengthening domestic manufacturing capacity, and increased government funding and international support.

2.5.2. National Mission for Enhanced Energy Efficiency

Current initiatives are expected to yield savings of 10,000 MW by 2012. Buildings

adhering to the Energy Conservation Act 2001 which plan recommends:

- Mandating specific energy conservation options decreases consumption in large energy-consuming industries, with a system for companies to trade energy-savings certificates;
- Energy incentives, including reduced taxes on energy-efficient appliances; and
- Financing for public-private partnerships to reduce energy consumption through demand-side management programs in the municipal, buildings and agricultural sectors.

2.5.3. National Mission on Sustainable Habitat

The National Mission on Sustainable habitat comprises of three components, viz.

- Promoting energy efficiency in the residential and commercial sectors
- Management of municipal solid waste and
- Promotion of urban public transport

To promote energy efficiency as a core component of urban planning, the plan calls for:

- Extending the existing Energy Conservation Building Code (ECBC);
- For addressing component 2 on management of municipal solid waste, a greater emphasis on urban waste management and recycling, including

power production from waste is envisaged;

- For addressing promotion of urban public transport the following are promoted:
 - o Strengthening the enforcement of automotive fuel economy standards and using pricing measures to encourage the purchase of fuel efficient vehicles; and
 - o Incentives for the use of public transportation.

2.5.4. National Water Mission

Some of the major areas of intervention identified in the Water Mission are to increase efficiency of water use, to explore options to augment water supply in critical areas and to ensure more effective management of water resources. With water scarcity projected to increase as a result of climate change, the Mission sets a goal of a 20% improvement in water use efficiency through pricing and other measures.

2.5.5. National Mission for Sustaining the Himalayan Ecosystem

The Himalayan Ecosystem is vital to the ecological security of the Indian landmass, through growing forest cover, feeding perennial rivers which are sources of drinking water, irrigation and hydropower, conserving biodiversity, providing a rich base for high value agriculture, and spectacular landscapes for sustainable tourism. Climate change may adversely impact the Himalayan ecosystem through increased temperature,

altered precipitation pattern, glacial melting and episodes of drought. The plan aims to conserve biodiversity, forest cover and other ecological values in the Himalayan region, where glaciers which are major sources of India's water supply are projected to recede as a result of global warming.

2.5.6. National Mission for a "Green India"

The Mission aims at addressing climate change by enhancing carbon sinks in sustainably managed forests and other ecosystems, enhancing the resilience and ability of vulnerable species/ecosystems to adapt to the changing climate, and enabling the adaptation of forest dependant local communities in the face of climatic variability. Goals include afforestation on 6 million hectares of degraded forest lands and expanding forest cover from 23% to 33% of India's territory.

2.5.7. National Mission for Sustainable Agriculture

The National Sustainable Agriculture Mission aims to focus on four areas crucial to agriculture for adapting to climate change, viz., dry land agriculture, risk management coupled with weather insurance, access to information and use of Biotechnology. The Mission aims to support climate adaptation in agriculture through the development of climate-resilient crops, expansion of weather insurance mechanisms and agricultural practices.

2.5.8. National Mission on Strategic Knowledge for Climate Change

To gain a better understanding of climate science, impacts and challenges, the plan envisions a new Climate Science Research Fund, improved climate modeling, and increased international collaboration. It also encourages private sector initiatives to develop adaptation and mitigation technologies through venture capital funds.

2.6 Mapping National Mission in the State

The following section maps the State missions according to the National Action Plan. Therefore it would have the following kind of linkages as detailed out in the table below:

Table 2: Relating State Mission under Climate Change and National Mission

National Mission	State Mission	Key Issues to be Addressed
Solar mission	Energy Mission	- Reduction of carbon foot print – Bringing about grid parity for solar PV and solar thermal technology including concentrator - Address energy security - R&D collaboration, technology transfer and Capacity building - Promotion of Renewable energy technology other than Solar - Address issues of Energy Security – Electrification and issues of rural energy - Green building and energy security
Enhanced Energy Efficiency Mission	Enhanced Energy Efficiency Mission	- Energy efficiency in industrial (MSME sector) facilities, commercial, agricultural and residential sector , - Fuel switch (shifting of Diesel based decentralized power generation option to Renewable based power generation option and also addressing the issue of lowering T&D losses - Financial mechanism (ESCO) - Renovation and modernization of existing hydro power project
Sustainable habitat Mission	Sustainable Habitat mission	- Climate resilient urban planning - Solid waste Management - Efficient public transport mechanism

National Mission	State Mission	Key Issues to be Addressed
Water Mission	Water Mission	- Water management and water use efficiency - Basin management - Management and regulation of ground water - Conservation of wetland
Mission on Himalayan Ecosystem	Mission on Sustainable Himalayan Ecosystem	- Academic research, glacial flow, delta management, local variability - Appropriate land use planning
Green India Mission	Green India Mission	- Forestry, Bio-diversity conservation - Livelihood promotions
Sustainable Agriculture Mission	Mission on Sustainable Agriculture	- Develop new varieties of crops - Flood resistant varieties and dry land agriculture - Methane management from rice cultivation - Soil and water conservation - Promotion of Horticulture , Floriculture and sericulture - Climate proofing of the animal husbandry sector
Strategic Knowledge for Climate Change	Strategic knowledge on climate change	- Enhanced research on Climate modeling - Capacity building and awareness generation
Cross Cutting	Tourism mission	- Climate proofing of the fragile and sensitive tourist destination - Climate resilient planning
	Health Mission	- Improving health infrastructure - Early warning system of health related stress - Reduce surface water contamination and prevention of water borne diseases
	Disaster Risk management	- Disaster management planning and infrastructure development - Awareness creation amongst the community on climate related vulnerability and preparedness

Apart from eight national missions the state has proposed three separate missions under cross-cutting themes. Cross-sectoral working groups are formulated to improve the climate related governance so that the issues can be addressed in a holistic manner. The state has decided to focus on a comprehensive strategy for adaptation to climate change in different sectors and for mitigation the State would follow the national policy and no sectoral targets have been set as a part of the action plan. Inclusion of market mechanism like CDM, REDD+ will be a co-benefit.

2.7 Institutional Mechanism

The State Missions are to be institutionalized by the respective departments with support of the inter-sectoral groups. Appropriate

mechanisms including public private partnership and civil society actions, are to be devised, as suited, for effective achievement of each individual Mission's objectives. The State has proposed constitution of J&K Climate Change Cell which would act as a nodal entity to address the climate change issue in a holistic manner.

The proposed institutional mechanism consists of the J&K State Council on Climate Change and Climate Change Cell. These bodies would coordinate among departments on cross cutting issues. It would also interpret the various national missions, objectives and the strategy in the context of the State. The body would forge partnership with private sector, academic institutions and civil society to facilitate the implementation of various proposed actions in the State.





3 Vulnerability Analysis

3.1 Introduction

Climate change has become the key environmental concern around the globe, and Governments, Non-Governmental Organizations, Environmentalists, Conservationists and Donors around the globe are focusing on this issue. Since the industrial revolution, the mean surface temperature of the Earth has increased by 10°C and temperatures are likely to continue to rise, with serious consequences for biodiversity, natural ecosystems, agricultural productivity, food security and human livelihoods. In the context of India, especially Jammu and Kashmir, which nestles in fragile Himalayan Ecosystem; there are natural fluctuations in climate, human induced changes due to large scale urbanization in the valley is driving the warming trend. The biodiversity loss and water stress owing to the climate change are the greatest challenge for the state over the coming decades. Climate change poses a serious threat to the species diversity, habitats, forests, wildlife, fisheries and the water resources in the region. Many wetlands in J&K that support 20% of the known range

of biodiversity in the region are adversely affected. The climatic changes resulting in changing weather movements are relatively unpredictable, but could be important in context of the primary incursion of different diseases like avian influenza. However, the Governments, local communities, NGOs, civil society, scientists and conservationists need to deliberate on the more important issues of the role that natural ecosystems, biodiversity and nature-based solutions can play in increasing resilience and reducing vulnerability to climate change.

The vulnerability and adaptive capacities are diverse and varies from state to state based on several sectoral and cross sectoral parameters. Sectoral parameters include key sectors of the state's economy and cross sectoral factors include (a) Poverty (b) inequality and social discrimination over property rights and (c) access to resources (d) social attrition/migration, and (d) unequal and unsustainable competition for scarce natural resources.



Figure 4: Vulnerability Profiling

3.2 Climate Sensitivity

The district level data to compute climate change risk for J&K is not available. The map below from UNEP describes relative vulnerability of the state spatially. Vulnerability map of Indian Subcontinent

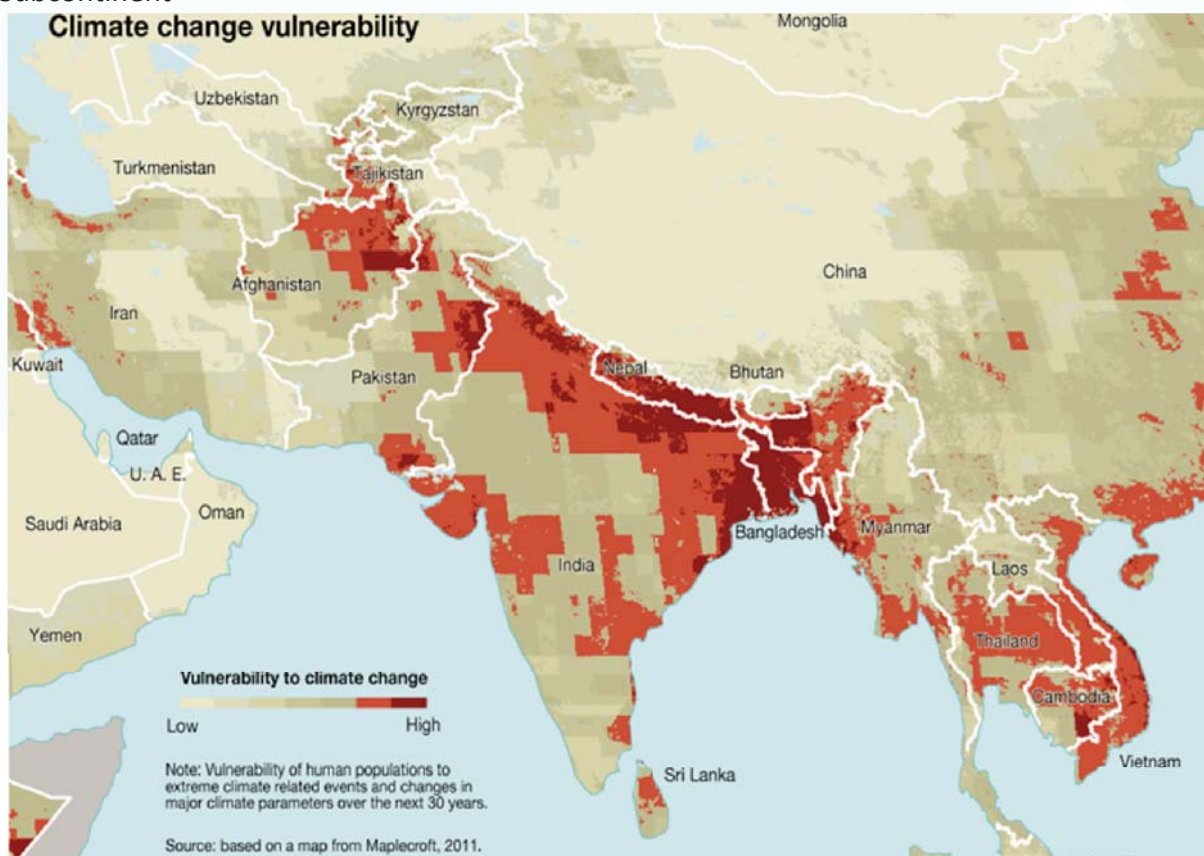


Figure 5: Vulnerability map of Indian Subcontinent

From the above picture, it is clear that some parts of the state are moderate to high vulnerability. A large part of the state are under seismic zone but, that cannot be strictly linked to climate change.

Climate profile of J&K indicates variation in climate characteristic over the years. According to IMD², there is increase in an average temperature in Jammu & Kashmir. Kashmir valley has shown rise of 1.45°C & Jammu region has shown a rise of 2.32°C over last two decades. Maximum temperature has increased by 0.5°C per year in Kashmir valley. In Jammu region, maximum temperature has risen by 0.08°C per year. Also amount of snow fall has reduced over the years. As per UNEP and ICIMOD, the temperature in Himalayan region has risen by 1°C since 1970s. This has caused meltdown of snow and glaciers at rate of 15 m/yr even in winter. As a consequence, the water availability is reduced. Further, changes in the rainfall pattern and relative humidity add to weather inconsistencies which exist at the time of plantation and harvesting.

As per IPCC 2007, glacial melt is projected to increase flooding, snow avalanche fall from established slopes and diminishing water supplies and decrease in river flows as glaciers recede. The impact of change in climate shall be severe on natural resources and the environment associated with urbanization, industrialization and economic development. In the context of climate change scenario in J&K, it is pertinent to study the climate change projections in J&K.

Projection for Climate Change for Jammu and Kashmir is predicated on 4x4 report of The Indian Network for Climate Change Assessment (INCCA), 2010 covering the Indian Himalaya including Jammu and Kashmir. In this report, the future period 2021–2050 under SRES scenario A1B (atmospheric CO₂ concentration of 490 ppm by 2035) is considered. The mid-year for this future period is 2035, and hence assessment period is referred as the ‘2035’ scenario. It should be noted that scenario ‘2035’ does not represent the exact year 2035: rather, it refers to the climate averaged over 2021–2050. Comparison of the results of 2035 scenario with the “baseline” scenario has been made, which represents the averaged observed climate over the period 1961–91. “Baseline” is also referred to as either the “reference” or “control” case.

3.2.1 Precipitation

The projections as per INCCA show the following:

Extreme precipitation: The number of rainy days in the Himalayan region in 2030s may increase by 5-10 days on an average, with an increase by more than 15 days in the eastern part of the Jammu and Kashmir region. The intensity of rain fall is likely to increase by 1-2 mm/ day. This is likely to impact some of the horticultural crops though rangelands and livestock are likely to benefit.

Projection of Precipitation: The PRECIS run for 2030’s indicate that annual rainfall in

IMD: India Meteorological Department, ministry of Earth Sciences, Govt. of India

the Himalayan region is likely to increase in 2030's with respect to 1970's range from 5% to 13% with some areas of Jammu and Kashmir showing an increase of up to 50%. All seasons in the Himalayan regions indicate an increase in rainfall, with monsoon months

of June, July, August and September showing the maximum increase in the rainfall. The winter rain in the month of January and February are also projected to increase whereas minimum increase is indicated in October, November and December.

3.2.2 Temperature

The temperature variations have been shown in the table below:

Table 3: Temperature variation at Srinagar (°C)

Sl. No.	Month	1985	1990	1995	1996	1997	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
1.	January	-1.3	-1.1	-4.7	-2.7	-3.2	-0.8	-2.0	-3.6	-2.8	-2.9	0.3	-0.3	-01.3	-02.5	-02.5	0.4	-01.5
2.	February	-1.3	1.5	-0.7	0.7	-0.8	2.2	-0.3	-0.3	-0.5	0.4	1.1	0.7	03.3	02.9	01.4	1.5	00.4
3.	March	4.7	2.5	3.4	4.9	4	4.6	2.9	3.2	4.5	3.7	5.6	5.3	04.7	03.0	05.3	5.0	06.5
4.	April	8.6	6.9	7.4	7.5	7.4	8.3	7.9	8.5	8.6	8.6	8.5	7.1	07.2	08.9	07.7	8.0	09.0
5.	May	10.9	11.9	10.5	10.0	9.6	11.4	13.2	14.5	11.3	9.1	10.7	9.9	13.3	11.9	11.5	10.8	11.1
6.	June	14.1	16.3	14.1	15.4	14.7	14.4	16.3	17.4	15.2	14.9	15.0	14.4	14.7	16.0	18.3	13.0	13.5
7.	July	19.1	18.5	18.7	17.8	20	18.4	18.7	19.5	17.1	18.9	16.9	18.0	19.3	17.8	19.1	16.9	17.7
8.	August	17.9	18.3	18.5	17.5	17.1	17.3	17.7	17.6	18.2	17.1	17.1	16.9	18.3	17.8	17.8	17.9	18.8
9.	September	12.5	14.1	11.4	13.8	13.8	14.5	11.3	11.2	11.6	13.6	12.6	13.7	12.3	13.3	11.3	11.9	13.3
10.	October	6.5	4.7	6.5	5.9	7.6	5.1	5.6	6.1	6.6	5.1	6.2	5.8	08.0	03.9	06.9	5.1	07.3
11.	November	0.3	0.3	-0.6	1.1	2.3	2.5	2.5	0.9	0.7	0.7	1.1	-0.02	03.5	-01.1	01.0	0.5	02.2
12.	December	-0.8	-1.7	-0.3	-3.1	0.5	-3.9	-0.7	-0.9	-0.8	0.0	-0.7	-0.03	00.0	-02.0	00.7	-0.8	-3.7

Source: Regional Meteorological Centre, Srinagar

Projections of Surface Temperature

The annual temperature is projected to increase from $0.9 \pm 0.6^{\circ}\text{C}$ to $2.6 \pm 0.7^{\circ}\text{C}$ in 2030. The net increase in temperatures is ranging from 1.7°C to 2.20°C with respect to 1970's. Seasonal air temperatures show rise in all seasons. However, one simulation (Q1 simulation) shows a decrease by 2.6°C in October, November and December.

Summary of data on Climate Change and its Impact indicate that since July (1901-1950), the maximum temperature in July (1976-96) has increased by $2-4^{\circ}\text{C}$. As per SASE report, Himalayan region has grown warmer by 2.2°C in last two decades. Between 1901 and 1996, rainfall has shown decline in Kashmir Valley. This has implication for change in cropping pattern and food security of the region.

3.2.3 Glacial Change and cold wave

The snowline and glacier boundary are sensitive to changes in climate conditions. Trend analysis of various glaciers from the past data gives some ideas of future projections. Glaciers are not well represented by PRECIS, modeling. Warming is likely to rapidly increase the rate at which glaciers are melting, leading to greater ablation than accumulation. Glaciers melt is expected to increase under change in climatic conditions, which would lead to increased summer flows in some rivers systems for few decades, followed by reduction in the flow as glaciers

disappear (IPCC 1998). According to IPCC, the increase in temperature by 0.74°C over 150 year have caused glaciers and snow cover to decline leading to reductions in melt water discharges. According to SASE report, Himalayas have grown warmer by 2.2°C in last two decades. Almost all glaciers reportedly show negative mass balance.

Glaciers in the context of Himalayan Region

Major glacier-fed Himalayan rivers, along with glaciated catchments, have regional importance the water from the glacier melt sustains stream flow in these rivers through the dry season. The "frozen water" in the Himalayas is crucial for the people inhabiting the mountain areas as well as those inhabiting the downstream regions. The Indus and the Ganges– the two major rivers in western Himalayan region – directly impact the lives of a large population living in the northern part of India, and even beyond the national boundaries.

The Indus basin has 7,997 glaciers with a total glacier cover of $33,679 \text{ km}^2$ and total ice volume of 363.10 km^3 . The Ganges basin has 968 glaciers with a total glacier cover of $2,857 \text{ km}^2$ and total ice volume of 209.37 km^3 . The contribution of snow to the runoff of major rivers in the eastern Himalayas is about 10% but more than 60% in the western Himalayas. Major glaciers in Indus basin are given in table 4.



Table 4: Major glaciers in the Indus basin of India

Basin	No. of glaciers	Glacier covered Area (km ²)	Ice volume (km ³)
Ravi	172	193	8.04
Chenab	1,278	3,059	206.30
Jhelum	133	94	3.30
Beas	277	579	36.93
Sutlej	926	635	34.95
Upper Indus	1,796	8,370	73.58
Shyok	2,454	10,810	NA
Nubra	204	1,536	NA
Gilgit	535	8,240	NA
Kishenganga	222	163	NA
Total	7,997	33,679	363.10

Table 5: Characteristic of glacial coverage in the state is given below.

Glaciers (no.)	Area (km ²)	Average Size (km ²)	Glacier%
526	29163	10.24	61.8

The rate of recession of glaciers is reportedly varying which is being attributed to winter precipitation climate warming and anthropogenic elements of landscape. Recent trend related to glaciers are given in table below.

Table 6: Recent retreat pattern of selected glaciers in Jammu and Kashmir

Name of the Glacier	Retreat of snout (m)	Observation Period	Trend	Avg. retreat rate (m/yr)
Drang-Drung	-	2004-07	No Change	-
Kangriz	-	1913-2007	No Change	-
	NA	1862-1909	Advancing Retreating	15.42
	--do--	1909-1929	Retreating No	2.5
	--do--	1929-1958	Change Retreating	14
	- NA	1958-1985	No change	-
	-	1985-2004		3
		2004-2005		-

Observed changes in glaciers in Jammu and Kashmir

Studies conducted during last three decades by the National Institute of Hydrology, Roorkee, reveal that in Ladakh, Zaskar and the Great Himalayan ranges of Jammu and Kashmir are generally receding, and the glacier volume changes range between 3.6% and 97%, with the majority of glaciers showing a degradation of 17%–25%. The 23 km long Drang-Drung glacier in the Zaskar valley is highly affected by western disturbances (ablation rate variations between 0.75 cm/day and 2.67 m/day during July and August); The studies, however, do not reveal any significant retreat during 2004–07.

The Nubra valley of Jammu and Kashmir has 114 small-sized glaciers varying between less than 5 km and 10 km in length. The glaciers of the valley do not show much change in their length and area during the period 1989–2001. However, variable decline in the glacial area of the Siachin glacier has been observed. The area has reduced from 994.99 km² in 1969 to 932.90 km² in 1989. However, small change in the area (932.90 km² to 930 km²) has been noticed during the following decade (1989–2001). Recession patterns of 466 glaciers in the Chenab, Parbati and Baspa basins of the western Himalayas have been studied for the period 1962–2008. Here, a reduction in the glacial area from 2,077 km² to 1,628 km² and an overall de-glaciation of 21% has been observed.

Most of the glaciers in western Himalayas are receding (except a few in Jammu and

Kashmir, which do not show any change or are advancing). The processes controlling the rate of retreat of glaciers are complex and vary with location and topography of the area. However, the impact of rising temperature and reducing snowfall on glacier mass balance is reflected in these findings, which may require a sound long-term database for precise climate change assessment.

Further, quantity of snow fall has reduced (Reported 0.60 m of snow today as compared to 3 m about 40 year ago in Fakir Gujri); Total loss in extent of glaciers is reported to be 17.8 km² between 1911 and 2001 e.g. reported retreats in Kolhai glacier from 112 km² to 94 km² between 1911 and 2004. Large glaciers are fragmented and small ones have diminished e.g. more or less disappearance of Nijwan Akal glacier in Sindh valley; Change in maximum and minimum rainfall in Kashmir valley (Minimum has fallen for 181 mm to 84 mm and maximum from 65.89 m and 21.7 mm between 1901 to 1994); Extent of wetlands decreased between 1911 & 2004.

3.3 Vulnerability of agriculture and allied activities due to climate sensitivity

Temperature, precipitation and cold wave significantly impacts the agriculture sector and enhances its vulnerability. This happens due to the early onset of rain and number of dry days. The valley has been receiving less amount of rainfall.

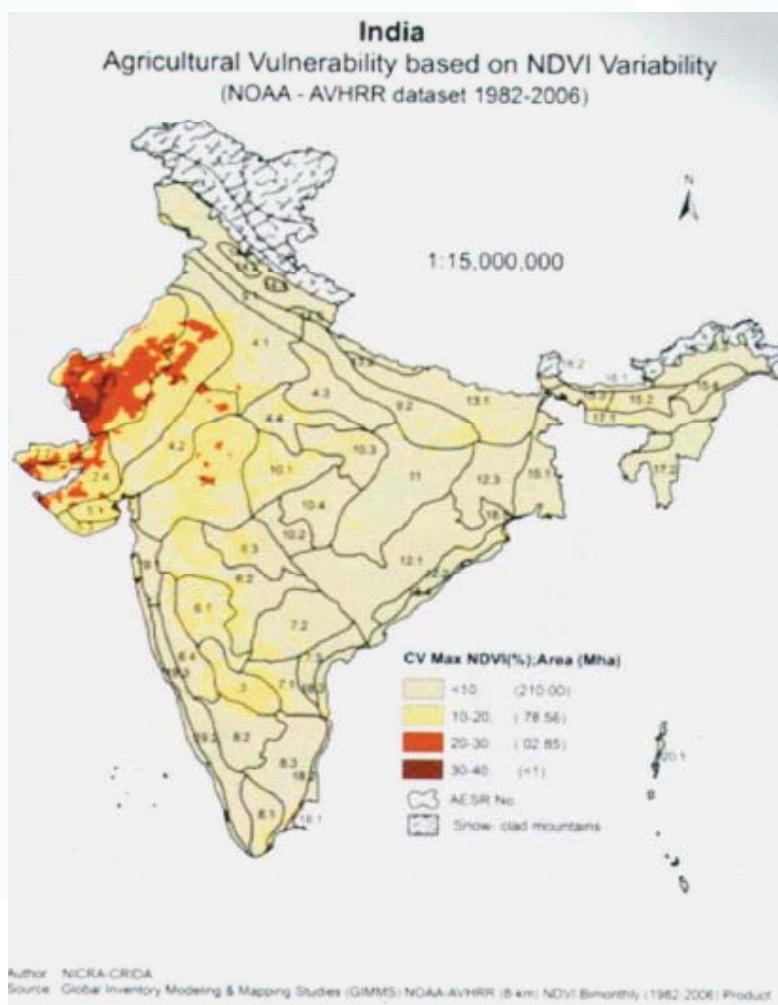
Agriculture and Allied Crop Production practices in the Himalayan region

In western Himalaya, the traditional farming operation is a complex product of crop husbandry, animal husbandry and forest resources constituting interlinked diversified production systems. However, with changing land use, the area under cultivation of many traditional crops has been reduced and some others are at the brink of extinction. Deficit in food production is growing in recent times in Jammu & Kashmir. With the reduction in rainfall, the rain-fed agriculture will suffer the most. Horticultural crops like apple are also showing decline in production and a real coverage particularly due to decline in snowfall.

Thermal stress effects on livestock productivity

The THI changes in four distinct regions- the Himalayan region has been analyzed for the months of January, February, March, May, June and July. The regions defined for this analysis include Jammu & Kashmir, Himachal Pradesh and Uttarakhand in the Himalayan region. The analysis for the baseline period and the 2030 scenario has shown the likely increase in THI in many parts of Himalayan region between March-September with a maximum rise between April- July. In the Himalayan region for 2030 scenario, thermal discomfort is likely to increase with THI > 80 compared to the baseline scenario, thereby indicating that in the 2030's, most places in this region are likely to remain under high temperature stress as compared to the baseline period.

Figure 6: Agricultural Vulnerability based on NDVI variability



The climate change indicators represent environmental conditions and increase our understanding of cause and effects of climate change and act as tools for evaluating on-going and future development programmes. Some of these indicators include: Increase in average temperature regime in State; Shift in pattern of rainfall & deficit snow fall; Decrease in water levels of rivers and streams; Reduced snow fall as a result of heat-trapping gasses; Drying up of springs & reduction in the flows; Shift in snow fall timing e.g. February & March winters receive heavy snowfall & December & January, the usual snow time receives less snowfall; Reported decline in glacial area for example in Chenab basin from 1 sq.km to 0.3 sq.km between 1962 – 2004. Length of glacier has reduced from 12 m to 4 m in State; Rising temperature & flash floods; Altitudinal movement of temperature sensitive plant species.

Plant species responding to high temperature tend to grow at higher altitudes e.g. red algal bloom related to water temperature rise increased in Dal Lake.

3.4 Bio-Physical Factors

1.3.1 Forest

Indian Institute of Science has conducted an

assessment of the impact of climate change on forest ecosystems in India. A dynamic vegetation model IBIS (Integrated Biosphere Simulator) was used to assess the impacts of climate change on forests in India. The study indicates that about 39% and 34% of the forested grids are likely to undergo shifts in vegetation type under A2 and B2 climate scenarios, respectively with a trend towards increased occurrence of the wetter forest types.

Approximately 47% and 42% of tropical dry deciduous grids are projected to undergo shifts under A2 and B2 scenarios respectively, as opposed to less than 16% grids comprising of tropical wet evergreen forests. Similarly, the tropical thorny scrub forest is projected to undergo shifts in majority of forested grids under A2 (more than 80%) as well as B2 scenarios (50% of grids). According to another study projected impacts of climate change, using a moderate A1B scenario and IBIS vegetation model for the period of 2030s and 2080s, are projected in figure 3.3 and 3.4, which show the impacts on current forested locations in India. The figure shows that Jammu and Kashmir is projected to undergo vegetation change in both A1B (2035 and 2085) scenarios.



Figure 7: All forested grids in India

All forested grids in India are shown in color (red or green): red indicates that a change in vegetation is projected at that grid in the time-period of 2021-2050, and green indicates that no change in vegetation is projected by that period.

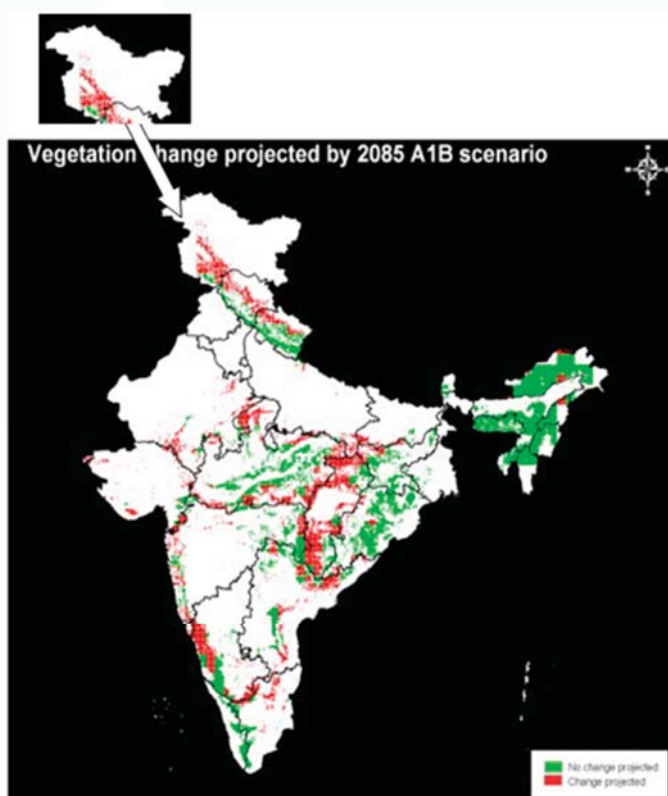
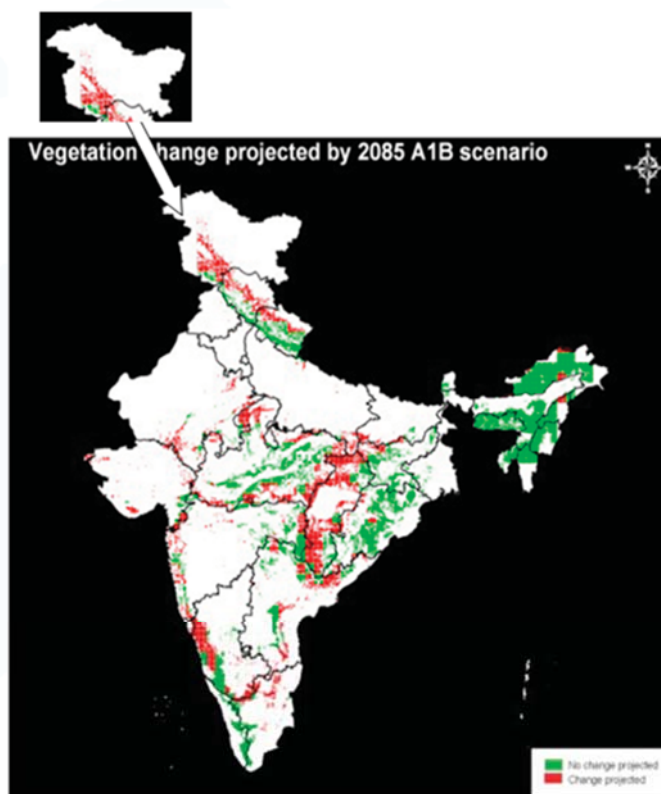


Figure 8: All forested grids in India (2071-2100)

All forested grids in India are shown in color (red or green): red indicates that a change in vegetation is projected at that grid in the time-period of 2021-2050, and green indicates that no change in vegetation is projected by that period. The timeframe is 2071-2100

Impact of climate change on forests

The impact is assessed with respect to the potential shift in vegetation types or Plant Functional Types and changes in NPP. In the study, this region is defined by forests that lie in the States of Jammu and Kashmir, Himachal Pradesh and Uttarakhand. Much of the dense forests of these areas are part of the Himalayan biodiversity hotspot

as defined by Conservation International. The model simulated vegetation as well as grids that undergo vegetation change for the Himalayan region. The entire Himalayan region is covered by 98 IBIS grids, out of which 55 (56%) are projected to undergo change. Thus, over half of the forests are likely to be adversely impacted in the Himalayan region by 2030's. NPP is projected to increase in the region by about 57% on an average by 2030s.

The impact is assessed with respect to the potential shift in vegetation types or Plant Functional Types and changes in NPP. In the study, this region is defined by forests that lie in the States of Jammu and Kashmir, Himachal Pradesh and Uttarakhand. Much of the dense forests of these areas are part of the Himalayan biodiversity hotspot as defined by Conservation International. The model simulated vegetation as well as grids that undergo vegetation change for the Himalayan region. The entire Himalayan region is covered by 98 IBIS grids, out of which 55 (56%) are projected to undergo change. Thus, over half of the forests are likely to be adversely impacted in the Himalayan region by 2030's. NPP is projected to increase in the region by about 57% on an average by 2030s.

3.5 Soil Water Regime

The Hydrologic Simulation with Climate Change Scenario in 2030s

The Himalayan region is mainly fed by the Indus river system. The whole area is exhibiting an increase in the precipitation

in the 2030s scenario. The increase varies between 5% and 20% in most areas, with some areas of Jammu and Kashmir and Uttarakhand showing an increase of up to 50%. The impact of the increase in precipitation in this region is reflected in an almost similar pattern of increase in the evapotranspiration (ET). This is expected to happen presumably on account of (a) the increase in the amount of moisture in the soil and land surface and (b) the increase in temperature.

Both these factors enhance the opportunity for ET. The water yield, which is the total surface runoff, is usually a function of precipitation and its distribution. The other factors that influence water yield are the soil profile and land use in the area. It may be noted that for the Himalayan region there has been a general increase in the water yield for the 2030s scenario. However, it may be noticed that the increase in the water yield is more for those areas where the increase in ET is less. The increase in water yield has been up to about 50% for some areas of Indus River for the 2030s.

The general impact of increase in the precipitation is reflected in the increase in sediment yield. This is quite evident in the Himalayan region as well. The other major factors that dictate sediment yield are the intensity of rainfall, land use and the soil type of the area. The increase in the sediment yield in the Himalayan region is up to 25%, which can be detrimental for the existing water resources projects and has the potential to cause considerable damage to the environment.

Aquatic ecosystems

Aquatic ecosystems have played an important role in the socio-culture of the State. The economy of many people is dependent on these systems as they provide livelihoods to boatmen, agriculturists, tourism industry etc. The aquatic systems in the State are however, confined to only approximately 0.8% of the total land mass, yet they have a tremendous impact on the lives of the people, the State also depending mainly on hydel power generation. The main aquatic systems found in the State are:

- a) Low altitude lakes which are fresh water and high in productivity. These are mainly found in the Jammu region.
- b) Valley Lakes: The valley abounds in lakes of varying sizes and shapes. These are all fresh water.
- c) Forest Lakes: Found within the hills, these lakes are small as compared to the valley lakes.
- d) Glacial Lakes: These are high altitude lakes and are found in and above the alpine areas. Some of

Pressures/ Threats to Aquatic eco-systems

1. **Encroachments:** This has been the main threat to lacustrine systems of the valley with the two major lakes i.e. the Wular Lake and the Dal Lake having shrunk to approximately half their size in the last 50 years or so. Encroachments along the Wular Lake are mainly from expansion in arable fields for paddy whereas those within the Dal Lake are

for housing and vegetable gardens. The Dal Lake traditionally has also had floating gardens (made from dug up weeds of the lake) along north-western edge of the lake. These have increased in size and extent and the older ones have been reclaimed to build permanent settlements. A direct consequence of encroachment into a lake is the degradation of another lake known as the Aanchar Lake on the north-western part of Srinagar city. A major part of this lake is now a residential colony and the remaining portion is a swamp. This lake, a few years ago used to supply the valley of Kashmir with a major supply of reeds for making mats and also of lotus stems (*Nelumbo* sp.) used as food in Kashmir during winters. This supply has now dried up.

2. **Pollution:** Three main sources of pollution to the water bodies of the State are identified. These are:

- a. **Solid waste:** non-degradable materials like polythene and plastics which are dumped into the lakes and rivers by the local people.
- b. **Organic pollutants:** Most of the human habitations are settled around the water bodies, i.e. the rivers and the lakes. They as a result become the main channel to carry the organic wastes and night soil from the city and villages. Mushrooming of hotels around the famous Dal Lake also discharge the night soil produced from these,

in absence of a proper drainage (sewage) system. The houseboats also have contributed in no mean measure to the organic pollution of the Dal Lake by transmitting organic effluents directly into the lake.

- c. **Inorganic pollutants:** The chemical fertilisers and pesticides used by agricultural farmers around the Dal and Wular Lake as also around most of the wetlands seep into the water. This has caused a build-up of inorganic pollutants and other heavy metals, detrimental to the life in the lake. Aspects of bio-magnification have not been studied but could be a major cause of worry regarding the valley lakes.

All this has been a cause of Eutrophication, especially of the lakes which are enclosed and have a weak water flow.

3. **Siltation:** Degradation of catchment areas of the lakes and water-bodies have brought down large sediment loads into the river and lake systems causing an appreciable increase in the suspended particulate matter. The water bodies lose transparency which may cause changes in the biota. Heavy particulate matter also causes siltation and subsequent choking of lakes and rivers. The Jhelum River is a case in point where the depth of the river is barely a few inches at certain points.
4. **Increased Resource Extraction:** The

water bodies have provided many resources to the people of the State. In addition to the vegetables growth in the lakes, the water bodies also provide large stocks of lotus stem and water chestnut. The lakes also provide reeds for making mats. In addition to this, fish abound in the water systems of the State providing cheap protein. However, over exploitation of these resources, especially fish resources have caused major declines in fish yields.

3.6 Socio-economic factors contributing to the vulnerability

3.6.1 Economic growth

Latest 2001 census reveals that the population of Jammu and Kashmir is 10,069,917 is about 0.98 percent of India's population. The population growth rate in the state has been consistently high and during 1981-2001 it was more than 2 percent. Its geographical area is 222,236 sq. km., with a share of 6.76 per cent of the country's size. About 54 percent of the population in the state is literate, as against the national literacy rate of 65.38 per cent. The age structure of the population indicates a high dependency ratio both in the 1971 and 1981 census. The trend in the development of Jammu and Kashmir is not encouraging. It has been lagging behind most of the states in regard to the growth of Net State Domestic Product (NSDP) at current prices. The average annual growth of Net State Domestic Product at current prices during 1980-81 to 1999-2000 was 12.45 percent for Jammu and Kashmir against 15.01 percent, 14.28 percent,

13.83 percent and 14.3 percent for Andhra Pradesh, Gujarat, West Bengal and Kerala respectively. In the case of the growth of Per Capita Net State Domestic Product at current prices also, the state of Jammu and Kashmir was lagging behind most Indian States. The average annual growth of Per Capita Net State Domestic Product at current prices during 1980-2000 was estimated as 9.63 percent for Jammu and Kashmir against 12.9 percent, 11.63 percent, 11.63 percent, and 12.86 per cent for Andhra Pradesh, Gujarat, West Bengal and Kerala respectively.

Impediments to growth in J&K

The slow growth of the state of Jammu and Kashmir can be attributed to various factors. The climate of armed militancy in Kashmir during the past decade has been a major factor. Low productivity in agriculture and allied sectors has impeded employment and income generation. Poor industrial infrastructure along with the poor investment climate has left the industrial sector in its infant stage. There has not been a suitable strategy for the potential

sectors to achieve higher economic growth. Lack of good governance and sound fiscal management has also been responsible for the poor economic growth of the state.

3.6.2 Poverty and Hunger

While the growth figures are not robust the state has not done badly in term of poverty alleviation. Food related vulnerability is not as high as compared to some other Indian states. Official data suggests that indeed levels of poverty are negligible in the valley. As compared with 28.3 percent people officially estimated to survive below the poverty line in India in the year 2004-05, the comparable ratio for the state of Jammu and Kashmir in the same year was pegged by the Planning Commission at a meagre 4.5 percent. Kashmir is one of the most egalitarian societies in the country, in which land reforms were implemented with greater vigor than in most other regions of India. In the first decade after India's Independence, big farms were abolished resolutely, and subsequently surplus lands were distributed among landless farmers.

Table 7: Estimates of Poverty of J&K

Year	Poverty Line (₹)		Poverty Ratio (Percent)			Population BPL (in millions)		
	Rural	Urban	Rural	Urban	Combined	Rural	Urban	Combined
1973-74	46.60	NA	45.51	21.32	40.83	NA	NA	NA
1977-78	NA	NA	42.86	23.71	38.97	NA	NA	NA
1983	91.80	99.62	26.04	17.76	24.24	1.311	0.249	1.560
1987-88	NA	NA	25.70	17.47	23.82	NA	NA	NA
1993-94	213.83	253.61	30.34	9.18	25.17	1.905	0.186	2.092
1999-00	367.45	420.20	3.97	1.98	3.48	0.297	0.0049	0.346

3.6.3 Livelihood scenario and extreme weather events

A careful examination of the disaggregated official data also suggests that although overall ratios of poverty are much lower in Jammu and Kashmir than in the rest of India, the state lags behind many others in several specific indicators of poverty. This is a predominantly agrarian economy, in which 80 per cent of the population of the state is dependent on agriculture directly or indirectly. Ninety-seven percent of the cultivators are small or marginal farmers, with average land holdings as small as 0.7 hectares. There has been a worrying deceleration of agricultural production in the state. The valley suffers from a 44 per cent deficit in food grain production, 33 per cent in vegetables and 69 per cent in oilseeds, all of which are imported into the state from the rest of India. These crises of livelihoods have been aggravated by the collapse of the carpet weaving industry, and setbacks to tourism. The per capita income of the state is only two thirds of the national average, at Rs. 17,174 against Rs. 25,907 in India taken as a whole. Its unemployment rate is 4.21 percent, against a national rate of 3.09 percent.

In recent years, the state has seen large scale migration to cities and resource congestion and pollution has increased significantly. Coupled with the fact that, there are a large number of people moving out of hill agriculture is affecting the ecological balance in the region. Especially, the Leh region has rapidly changed in last few years. The growing

numbers of tourists has led many Ladakh is to abandon agriculture and move to the city for tourism-related businesses. The rapid urbanisation is also leading to overcrowding with an influx of migrant workers. Day-to-day life has been impacted as young Zakhakis increasingly adopt new and mainstream ideas of development. Local elders are concerned that these changes are resulting in the breakdown of their traditional value system and an abandonment of the longstanding ecological approach and indigenous wisdom. The negative impact of these changes has become more apparent in recent times. Two severe flash floods, a phenomenon uncommon for the region, devastated Leh in 2006 and 2010. The damage was exacerbated by the overcrowding and encroachment on water channels. At the same time, Ladakh is have been witnessing a fast depletion of glaciers and snow cover; erratic water supply in the glacier fed streams; warmer temperatures; and new pest species that often destroy annual yields. These new realities have left Zakhakis increasingly insecure and uncertain about their future. Changes are happening at multiple levels and at a rate faster than can be managed and understood. According to scientists at the Indian Institute of Tropical Meteorology at Pune, the August 2010 phenomenon in Ladakh is due to the impact of global warming. With higher average temperatures in high regions, monsoon winds have reached Leh. Geologists of Kashmir University have also identified human folly as a factor. Unplanned urbanization and drainage congestion leads to such disasters. The Mithi river flood of Mumbai cloudburst of 2005 or the failed engineering solution of

embankments of Kosi flood of 2008 are clear examples of human abetted disasters.

The problem in Ladakh is unique as it is a high altitude cold desert. Traditional material of sun-baked mud bricks is ecologically sound in one fundamental respect. In winter with sub-zero temperatures outside, it keeps the interior warm. Over the years, cement has come to be used which freezes the interior in winter. Much more heat and energy is needed to keep it warm inside. With change in precipitation to rain, new material has to be thought of with a design that can tolerate rain and also provide the space heating that is possible with baked mud bricks. Getting the right material that is affordable is the new challenge to cater for both change in precipitation and allow natural space heating in winter.

Similarly, the earthquake and flood in northern and north western parts of Kashmir caused significant devastation. The economic was remittance and border trade driven, collapse of traditional routes and nallas and security related restriction restoration enhanced the vulnerability of the affected population in both 2005-06 and 2010.

3.6.4 Human Development

Economic Survey of J&K 2010, mentioned that more than 42 per cent population of Scheduled Tribes, majority of them Gujjars and Bakerwals, lived below poverty line in the state and are highly vulnerable to climate change. A study by Mr Harsh Mander also highlights the vulnerability in the region due to food deficit. Given that this is a

food deficit state, the contribution of the Public Distribution System to food security of the residents of Kashmir cannot be overstated. The researchers found functioning ration shops even in the deep interior, and less than four per cent people did not have ration cards. Most reported that they were able to access the subsidized grain, even if sporadically. But the shops are opened only one or two days in a month, and if they miss their chance, their allocations of food lapse, and are presumably sold in the black-market. The study found gaps in the opening of ICDS centers in some remote locations, but the supply of hot cooked food to children was heartening. However, in most locations, the centers functioned as little more than feeding points. Children were not weighed, and malnourished children not identified or treated. Few centers run pre-school classes, and expectant mothers are not examined or advised about their nutrition and that of their children. Ninety-eight per cent children reported that they ate hot cooked meals at school, although there are many months in which the meals are not served as supplies to do not reach. Teachers are burdened with this work, instead of this being entrusted to women's groups as in some other parts of the country.

The scenario is even worse in some northern and northwestern areas in greater Kashmir region, where poor land man ratio, feudal nature of the society and poor out reach of the public services due to the armed conflict in the region have made the population highly vulnerable. Even the livelihood programmes proposed under National Rural Livelihood

Mission for the state that is supposed to cover all the villages is doubtful to benefit these areas.

3.6.5 Impacts of climate change on human health

The impacts of climate change on health have been deduced in quantitative as well as qualitative terms. The quantitative aspects that have been studied are only for the transmission of malaria in the 2030s. The transmission windows have been determined in terms of (a) temperature only and (b) temperature plus the relative humidity requirements for transmission.

Based on minimum required temperature for ensuing transmission of malaria, a district-wise map of India was generated to show the distribution of different categories of transmission windows under baseline and by the year 2030s. The climate projections for 2030s are derived from the PRECIS, regional model developed by the Hadley centre and forced by the GHG emission scenarios arising out of the IPCC defined A1B socio-economic scenario for the future. The transmission windows in some parts of the Himalayan region i.e. Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim and Arunachal Pradesh however, continue to have only 0-2 months open for transmission even in the 2030's.

Himalayan region: The increase in temperatures may lead to increasing morbidity due to heat stress. Flash floods due to Glacial Lake Outburst Floods (GLOF) may lead to large scale landslides and affect

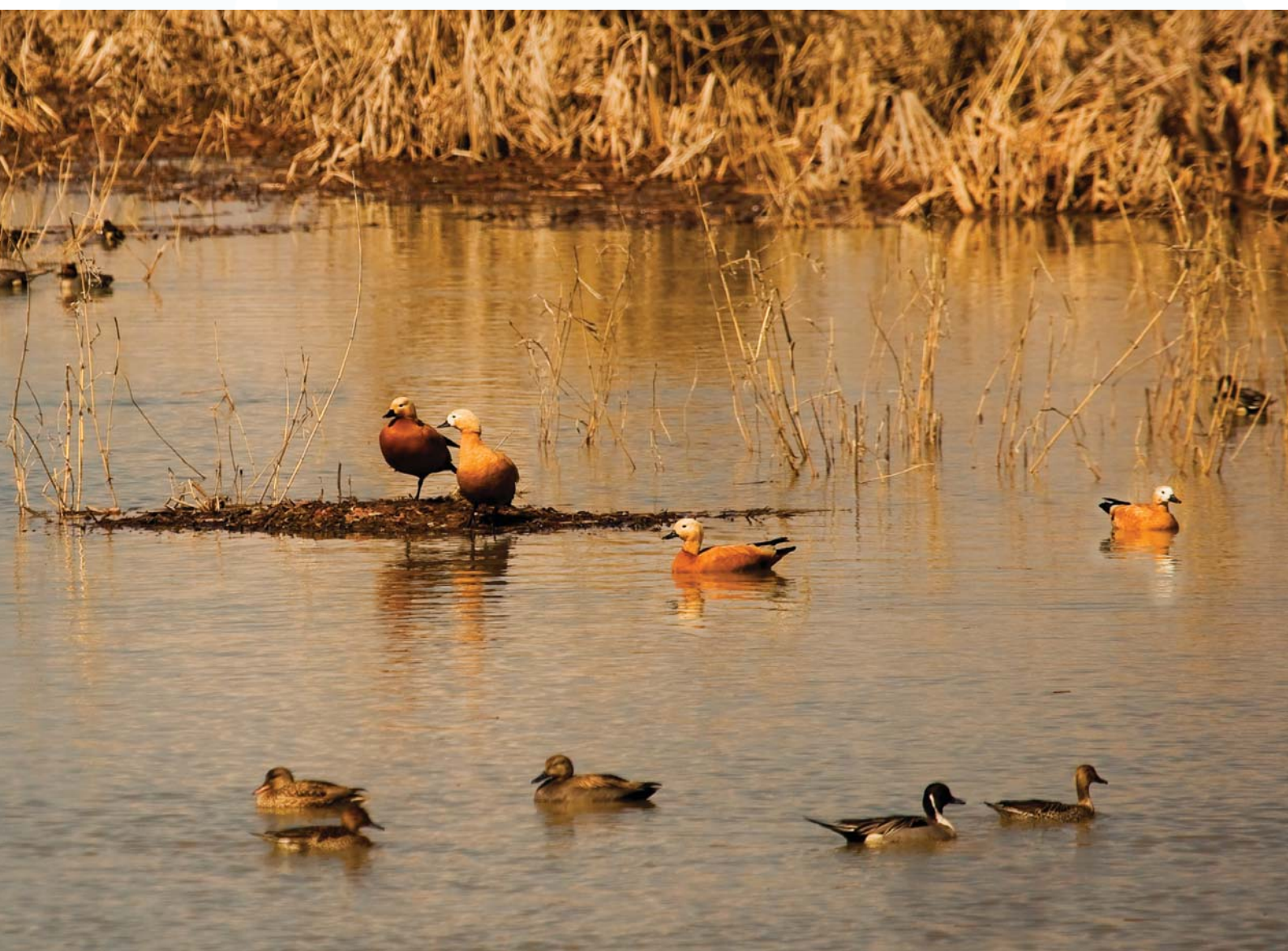
food security and hence nutritional health.

It is important to note that the entire region of Jammu & Kashmir is eco-sensitive and contiguous and any development to adapt to climate change requires a holistic approach rather than piecemeal and regional basis. Ultimately the vulnerability of the ecosystem will affect not only the nomads, rare faunas and flora in the region but also everyone in the region. The region is in seismic zone, several areas are flood prone and Indus water treaty still many times poses developmental challenge in the valley. Similarly, geo-strategic issues with the neighboring countries like Pakistan and China initiates much developmental trade-offs for strategic region that may not be ecologically sustainable.

Some tentative conclusions are:

- (a) There have been extreme weather events in North Pakistan, Jammu and Kashmir and even parts of the high altitude region of Himachal Pradesh. The need is to study this phenomenon to better understand the science of climate change as its impacts the region.
- (b) Adaptation to extreme weather events must be done in a scientific manner. Local governments, non-governmental organizations and the people must be empowered with resources. Local solutions may provide best answers. Documentation and mainstreaming of indigenous practices that is highly climate resilient need to be taken up for the region.

- (c) Study and understanding of relationship of trees with rainfall needs immediate attention. The phenomenon could well be due to global warming and climate change. Local narratives, though anecdotal, must be merged to get a scientific picture from the bottom. Relying on computer simulation may not be enough.
- (d) The observed data in the region is rather limited and makes planning quite difficult and district level vulnerability assessment and extent of exposure needs to be estimated to compute the index.
- (e) The emission and pollution index for major cities need to be computed to estimate the extent of vulnerability.
- (f) Drainage congestion due to urbanisation must not be allowed to happen. Infrastructure building must be given priority to build resilience. The design and material of building should be such so as to cater for expected rain in future. Scientists, architects and construction engineers in India must rise to the occasion to suggest the most cost effective architecture. Climate resilient buildings (GRIHA/IGBC) standard may be incorporated in the building codes and development regulation.







4 Sustainable Energy Mission

4.1 Introduction

Energy is a vital component for sustained economic growth and the energy mix is an indicator of sustainable development of a state and the country. Also proportionate growth in power sector is essential to maintain the economic growth. With enhanced modernization, industrialization and change in life style; the energy demand has increased several folds in across the last decade. Energy demand across the country is met primarily through conventional energy resources which not only raises the concern on energy security but also results in increased accumulation of green-house gas in the atmosphere. As per IPCC Fifth Assessment Report Green House Gas emissions resulting from provision of energy services have contributed significantly to the historic increase in atmospheric concentration of Green House Gas concentration. Thus, the states with high dependency on conventional energy sources for meeting the energy demand contribute significantly to environmental degradation. However, the demand for energy to meet

economic development exists but a strategy should be developed towards lowering of Green House Gas emissions from the energy system besides satisfying the global demand for energy services. Increasing the share of renewable sources in the existing energy generation mix is assessed as the best possible option towards meeting the global energy supply challenge.

The action plan proposed under State energy mission while catering to the objective of national solar mission and also the renewable energy is also aimed at enabling communities to understand the uncertainty of future climatic conditions and engage them effectively in a process of developing adaptation and mitigation programmes. The Climate Change Action Plan for solar & renewable energy technology is strategized in line with the National Action Plan on Climate Change with objectives of meeting the country's development goals while simultaneously yielding co-benefits to address climate change effects.

The social, cultural, economic and other developmental aspects of any region are influenced by climate. The state of Jammu & Kashmir has a diverse physiographical characteristic and climatic condition ranging from extreme hot zones/areas to snow-capped hills to cold arid deserts. With the presence of several glaciers; hilly and rugged topography of the land and extensive network of canals and streams the state has immense potential for mini, micro and small hydro projects.

The state is currently focused on generation though big hydro power project. Potential of micro hydel power and Solar are increasingly being tapped. Grid electricity penetration in remote hilly areas of J&K is techno-economically unviable by virtue of geographical disadvantages and scattered

household pattern. Moreover, the power sector of the state is already facing difficulties like slow rate of capacity addition, poor power evacuation facility, high AT&C losses and mismatch in load profile and power utilization mix is not commensurate with the state climatic conditions. Promotion and utilization of renewable energy is the most feasible solution which would also promote low carbon growth and can meet the decentralized energy requirement of the remote locations.

The state population has increased by 23.71% in the period of 2001- 2011 and reached 125, 48,926 numbers. Interestingly the urban population has increased at a rate of 27.21% in the same period which has certainly increased the energy demand.

Table 8: population distribution of Jammu & Kashmir³

Sl. No	State	Population 2011	Percentage decadal growth rate	Population density per sq. km.	
		Total	2001-2011	2001	2011
1	Rural	91,34,820	19.77	100	124
2	Urban	34,14,106	35.66		
	Total	125,48,926	23.71		

³Source: Digest of Statistics for 2010 -11, Directorate of Economics & Statistics, Govt. of J&K



The power available in 2011-12 from all sources was able to satisfy only 64% of total power demand of the state. Due to the shortfall of 6,231.74 million units during 2011-12, the state faced problems of frequent power cuts for 8- 10 hours a day which has a detrimental effect on the growth potential of the state. To meet its power demand of 17,323.00 million units during 2011-12, the state has procured 66.76% of its total power supplied (i.e. 11,091.26 million units) from the NEWNE grid which is highly fossil fuel dominated.

4.2 Key Trends in the sector

The state is largely dependent on the power generated from hydropower projects and thermal power plants besides supplements from DG set and NEWNE grid power. Jammu & Kashmir has a total power generation capacity of 2,648.46 MW under central and state sector. The state is heavily relying on power purchase from the NEWNE grid and thermal power generation units particularly gas and diesel based power units during winters when its own hydro power generation reduces and power demand rises. The state is facing power crisis owing to untapped renewable energy, high rate of AT&C losses including pilferage.

Table 9: Electricity Demand in the state

Peak Power Demand ⁴	2,500 MW
Peak Power Met	1788.9 MW
Energy Requirement	17,323.00 Million units

⁴Source: http://www.jandkplanning.com/index.php?option=com_content&view=article&id=2307&Itemid=295&lang=en

Out of total power demand of 17,323 million units, power generation from the state owned power houses is only 2562.723 million units. Bulk of electricity consumption in the state is by the domestic sector. With modernization and increased urbanization, per capita energy consumption of the state has increased from 849.98 kWh in 2010-11 to 882.82 kWh in 2011-12. The energy demand has gradually increased during last five years at an annual rate of 5 to 6%. According to the sixteenth All India Power Survey, the power requirement of the state is expected to reach 19,500 million units during 2020 -21.

In addition to in-sufficient power generation capacity of the state, high transmission and distribution losses of 61.61% and loss due to collection inefficiency accumulating to 72.68% of total power demand as AT & C loss adds to the power crisis.

The state of J&K has vast hydro potential estimated at 20,000 MW out of which only 16,480 MW is identified till date mainly due to resource constraints. Approximately, 15% of total identified hydro power potential i.e. 2457.96 MW has been exploited so far out of which the states sector projects is only 760.46 MW.

The primary sources of cooking fuel in rural areas of Jammu & Kashmir are firewood and chips followed by LPG. Urban cooking fuel demand is mainly met through LPG. Electricity is the main energy source for urban lighting. The scenario is same in the

rural areas too; kerosene is being seldom used as an alternative. Although the state is fairly electrified with 97% of villages lit, but providing continuous power at cheaper rate is difficult.

4.3 Vulnerability of the sector

Climate change can adversely impact water resources and other sectors of the state. The source of water is mainly summer and monsoon precipitation and melting of the large reserve of snow and glaciers in the Himalayan region. The high rate of warming will result in rapid shrinking of majority of glaciers in Himalayan range. Formation, growth and likely outburst of glacial lake are directly related to climate change and deglaciation.

Strategies for linking climate change and energy sector are usually centered on mitigation efforts as, the prevalent fossil fuel based energy generation methods contribute to climate change in a big way. Developing options for low carbon growth and reducing carbon footprint are important to limit the degree of future climate change.

Energy and water sector are closely and dynamically associated. Research by World Bank⁵ establishes that, the climate change might impact hydropower generation in three possible ways. Firstly, the available discharge of a river may change, since hydrology is

usually related to local weather conditions, such as temperature and precipitation in the catchment area. Secondly, an unexpected increase in climate variability may trigger extreme climate events, i.e., floods and droughts and thirdly, changing hydrology and possible extreme events may increase sediment risks. More sediment, along with other factors such as changed composition of water raise the probability that a hydropower project suffers greater exposure to turbine erosion. Moreover; an unexpected amount of sediment will also lower turbine and generator efficiency, resulting in a decline in energy generated. Since, the majority of power is generated from hydropower sources; hence, there are high chances that, the state may face power crisis if the projected impact of climate change happens. As a result, the state will become highly dependent on thermal power plants and also higher carbon intensive power procurement from the NEWNE grid.

The projected impacts of variation in cold wave level would result in more intense winters in some areas. The peak winters would enhance the electricity demand since the consumption rises due to usage of power for heating purpose. The power shortage in winter would also lead to exploitation of other energy sources and fuel wood in rural areas and will lead to affecting the forest cover of the state.

⁵ Policy Research Working Paper on Estimating Global Climate Change Impacts on Hydropower Projects: Applications in India, Sri Lanka and Vietnam –The World Bank

Regions with increased temperature would face rise in electricity demand because of higher use of electrical cooling and would thereby enhance the pressure on electricity distribution network through increased seasonal demand.

Higher demand of energy due to climatic variability and lower generation due to projected impact of climate change would widen the power supply-demand deficit in Jammu & Kashmir. Extreme weather events are also likely to pose threat to physical infrastructure assets. Impact of extreme weather events on energy sector due to

climate change can harm economic and social infrastructure because of the fact that, centralized power plants tend to serve large section of population and are also sensitive to climate change.

Biomass still dominates the state domestic energy profile e.g. fire wood, dung, etc. which are more vulnerable to effect of climate change. With the increased cold wave, the efficiency of bio-digesters also deteriorates.

The envisaged climate change could impact different components of the electricity sector as outlined below.

Table 10: Possible Climate Change Impacts on the Energy Sector

Climate Change Indicators	Impacts on Energy Sector
Variation in Hydrological cycle	• Availability of water in hydro power plants. • Increased possibility of sediment and climate extreme events
Reduced Temperature	• Higher Demand of electricity resulting to increased drawl resulting to high dependency on fossil fuel based energy generation and increase in GHG emission • Energy demand will increase in domestic sector for room heating, hot water requirements and as well as in commercial sector. • Increase in use of firewood in rural area leads to affecting the forest cover of the state. • Could impact renewable energy generation potential like, solar, biogas, hydro, etc.
Increased Temperature	• Increased need of energy in household sector for cooling particularly in areas like Jammu. • Could impact renewable energy generation potential, especially solar
Extreme events (landslide, cloudbursts, etc.)	• Could affect the transmission and distribution infrastructure in the state. • Effect power plant infrastructure including transmission and distribution network

Assessing the vulnerability of energy supply to climatic events and longer term climate change needs a strategic approach to ensure that timely and effective adaptation measures are adopted for coherence across different sectors and effective governance to reduce the sector's vulnerability to the impact of climate change.

4.4 Key issues in the sector

Though around 17,426 villages/hamlets⁶ of the state are electrified but providing continuous power supply particularly during winter when the energy demand of the state is very high is still a challenge for the power department of Jammu & Kashmir. With 73% of population living in rural areas, the extensive use of firewood is leading to forest degradation. Infrastructure constraints are also rampant in both rural and urban areas.

The power supply scenario of the state is characterized by persistent power shortage. The state has very low electricity generation capacity as majority of hydro power potential is still untapped due to resource constraint and lack of interest in project investment amongst local entrepreneurs.

As power plants located in Jammu & Kashmir are largely run-off the river based hydro power plants, change in climate systems and variation in hydrological cycle at the both basin/catchment level across Himalayan region has direct impact on energy generation. Temperature variability and rise

in humidity index leads to enhanced energy consumption across the states, thereby increasing the supply-demand deficit.

During winter the problem is aggravated as the discharge rate in the rivers falls subsequently leading to generation drop to one-third of that of the generation during summer whereas energy demand increases due to heating loads. Thus, the overall impact in the state would be manifested through the supply-demand deficit.

However, the supply from the NEWNE grid is unable to meet the entire demand of the state especially during the long winter when the state faces deficiency in both peak and off peak hours. This makes the population suffer from long hours of load shedding, with the rural areas being the worst hit. The supply is also affected in terms of low voltage. The shortage during peak hours for sustained periods also calls for drawl of power resulting in higher cost of grid electricity.

The geographical condition of the state with many villages in remote and far-flung areas makes access difficult, in addition to the frequent natural calamities such as landslides, cloudbursts, etc.

Due to hilly and difficult terrain and dispersed population in rural areas, the transmission and distribution network systems are still unviable and moreover the operation and maintenance cost are also high.

⁶ 2010-2011 information from Digest of Statistics 2010-2011

The low-temperature condition lowers the efficiency of the biogas projects, one of the best choices to meet the cooking energy requirement in rural areas. Though areas like Ladakh, etc. receive heavy sunrays and have high potential of solar energy generation but the remoteness and difficult terrain of the location becomes the major barrier for the grid connection establishment. The distance of the Ladakh -Kargil region from the other main areas like Jammu-Srinagar would result in high T&D loss in power evacuation from the solar power plants. The remoteness of villages and transportation problem in winter due to snow fall is the major barrier in providing access to maintenance service for the users or beneficiaries of the remote villages for almost ¼ of the year.

4.5 Programme and Policies in the sector

JAKEDA since its inception in 1989 has been making stride in promoting renewable energy technology across the state and meeting the power demand of remote locations through decentralised renewable energy generation under various schemes and programmes of the Ministry of New and Renewable Energy (MNRE). Along with JAKEDA, Ladakh Renewable Energy Development (LREDA) and Kargil Renewable Energy Development Agency (KREDA) are also actively involved in renewable energy implementation projects. The renewable energy project implementation is undertaken under following programmes-

1. Remote Village Electrification Programme (RVEP)

2. Jawaharlal Nehru National Solar Mission (JNNSM) –
 - a) Solar Lanterns
 - b) Solar Home Lighting
 - c) Off-grid Rooftop Solar Power Plants
 - d) Grid connected Rooftop Solar Power Plants
 - e) Grid connected Solar Power Plant
 - f) Solar Street Lighting
 - g) Solar Water Heating
 - h) Solar Cookers
3. National Biogas & Manure Management Programme (NBMMP)
4. National Biomass Cook stove Initiative Programme
5. Small hydro power programme
6. Small Wind Energy & hybrid systems programme
7. Biomass Co-generation (non-bagasse) in industry Programme
8. Special Area Demonstration Programme (SADP)

Achievements:

RVEP:

JAKEDA, since 2004, is electrifying un-electrified remote census villages and hamlets as approved by Rural Electrification Corporation (REC) where grid connectivity is either not feasible or not cost effective. It has electrified 440 villages and hamlets so far. Around 37,700 solar home lighting systems are distributed across 16 districts of

the state under remote village electrification programme.

JNNSM:

- 13,000 solar lanterns are being distributed under the solar lantern programme of MNRE to 73 Remote Gujjar&BakerwalBastis located across Jammu & Kashmir province. The state govt. has already distributed 19,531 SPV home lighting systems till date to general category across the state.
- **Off-grid Solar Power plants:** 13 solar power plants of capacity ranging from 5 – 25.50 kW are being installed under JNNSM scheme in district / sub-district hospitals with cumulative capacity of 123.90 kW. Apart from central schemes, the state under BADP programme is also implementing 15 solar power projects of 5 kW capacities in Community Information Centres (CICs) accumulated to 75 kW.

Solar power projects under commissioning:

Total 33 projects are under installation

cumulating to 550.00 kW capacities in District/ Sub-District Hospitals, stadium, shrines etc. Simultaneously with the implementation of JNNSM scheme, JAKEDA is also undertaking implementation of 5 solar power plants under BADP programme of 5 kW capacities across the state.

JAKEDA is presently undertaking implementation of 6 solar power projects cumulating to 550.00 kW capacity in Sheep Breeding Farm - Daksum, Anantnag, University of Kashmir - Main Campus, Srinagar, Sainik School Manasbal, Ganderbal, Kashmir, Village Trehgam, Kupwara, J&K Institute for Management & Public Administration, Sidhara, Jammu and Panjbakhtar Temple, Jammu.

- **Solar Water Heating System:** Under JNNSM scheme, total 25,000 LPD⁷ of Solar water heating system is installed at 11 sites across the state. In addition to this, 51 solar water heating systems are installed in commercial complexes, institutions, district/ sub-district hospitals, municipal corporations, etc. accumulating to 64,500 LPD.

⁷ LPD refers Litre Per Day



- **Solar Cookers:** JAKEDA, considering high solar energy potential of the state, has also promoted and implemented 6000 nos. of Box type solar cookers in Kishtwar area.

Biogas Programme: To reduce climate change impact and prevent black carbon and GHG emission, the state has installed 2,063 biogas plants of 2 cum capacity. Under the programme of National Biogas & Manure Management Programme (NBMMP), the state is implementing 294 biogas plants.

National Biomass Cook stove Initiative

Programme: To achieve the objective of low emission cooking and conservation of forests by reducing firewood consumption, the state has distributed 4, 50,000 improved cook stoves under the NBIC programme.

Micro Hydel projects and Water Mills:

6 Mini Hydro projects are considered for implementation under IPP Mode (Phase-I).

Table 11: List of Mini Hydro Project Considered Under Implementation

Sl. No	Project Name	Project Capacity
1	Paristan MHP, Ramban	2 * 1 MW = 2.00 MW
2	Ukhral MHP, Ramban	2 * 0.875 MW = 1.75 MW
3	Buta-kulan MHP, Ganderbal	2 * 0.875 MW = 1.75 MW
4	Rayil MHP, Ganderbal	2 * 0.725 kW = 1.45 MW
5	Surasyar MHP, Budgam	2 * 0.60 MW = 1.20 MW
6	Dandipora Daksum MHP, Anantnag	2 * 0.90 MW = 1.80 MW
	TOTAL	9.95 MW

Water Mills: 50 traditional water mills have been upgraded for the mechanical and electrical output of 3 -5 kW.



Wind Power Projects:

A project of 8 MW is being implemented in Bidda (Reasi). Apart from standalone wind power project, the state is also implementing 5 Wind Hybrid Systems at following sites –

Table 12: List Of Wind Power Project Under Implementation

Sl. No.	Site	Capacity (kW)
1	PHC, Narbal	7.0 (4.2 kW from Wind + 2.8 kW from SPV)
2	SDH, Ganderbal	8.8 (6.6 kW from Wind + 2.2 kW from SPV)
3	SDH, Akhnoor	11.8 (8.4 kW from Wind + 3.4 kW from SPV)
4	Deera ki Gali Rajouri in lieu of SDH, Sunderbani	8.8 (6.6 kW from Wind + 2.2 kW from SPV)
5	SDH, Banihal	10.0 (6.6 kW from Wind + 3.4 kW from SPV)
TOTAL		46.40

4.6 Key Priorities

With the vision of providing energy access to all and improved energy security for the state, key elements for mitigation and adaptation were identified after detailed deliberation with the working group. With a very little or no net emission from energy sector, the state continues to be “carbon neutral”, despite the steep growth in energy consumption.

To promote low carbon sustainable growth, the state Government has planned to promote green energy by increasing share of renewable energy in generation mix, scaling up decentralized renewable energy application, etc. The strategies towards increasing resilience of the state’s energy sector lie in up scaling of current programmes and activities along with new interventions. The priorities are in line with the concerns raised due to impact of the climate change and the state’s response.

1. Promotion and implementation of solar city project

Around 100% of the urban population of Jammu & Kashmir is dependent on grid power or energy from DG set for lighting purposes. Urbanization and economic development are leading to a rapid rise in energy demand in urban areas leading to increase in greenhouse gas emission. The state is also experiencing rapid growth in peak electricity demand and the power department is not able to cope up with the rise in peak demand. Incorporation of the solar city programme would prove crucial towards supporting the development of renewable energy application and energy efficiency measures, thereby curbing grid power demand by 10% in next five years. The activities to be undertaken are –

- A. Preparation of a master plan for all district headquarters under the “Solar City project” for increasing renewable energy supply and energy efficiency

measures in the city.

B. Set-up institutional arrangements for implementation of the master plan.

- a. Formation of a Solar City Cell.
- b. Approval and Monitoring of projects implemented under “Solar City” plan.

C. Awareness generation and capacity building activities by managing publicity programmes, conducting training programmes/business meets for stakeholders like architects, builders & developers, engineers, RWAs, manufacturers/suppliers, financial and technical institutions, NGOs.

D. Promotion of cities as Solar and carbon neutral city in next five years through implementation of pilot solar city project in phases-

- Phase I - Jammu, Srinagar and Katra
- Phase II - Kupwara&Udhampur
- Phase III - Rajauri, Poonch and Baramula,

The following measures are proposed under solar city project -

- Installation of solar street lighting and load management of municipality
- Solar power packs, solar water heaters, solar air heaters/ coolers, rooftop solar plant for common area lighting, energy

efficiency measures to be undertaken in government owned and government aided institutions and organizations.

- Solar water heating systems for residential buildings, institutions, commercial buildings, hospitals and industries.
- Municipality waste or kitchen waste utilization for energy generation.

2. Promotion and facilitation of Off-grid and decentralized renewable energy generation for electrification, cooking and other thermal energy requirement.

Around 73% of the state population resides in rural areas⁸. More use of renewable energy technology would ensure energy security, reduce local pollution and increase access to energy in areas where distributed and decentralized forms of energy production would be economically convenient. Hence, renewed efforts to set up decentralized renewable energy solutions, would be undertaken primarily through a combination of off-grid solar power plants, small solar-wind hybrid generators, bio-digesters. This would ensure cooking energy supply for 50 -100 people in institutions, hospitals, industries, etc. and fulfill the energy demand for lighting of rural households. The activities planned are

A. Facilitating deployment of stand-alone off-grid solar power plants of capacity within 50 - 100 kW for fulfilling the

⁸ Economic Survey Report of Jammu & Kashmir -2012-13

power demand of 5 MW in dispersed locations by 12th plan period. The electrical energy from the off-grid power plants would empower the population of the higher terrain and far-flung areas in the country to sustain under extreme weather conditions and severe impact of climate change.

- B. Installation of 300 kW solar-wind hybrid power plants with capacity of 20 kW to meet the power demands in 12th plan period.
- C. Maximizing use of stand-alone Solar Home Lighting System (SHLS) in rural household through distribution of 20,000 SHLS of 40 Wp capacity annually.
- D. Implementation of pilot scale 15 nos. bio-digester systems annually for cooking energy generation for 50 -100 people from kitchen waste and wastes from hostels, hospitals, small scale industries and institutions etc.

The energy requirement for cooking purposes in the commercial establishment and institutions is met with firewood gathered mostly from forest areas or LPG. This not only reduces the area under carbon sink and depletes the conventional fuel reserve but also enhances the vulnerability of the area in light of the proposed climate change impact through top soil denudation in case of flood or other impact, greenhouse gas emission, etc. The use of kitchen waste or other industrial wastes in bio-digester would help in reducing waste generation and dependency on conventional fuel sources.

Moreover, the utilization of wastes would help in keeping the environment clean.

3. State Govt. by amendment of building bye-laws suitable to state condition would promote and mandate use of solar water heating and/or lighting systems.

The state govt. in line with the objective of national solar mission would promote and mandate the use of solar water heating and/or lighting systems according to the state geographical condition, climate scenario and energy usage pattern. Use of Solar Energy for water heating has high potential in the state because of high solar incident radiation and longer daytime. A large amount of energy is consumed for water heating in places like hotels, government offices hostels, guest houses, nursing homes, hospitals, residences due to long winters and extreme cold. Enhancing usage of solar water heating systems will replace use of crude type inefficient electric heaters and address problem of inadequate power supply. Solar driers can be promoted across small scale as well as medium scale drying applications like agro processing units etc. The Forest Rest Houses, Circuit Houses, PWD Bungalows, District Collectorate buildings, Govt. Guest House and Primary Health Centre's, Hospitals using electric geysers contribute higher power demand and green-house gas emission as the power demand is majorly met either using electrical energy or liquid fossil fuel. The activities planned are –

- A. Amendment of building bye-law considering state demographic profile for mandatory use of solar water

heating systems in

- All District Head Quarters of the state by year 2015
- All commercial and institutional buildings of urban and semi urban areas by 2017
- All private residential houses of urban areas with more than 1000 sq. ft. area by 2018
- By 2020, in all towns and small cities; solar water heating system usage by the private households and commercial buildings.

B. Promotion & facilitation for setting up of Solar Water Heating (SWH) systems in all Govt. establishments by undertaking pilot projects –

- Pre-feasibility study for SWH system in Govt. establishments.
- Preparation of bankable DPR
- Arrangement and management of project fund for implementation.
- Facilitating deployment of SWH system in Govt. establishments –
 - o Primary Health Centre's - 10 Nos. of 1,000 LPD each
 - o Hospital - 10 Nos. of 3,000 LPD each
 - o Govt. Higher Secondary School - 10 Nos. of 500 LPD each
 - o District Collectorate Building and other govt.

establishments -20 Nos. of 1,500 LPD each

- Training to users on operation and maintenance of the system

4. Institutional development and strengthening of JAKEDA, LREDA and KREDA for promotion of Renewable Energy technology applications.

To ensure proper implementation of climate friendly sustainable initiatives and to sustain the benefits for longer duration, a detailed strategy for project implementation and rigorous monitoring protocol of implementation initiatives under the state climate change action plan needs to be developed. Therefore, a Programme Management cell (PMC) is to be formulated under the Chairpersonship of Administrative Secretary Science and Technology for a cumulative review and to propose future directions including identification of financial mechanism and bridging of the investment gap for promoting renewable energy technology. This Cell, besides monitoring oversight would also issue guidelines with respect to renewable energy usage, green building concepts etc. and would issue green practices certificates to various projects through regular check mechanisms. A corpus fund will be created with inclusion fund from Govt. of India schemes and state government schemes which will be utilized for following activities -

- A. Training of the working group members and their representatives from JAKEDA, LREDA and KREDA and other concerned departments and organizations on

sector specific climate change issues and enhance knowledge about the policy measures.

- B. Introduction of e-governance system and enabling IT based operation in the JAKEDA, LREDA and KREDA and capacity building of all officers at district level.
- C. Development of Projects in JAKEDA, LREDA & KREDA by way of:
 - Identification of projects,
 - Development of Pre-Feasibility Report, Detailed Project Report,
 - Arrangement of fund,
 - Monitoring of project implementation and operation.

All the programs and initiatives proposed under state climate change action plan for solar energy sector will be implemented through these agencies and other line departments in coordination with PMC. The PMC would report to the Central Advisory Council for Climate Change constituted by the state government. The PMC would help in effective project implementation, monitoring & evaluation, easier documentation, learning, sharing & dialogue on climate change initiatives by the state on solar, small hydro and wind sector.

5. Harnessing renewable energy potential scenario of the state for power generation

The main objective of the action plan is to improve the productive potential of natural and renewable energy sources and reduce the huge dependency on thermal energy

supply to the state. Each of the renewable energy sources can be an eco-sustainable practice; hence this needs to effectively manage for increased supply of electrical and thermal energy and achievement of consequent economic and social growth. The activities planned are -

- A. Assessment of Biomass potential and availability for energy generation in the state and demarcation of biomass potential sites in the map:

The state particularly Jammu region with large no. of agro-processing units has huge potential for biomass based energy generation. To assess the bio-energy available in the state and initiate biomass project implementation following sub activities are planned –

- i. Identification of agency for assessment study
- ii. Study on biomass availability for energy generation in the state for demarcation of biomass potential map.
- iii. Risk assessment of bio- energy sources in anticipated climate change situations (variable rainfall, snowfall, temperature, extreme events)
- iv. Risk assessment of bio- energy infrastructure in climate change situations including extreme events.
- v. Projection and risk assessment of energy demand.
- vi. Conducting detailed feasibility study and identify viable biomass power project

vii. Awareness programmes and capacity building of nodal agency on technological and regulatory aspect

B. Assessment of Solar irradiation, temperature, wind speed at district level across the state for solar mapping.

The state having high incident solar radiation and having longer daytime, has tremendous solar energy potential particularly in the Ladakh region. But, due to rugged hilly terrain and extreme weather conditions in some areas of the state it is needed to identify the solar project sites and assess the energy generation potential of the sites for promotion of solar energy project in the state.

i. Assessment of Solar irradiation, temperature, wind speed at district level across the state for solar mapping.

ii. Risk assessment of solar energy sources in anticipated climate change situations (variable rainfall, temperature, extreme events).

iii. Risk Assessment of solar energy infrastructure in climate change situations including extreme events.

iv. Projection and risk assessment of energy demand.

v. Identification of appropriate sites for solar energy projects

vi. Awareness programmes and capacity building of nodal agency on technological and regulatory aspect.

C. Assessment of Wind energy potential of the state, wind mapping and

identification of wind project sites.

The state is experiencing high wind speed suitable for wind power projects but due to high rugged hilly terrain and large forest area, the wind power projects have not yet scaled up in the state. Moreover, the detailed assessment of wind potential is not carried out and sites appropriate for project set up is also not identified. To promote wind power projects it is therefore needed to conduct a detailed study for wind project site identification and assess its power generation potential.

D. Detailed assessment of Waste to Energy projects from Municipal Waste, Industrial and other wastes in the state.

Waste disposal in today's urban areas is a major problem. It not only requires a large land area but also leads to major environmental concern. Therefore, there is a great potential for diversification of waste to energy projects that would contribute to mitigate power demand – supply gap in urban areas as well as help to overcome the problem of waste handling. While significant potential exists for waste to energy projects, geographic location, end-use function, technology diffusion and grid-parity pricing are the key determinants in technology penetration. To promote waste to energy projects and publicize its importance, following actions planned -

i. Identification of agency for assessment study

ii. Study on waste availability for energy

generation in the urban areas and industrial areas for demarcation of waste mapping.

- iii. Risk Assessment of waste to energy projects viability in anticipated climate change situations (variable rainfall, snowfall, temperature, extreme events)
- iv. Projection and risk assessment of energy demand.
- v. Conducting detailed feasibility study and identify viable project
- vi. Awareness programmes and capacity building of nodal agency on technological and regulatory aspect
- E. Undertake R&D work to improvise the performance of biogas plants in low temperature condition and with reduced livestock waste in high temperature zones like Jammu.

The state is way behind its target for implementation of biogas plants under central scheme. Due to low temperature condition and collection inefficiency of livestock waste, the biogas plants installed in low temperature areas are not running successfully. Whereas collection of livestock is a constraint in high temperature zones; hence, technology innovation is needed to operate biogas plants efficiently with reduced livestock wastes for high temperature zones. To harness and implement biogas technologies R&D initiative needs to be undertaken for improved biogas plant technology and operations.

6. Strengthening of technical competency of various

stakeholders of renewable energy which includes O&M persons, technicians, installers, manufacturers and other relevant service providers as per international standard

Even though the Govt. of India is running several promotional programmes and schemes for renewable energy implementation but one of the main reasons for failure of renewable energy sources is the lack of technical support in terms of proper and adequate installation, maintenance and repair of renewable energy systems due to insufficient or technical competency. The state of Jammu & Kashmir being the extreme northern state and the scattered population distribution, the renewable energy technology manufacturer has low interest level in hilly regions. Therefore, the introduction of training courses in all ITI's, engineering colleges would be beneficial for successful implementation and operation of the renewable energy projects and reduce dependency on external technical experts for operation and maintenance, repair, etc. The activities designed are –

- A. Introduction of curriculum and assessment on Renewable Energy system installation, Operation & Maintenance, repair, etc. in all ITIs of the state to meet the local demand of technician and installer for RE.
- B. Introducing technical course on Renewable energy technology and Energy management at Poly-technic/ Engineering colleges of the state to achieve increase in availability of

technically qualified manpower.

- C. Conducting certificate courses for strengthening of technical competency of the existing solar and other renewable energy technology service providers.
 - D. Supporting state level entrepreneurs to become RESCOs, Channel Partners under JNNSM scheme and renewable energy device manufacturers, distributors, installers, etc.
 - E. Linking promotions of technical personnel with certification in renewable energy technology, O&M, repairs etc.
7. Enhancing state's own power generation capacity through hydro power plant set up and provides support to private/public investors in projects implementation and undertakes micro/mini hydro projects for remote area to meet up local demand.

The state energy sector has taken a lead in creating awareness and committed to go for hydro technologies for improving the scientific knowledge regarding the direct as well as indirect parameters of environmental impact. The state has hydro potential of 20,000 MW but identification of only 16,480 MW has achieved. Therefore, a detailed reconnaissance study is needed to assess the present hydro power potential and to identify project sites. These include analysis of hydrological data of major rivers of Jammu & Kashmir and its tributaries and regular petrographic analysis of major rivers of the

state for assessment of the rate of erosion and analyzing the causes, apart from regular monitoring of the isoceraunic conditions of the region. The above data shall be analyzed on regular basis and study of its impact on the efficiency of the system can be estimated. This would also addressing the issue of “energy access” to people by providing them with “clean energy” and thereby reducing usage of highly polluting thermal power and reducing power procurement from northern grid is the prime focus of the state govt. The following activities are planned –

- A. Detailed reconnaissance study on water availability and hydrology data evaluation for identification of new hydro projects and demarcation of hydro power sites with specific capacity mapping.
- B. Promotion & facilitation of hydro power project implementation by providing adequate support from the state government in terms of clearance, land acquisition, power transmission network development and funding. Encouragement of greener construction technologies in new Hydro power project implementation.
- C. Declaration of water policy and mandate siltation and pollution control in water bodies of hydro power projects.
- D. Implementation of 50 MW mini, micro hydro project in already identified project sites within next 5 years through IPP mode and 50 MW through IPP mode.-

Following sub-activities will be undertaken

to achieve the target of 60 MW hydro power project -

- Identification of project sites
- Bankable DPR preparation
- Identification and selection of project investor and fund
- Undertake project clearances, grid connectivity
- Implementation of hydro power project

8. Promotion of grid connected Rooftop and Small Solar Power projects in the state of J&K

Since, the state does not have adequate electricity generation capacity to meet the growing power requirements, renewable energy generation capacity addition particularly solar energy needs to be considered. Though multiple renewable energy projects have been implemented; still the grid interactive solar power generation option has not been explored so far. To mitigate power demand –supply gap of the state, particularly to meet the high peak demand, the state is focusing to explore its solar power potential for grid connected power generation and set the target for 25 MW grid connected solar power project.

Implementation of solar power grid connected projects will address the livelihood concerns of related stakeholders, strengthening supply chains of solar energy products in the state. The actions planned are:

- A. Implementation of 25 MW grid connected Rooftop and small solar

power plant up to 2 MW capacities across the state within 2017-18 by Independent Power Producers (IPP) and also through PPP mode.

Participation from all sectors investors is necessary to emerge the solar technology based power project which is new concept to the state. Hence, investments from private sector entities, central and state governments and also public sector entities should be encouraged and following activities to be executed -

- Identification of project sites
 - Bankable DPR preparation
 - Identification and selection of project investor and fund
 - Undertake project clearances, grid connectivity
 - Implementation of solar power plant
- B. Development of disposal policy towards proper decommissioning and recycling of solar panels installed across the state.

Even though solar photovoltaic based power generation is clean and environmentally sustainable; but use of harmful chemicals like mercury and chromium in solar cell manufacturing may prove to be detrimental if is not disposed properly. With the increase in solar project implementation environment degradation due to solar cell disposal will also increase which may affect the environment as well as the ecosystem of the state. Hence, policy needs to be developed for adequate disposal measures including recycling of

disposed solar cells. Separate solar cell disposal site and training of recycling of solar cells is also needed.

9. Promotion of Green buildings in the state—

The climatic condition of the state varies from region to region with hot climate conditions in Jammu and cold climate in Kashmir. This leads to higher power demand throughout the year. Due to land shortage in urban areas for construction and with ever increasing urban population it is much more crucial to have scientifically built and climate friendly buildings are inevitable. Conventional method of construction uses large quantity of material and many of them are non-renewable and toxic leading to environmental degradation. Green buildings would not only reduce the impacts but would consume less energy and water, generate less waste, conserve natural resources, would be climate friendly. The activities planned are –

- A. Development of policy framework on green building considering state geographical location, prevailing climatic condition.
- B. Introducing mandatory norms for undertaking green building measures in all new govt. buildings and residential buildings of more than 1500 sq. ft. in cities of Jammu & Srinagar in line with

Green building standard and Energy Conservation and Building Code (ECBC) of the Bureau of Energy Efficiency, Govt. of India and mandatory compliance of energy efficiency measures, rain water harvesting measures, recycling measures for water and wastes in all big hotels. The policy would also mandate the buildings to implement green building standard and should have green building certification. It is also recommended that till the point compliance is being made mandatory for the residential building, special incentives may be offered so as to encourage the technology and increase awareness about the potential benefit of using the green building concept.

- C. Capacity Building of authorities (Housing & Urban Development Department, PHE, Irrigation & Flood Control Department, Public Works (R&B) Department, JAKEDA, LREDA, KREDA) on green building policy and its requirement. Awareness generation through workshop and advertising media on green building standard and involve NGO, community groups, social Organisation for creating awareness.

The green building would contribute in energy security of nation and also reduce the overall global carbon footprint.



4.7 List of Key Priority Action

Table 13: List of Key Priority Action Proposed under Sustainable Energy Mission

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
1	Promotion and implementation of solar city project by	JAKEDA, LREDA, KREDA, Housing & Urban Dev. Dept.	20.0	20.0	40.0	MNRE, Govt. of J&K, Dept. of Urban Development
A)	Preparation of a master plan for all district HQs under the "Solar City project"					
B)	Set-up of institutional arrangements for implementation of the master plan					
C)	Awareness generation and capacity building activities by managing publicity programmes, conducting training programmes/business meets for stakeholders					
D)	Implementation of pilot solar city project in phase wise- Phase I - Jammu, Srinagar and Katra, Phase II - Kupwara & Udhampur, Phase III - Rajauri, Poonch, Baramula,					
2	Promotion and facilitation of Off-grid and decentralized renewable energy generation for electrification, cooking and other thermal energy requirement.	JAKEDA, LREDA, KREDA, Rural Dev. Dept.				MNRE, Govt. of J&K, User or Beneficiary Department or Organization, External Agencies
A)	Facilitating deployment of 5 MW stand-alone off-grid solar power plant of capacity within 50 - 100 kW for fulfilling the power demand in dispersed locations by 12th plan period.		1125.0	125.0	1250.0	
B)	Installation of 300 kW solar-wind hybrid power plants with capacity of 20 kW in 12th plan period.		3.4	1.1	4.5	

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
C)	Maximizing use of stand-alone Solar Home Lighting System (SHLS) in rural household level through distribution of 20,000 SHLS of 40 Wp capacity annually.		324.0	756.0	1080.0	
D)	Implementation of pilot scale 15 nos. bio-digester systems annually for cooking energy generation for 50 -100 people from kitchen waste and other waste in hostels, hospitals, small scale industries and institutions etc.		1.5	1.5	3.0	
3	State Govt. by amendment of building bye-laws suitable to state condition will promote and mandate use of solar water heating systems and/or lighting	JAKEDA, LREDA, KREDA, Housing & Urban Dev. Dept.				MNRE, Govt. of J&K, Dept. of Urban Development, Municipality of J&K, User or Beneficiary Department or Organization, External Agencies
A)	Amendment of building bye-law considering state demographic profile for mandatory use of solar water heating systems in - All District Head Quarters of the state by year 2015. - All commercial and institutional buildings of urban and semi urban areas by 2017. - All private residential houses of urban areas with more than 1000 sq. ft. area by 2018. - By 2020, in all towns and small cities; solar water heating systems usage by the private households and commercial buildings.		0.00	0.00	0.00	

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
B)	Promotion & Facilitation for setting up of Solar Water Heating (SWH) systems in all Govt. establishments by undertaking pilot projects in – • Primary Health Centre's - 10 Nos. • Hospital - 10 Nos. • Govt. Higher Secondary School - 10 Nos. • District Collectorate Building and other govt. establishments -20 Nos.		9.9	6.6	16.5	
4	Institutional development and strengthening of JAKEDA, LREDA and KREDA for promotion of Renewable Energy applications	JAKEDA, LREDA, KREDA, State Climate Change Cell, J&K				MNRE, Govt. of J&K, JAKEDA, LREDA, KREDA,
A)	Training of the working group members and their representatives from JAKEDA including LREDA and KREDA and other concerned departments and organizations on sector specific climate change issues and enhance the knowledge about the policy measures.		0.00	10	10	
B)	Introduction of e-governance system and enabling IT based operation in the JAKEDA, LREDA and KREDA and capacity building of all officers at district level.		1.3	3.7	5.0	
C)	Development of Programme Management Cell in S&T Dept. for identification of projects, development of Pre-Feasibility Report, Detailed Project Report, Arrangement of fund, Monitoring of project implementation and operation.		0.00	5.0	5.0	

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
5	Harnessing renewable energy potential scenario of the state for power generation	JAKEDA, Housing & Urban Development Dept. -J&K				MNRE, Govt. of J&K, Funding Agencies
A)	Assessment of Biomass potential and availability for energy generation in the state and demarcation of biomass potential sites in the map.		0.00	3.0	3.0	
B)	Assessment of Solar irradiation, temperature, wind speed at district level across the state for solar mapping.		0.00	5.0	5.0	
C)	Assessment of Wind Energy potential of the state, wind mapping and identification of wind project sites.		0.00	5.0	5.0	
D)	Detailed Assessment of Waste to Energy projects from Municipal Waste, Industrial and other wastes in the state.		0.00	5.0	5.0	
E)	Undertake R&D work to improve the performance of biogas plants in low temperature condition and with reduced livestock waste in high temperature zones.		0.00	5.0	5.0	
6	Strengthening of technical competency of various stakeholders of renewable energy which includes O&M persons, technicians, installers, manufacturers and other relevant service providers as per international standard.	JAKEDA, Technical Education Dept. -J&K, Industry & Commerce Dept. -J&K				Govt. of J&K, Dept. of Higher Education, Dept. of Technical Education, MNRE
A)	Introduction of subject or paper on RE system installation, O&M, repair, etc. in all ITIs of the state to meet a local demand.		1.5	0.00	1.5	

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
B)	Introducing technical course on Renewable energy technology and Energy management at Poly-technic /Engineering colleges of the state to achieve increase in availability of technically qualified manpower.		1.5	0.00	1.5	
C)	Conducting certificate courses for strengthening of technical competency of the existing solar and other renewable energy technology service providers.		0.00	1.0	1.0	
D)	Supporting state level entrepreneurs to become RESCOs, Channel Partners under JNNSM scheme and renewable energy device manufacturers, distributors, installers, etc.		0.00	0.5	0.5	
7	Enhancing state's own power generation capacity through hydro power plant set up and provides support to private/public investors in project implementation and undertakes micro/mini hydro projects for remote area's to meet up local demand.	JAKEDA, JKSERC, JKSPDC, LREDA, KREDA, PHE, Irrigation & Flood Control Dept. -J&K				Govt. of J&K, Funding Agency, Govt. of India, Dept. of Water, MNRE
A)	Detailed reconnaissance study on water availability and hydrology data evaluation for identification of new hydro projects and demarcation of hydro power sites with specific capacity mapping.		0.00	10.0	10.0	
B)	Promotion & facilitation of hydro power project implementation by providing adequate support from the state government in terms of clearance, land acquisition, power transmission network development and funding. Encouragement of greener construction technologies in new Hydro power project implementation.		0.00	1.0	1.0	

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
C)	Declaration of water policy and mandate siltation and pollution control in water bodies of hydro power projects.		0.5	0.00	0.5	
D)	Implementation of 10 MW mini, micro hydro project in already identified project sites by next 5 years through PPP mode and 50 MW through IPP mode.		0.00	1000.0	1000.0	
8	Promotion of grid connected Rooftop and Small Solar Power projects in the state of J&K	JAKEDA, JKSERC, JKSPDC, LREDA, KREDA				MNRE, Private or Public Agency, Govt. of J&K, Funding or Donor Agency
A)	Implementation of 25 MW grid connected Rooftop and small solar power plant up to 2 MW capacities across the state within 2017-18 by Independent Power Producers (IPP) and also through PPP mode.		0.00	2000.0	2000.0	
B)	Declaration of policy measures for proper decommissioning and recycling of all solar panels installed across the state.		0.00	2.5	2.5	
9	Promotion of green buildings in the state	JAKEDA, LREDA, KREDA, PHE, Irrigation & Flood Control Dept. - J&K, Housing & Urban Development Dept. - J&K, Public Works (R&B) Department				MNRE, Govt. of J&K, Govt. of India, BEE, Private or Public Agency, JNNURM
A)	Development of policy framework on green building considering state geographical location, prevailing climatic condition.		1.5	0.00	1.5	
B)	Introducing mandatory norms for Undertaking green building measures in all new govt. buildings and residential buildings of more than 1500 sq. ft. in cities of Jammu & Srinagar. Mandatory compliance of energy efficiency measures, rain water harvesting measures, recycling measures of water and wastes in all big hotels.		0.00	20.0	20.0	

Sl. No.	Title	Organizations	Budget (Rs. In Million)			Source of Funding
			Existing	Additional	Total	
C)	Capacity Building of authorities (Housing & Urban Development Department, PHE, Irrigation & Flood Control Department, Public Works (R&B) Department, JAKEDA) on green building policy and its requirement.		0.00	3.5	3.5	
	Total		1470.1	3970.4	5440.5	



5 Enhanced Energy Efficiency

5.1 Introduction

Jammu & Kashmir is the northernmost state of India. It is situated mostly in the Himalayan Mountains. Jammu and Kashmir shares a border with the states of Himachal Pradesh and Punjab to the south and internationally with the People's Republic of China to the north and east and Pakistan to the west. The State has a population of 1,25,48,926 with 20,15,088 households (14,97,920 rural and 5,17,168 urban) as per the 2011 Census.

The state is traversed by three main rivers i.e. Indus, Jhelum and Chenab. The Indus traverses through Ladakh, while the Jhelum and Chenab flows through Kashmir & Jammu respectively. The average rainfall is about 100 cm. Huge glaciers in the state and existence of high mountains with glacial cover and heavy rainfall makes it a haven for hydro power generation.

As per preliminary survey an estimated potential of only 16,480 MW of power has been identified, much less than the actual potential. The state of Jammu and Kashmir was the second, next to Mysore to use Hydro Power much back in the 1st decade of this century.



Figure 9: Location of Hydro Power Project

5.2 Key trends in the sector

Power sector plays a major role to any developmental effort. A sustained growth in power sector is crucial to ensure economic growth. The gradual increase in demand of power is a sign of improvement in human development as well as industrialization. Hydro power is recognized as an ecofriendly, economical and green renewable source of energy. State of Jammu and Kashmir has huge hydel potential which if utilized fully would provide a strong impetus to the growth of the State's economy. The tapping of hydro power potential calls for huge resources, technical expertise, administrative reforms, congenial environment, proper regulation and management besides competitive marketing, policy formation and private participation. Optimal exploitation of the

available hydel resources would also suffice for power supply to Northern grid along with meeting the state's internal demands.

Total estimated potential for hydro power generation in the state is 20,000 MW out of which 16,480 MW (excluding 251 Projects up-to 25 MW generation potential identified totaling to 986 MW Capacity) has been identified as the actual generation potential. Out of the identified potential only 2457.96 MW or 15% has been harnessed so far.

5.2.1 Electric Power Installed Capacity⁹

The power generation in the state is mainly governed by hydro power with a little share of fossil fuel based grid interactive power plant. The list of hydro power generating station in the state is as follows:

Table 14: Capacity wise installed hydro power project State Sector

Under Power Development Corporation	
Hydro	759.96 MW
Total No. Of Hydel Projects	20
Thermal	175MW
Total No. Of Thermal Projects(Operated only in case of emergency in view of high per unit generation cost)	2
Under Power Development Department	
Diesel	33.146 MW

Table 15: Capacity wise installed hydro power project Central Sector

Hydro	1680 MW
Total No. Of Hydel Projects	4

⁹Year book 2011-12_PDD J&K_12th Oct'12 & Economic survey J & K 2012-13-Power

Table 16: Capacity wise installed hydro power project Independent Power Producers

Hydro	17.5 MW
Total No. Of Hydel Projects	4

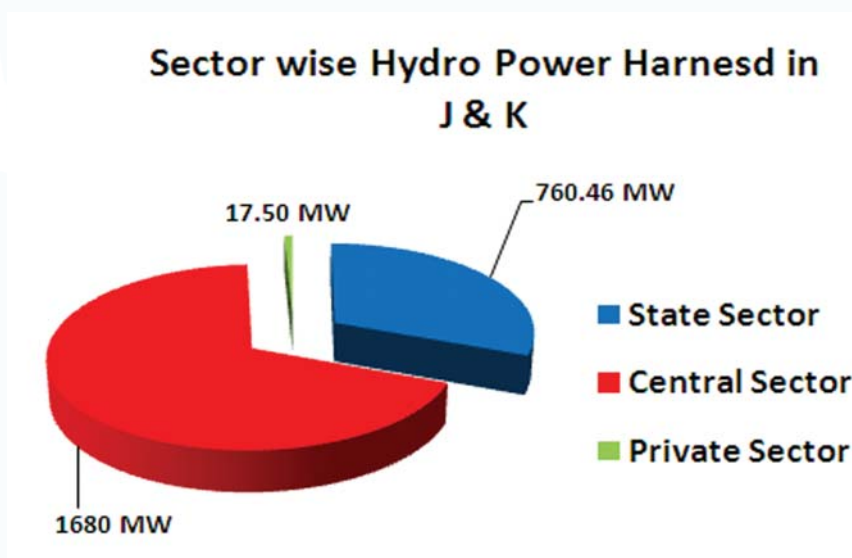
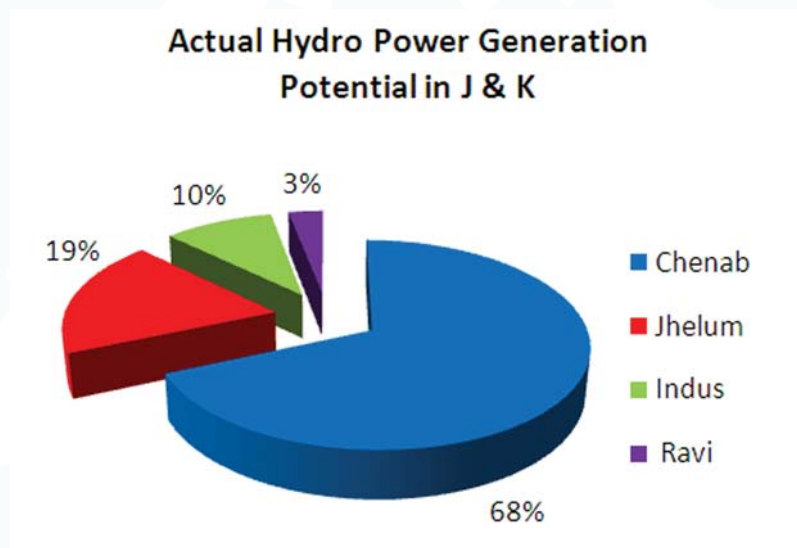


Figure 10: Source wise hydro power potential and sector wise hydro power resource harnessed

Table 17: List of Key Hydro Power Project and their capacity

Sl. No.	Name of Project	Basin	Capacity in MWs	Configuration (MW)
1	Lower Jhelum	Jhelum	105.00	3 x 35
2	Upper Sindh-II	Jhelum	105.00	3x35
3	Upper Sindh-I	Jhelum	22.60	2 x 11.3
4	Ganderbal	Jhelum	15.00	2x3+2x4.5
5	Pahalgam	Jhelum	3.00	2 X 1.5
6	Karnah	Jhelum	2.00	2 x 1
7	Baglihar Stage I	Chenab	450.00	3x150
8	Chenani-I	Chenab	23.30	5 x 4.66
9	Chenani-II	Chenab	2.00	2 x 1
10	Chenani-III	Chenab	7.50	3 x 2.5
11	Bhaderwah	Chenab	1.50	3 X 0.5
12	Iqbal	Indus	3.75	3 x 1.25
13	Sumoor	Indus	0.10	2 x 0.50
14	Hunder	Indus	0.40	2 x 0.20
15	Bazgoo	Indus	0.30	2 x0.15
16	Igo- Marcelloung	Indus	3.00	2x1.5
17	Marpachoo	Indus	0.75	3x0.25
18	Haftal	Indus	1.00	2x0.5
19	Satakna	Indus	4.00	2 X 2
20	Sewa-III	Ravi	9.00	3 x 3
21	Sanjak	Indus	1.26	3 x 0.42
Sub Total State Sector(in MW)			760.46	
1	Salal	Chenab	690.00	6x115
2	Dulhasti	Chenab	390.00	3 x130
3	Uri-I	Jhelum	480.00	4x120
4	Sewa-II	Ravi	120.00	3x40
Sub Total Central Sector(in MW)			1680.00	
Private Sector				
1	Athwattoo	Jhelum	10.00	2x5
2	Brenwar	Jhelum	7.50	3 X2.50
Sub Total Private Sector(in MW)			17.50	
Total potential in operation(in MW)			2457.96	

5.2.2 Sector Wise Energy Consumption

Domestic consumers as a whole are the major electricity consuming sector in the state with cumulative consumption in tune of 34%. The category-wise energy consumption is given in the table below.

Table 18: Category wise Electrical Energy Consumption

Sl. No.	Customer category	2009-10		2010-11		2011-12	
		Energy Consumption (In MU)	%age	Energy Consumption (In MU)	%age	Energy Consumption (In MU)	%age
1	Domestic	1,380.00	36.00	1,255.93	31.08	1,431.98	33.56
2	Non- Domestic/ Commercial	295.00	7.70	349.1	8.64	333.26	7.81
3	Industrial	856.00	22.33	816.54	20.21	849.53	19.91
4	Govt.						
4.a	Irrigation/ Agriculture	295.00	7.70	198.1	4.90	140.67	3.30
4.b	Public Lighting	14.00	0.37	29.07	0.72	35.66	0.84
4.c	Public Water Works	383.00	9.99	586.04	14.50	636.14	14.91
4.d	State Central Dept.	532.00	13.88	711.2	17.60	695.65	16.30
4.e	General Purpose Bulk supply	78.00	2.03	95.1	2.35	144.10	3.38
Grand Total		3,833.00	100.00	4,041.08	100.00	4,266.99	100.00



From above statistics it is evident that although the demand of electricity across public utility is low but the demand is increasing. However this increasing demand can also be met without increasing the load by introducing energy efficiency measures. A huge potential of energy saving lies with the domestic sector that accounts for the major share of energy in the state. With a potential of 20-25% saving in the domestic sector, introduction of energy conservation measures can result in substantial energy saving which can result in meeting of unmet demand and energy shortage in the state.

5.2.3 Demand Supply Scenario Analysis

The energy demand has seen a gradual

increase during 11th Plan period at an annual rate of 5 to 6%. To bridge the widened gap between demand and supply, the State Power Department has enforced power cuts of the order of 8 hours a day in summer and 10 hours a day in winter (under most critical scenario). Even with scheduled load shedding activity, the restricted demand turns out in tune of 11,091.26 MUs in 2011-12, which necessitated banking of power during summer with other state utilities and using of the same during winter.

The total availability of power in J & K from all the sources was just around 64 percent of the total requirement in FY 2011-12. The State is forced to purchase power from various sources besides, drawing from Northern grid.

Table 19: Peak and Power Availability Scenario in the State

Energy Scenario (in MW)	2004 -05	2005 -06	2006 -07	2007 -08	2008 -09	2009 -10	2010 -11	2011 -12	2012 -13 (E)
Peak Demand	1706	1800	1925	2020	2120	2290	2369	2500	2600
Peak Met	1350	1242	1306	1350	1340	1407	1490	1788	1900
Peak Deficit	356	558	619	670	780	883	877	711	700
Peak Deficit (%)	20.87	31.00	32.16	33.17	36.79	38.56	37.02	28.44	26.92

Table 20: Energy Demand and Availability

Energy Scenario (in MU)	2004 -05	2005 -06	2006 -07	2007 -08	2008 -09	2009 -10	2010 -11	2011 -12	2012 -13 (E)
Energy Requirement	9604	10503	11343	14037	14750	15656	16544	17323	17842.69
Restricted Energy	7306	7916.38	8236.53	8743.96	9147.21	10370	10667	11115.41	11560.03
Energy Deficit	2334	2586.62	3106.47	5293.04	5602.79	5286	5877	6207.59	6282.66
Energy Deficit (%)	24.21	24.63	27.39	37.71	37.99	33.76	35.52	35.83	35.21

Per Capita Power Consumption (kWh)

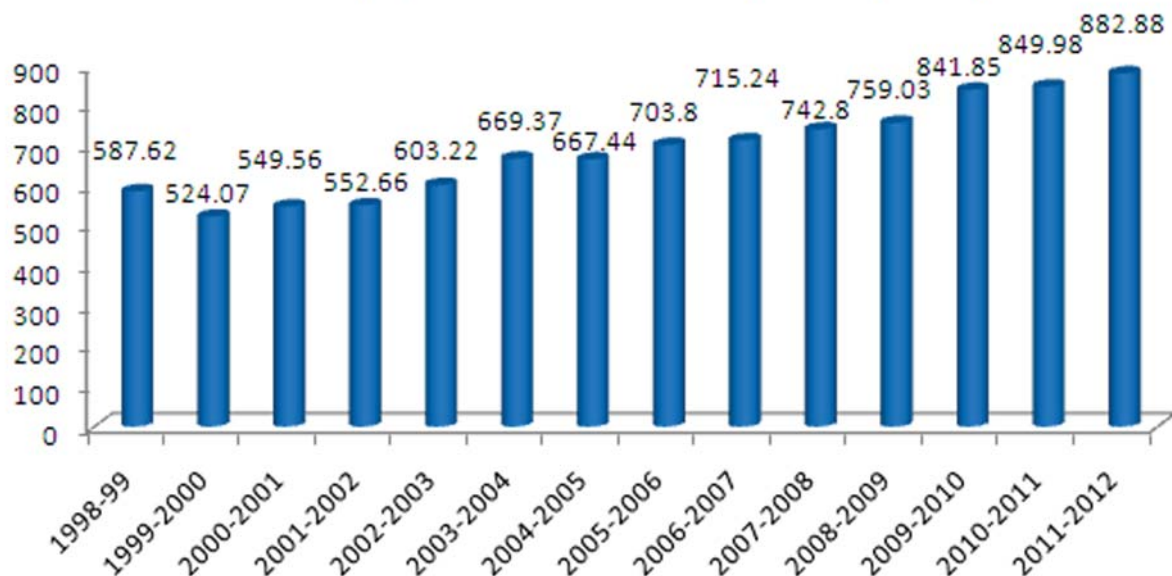


Figure 11: Gross Per capita Electricity (kWh) consumption in J & K¹⁰

5.3 Key issues in the sector

5.3.1 AT & C Losses

Once electric energy is generated, it is transmitted to areas where it will be used. Transmission refers to delivering electric energy from point of generation to distribution network. Distribution encompasses the substations and feeder lines takes power from the high voltage grid and progressively step down the voltage before it finally reaches the premises of different categories of consumers. The entire process of transmission from generating station to users results in technical losses of energy known as Transmission and Distribution losses. Over and above the T&D losses, losses

are also accounted on commercial ground due to low metering efficiency, theft, pilferages, non-metering or non-recovery of payments. The transmission and distribution losses coupled with commercial losses are collectively known as Aggregate Technical and Commercial (AT&C) loss. The AT&C losses in the State are very high of the order of 61% as recorded in FY 2011-12. Technical losses pertain to outdated and old electrical device in the network. To minimize such losses, the system needs up-gradation and improvements. Commercial losses include power theft, unaccounted and uncontrolled consumption beyond agreement load, unregistered consumers, lesser contract demand etc.

¹⁰Year Book 2011-12_PDD_12th Oct'12

Table 21: T&D and AT&C Losses (from 2006-07 to 2011-12)

Sl. No.	Particulars	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12
1	Energy Input (MU)	8236.53	8743.96	9148.07	10370.65	10667.20	11091.26
2	Energy Billed (MU)	4030.84	3331.64	3538.71	3833.00	4041.10	4267.00
3	T&D Losses (MU)	4205.69	5412.32	5609.36	6537.65	6626.10	6824.26
4	%age T&D Losses	51.06	61.90	61.31	63.04	62.12	61.61
5	Collection Efficiency (%age)	66.59	69.74	75.70	75.67	68.00	70.99
6	Energy realized (MUs)	2676.26	2416.20	2633.12	2902.07	2749.38	3029.14
7	AT&C Losses (MUs)	5560.27	6327.76	6586.85	6537.65	7917.82	8062.12
8	%age AT&C Losses	67.51	72.37	70.69	72.03	74.23	72.68

5.3.2 Sector specific energy conservation and saving potential

The energy consumption patterns are changing with invention of new technology and initiation of energy conservation measures by State Government. Efforts have been made to collect data on electricity conservation measures in agriculture, commercial, municipalities, SMEs, etc. Irrespective of the initiatives being taken up a considerable potential of energy saving lies in each sector. "State-wise Electricity Consumption & Conservation Potential in India" carried out by NPC in 2009 provides the following estimation of possible saving in energy consumption in J & K.

The possibility of energy savings is high in the following sectors of the State.

- Agricultural Sector
- Power Distribution System
- Lighting Sector
- Building Sector
- Industrial Sector
- Municipalities, etc.
- **Energy saving potential in Agricultural Sector**

A large numbers of Agricultural pump sets and Agricultural machineries consume significant quantum of fossil fuels and electricity. Inefficient agricultural pumps may be replaced by star rated efficient ones. Based on several studies on agricultural pump set efficiency, it has been found that the efficiency varies from 25-35% due to various factors. With a BEE star labeled agricultural pump set, the efficiency can be enhanced upto 50-52%¹¹.

¹¹ Combined Summary Report_ NPC 2009

- ***Energy saving potential in Power distribution system***

Jammu & Kashmir Power Development Department has an important role to play in making the state level Electricity grid more climate-friendly. The transmission and distribution loss in Jammu & Kashmir is more than 61% and this can be reduced up to 28.6% on a time bound programme, say by the financial year 2018-19. Appropriate strategic actions are required to reduce the T&D Loss of the State.

- ***Energy saving potential in Lighting & Domestic Sector¹²***

30% of the power is consumed for lighting purposes. Thus use of efficient light could be made mandatory particularly for commercial organizations. Star rated home appliances should be encouraged in domestic sector. Usage of CFLs and BEE star rated products can save up-to 40-50% of power in rural sector and 15-20 % in urban areas. On the whole, the energy savings potential in domestic sector is estimated 20-25 %.

- ***Energy saving potential in Building sector***

Energy Conservation Act, 2001 empowers the Government to prescribe the ECBC (Energy Conservation Building Code) for efficient use

of energy and its conservation in buildings or building complexes. The ECBC sets minimum energy performance standards for design and construction of non-residential buildings. Studies conducted by NPC reveal that energy savings potential in commercial buildings varies from 20-30%.

- ***Energy saving potential in Industrial Sector***

The annual electricity sale to the industry sector including low & medium voltage consumers (SME) and high voltage consumers (large industries) is 953 MU which is 23.62 % of the total electricity sold. The larger industry segment is covered under the mandates of EC Act as designated consumers for energy efficiency, while SME segment is being addressed through cluster based initiatives by Bureau of Energy Efficiency. Based on several studies & energy audits by NPC, the electrical energy saving potential in industry sector varies from 7-10%.

- ***Energy saving potential in municipalities***

Based on sample studies by NPC, the energy saving potential for municipalities & corporations from street lighting is assessed to be 25% and that from water works & sewage is 20%.

¹²Combined Summary Report_ NPC 2009

Table 22: Sector wise Energy Conservation Potential in Jammu & Kashmir (2007-2008)

Sl. No.	Sector Reference	Annual Energy Consumption		Estimated annual Energy Saving Potential	
		In MU	In TOE	In MU	IN TOE
1	Agricultural	271	-	58.8	-
2	Commercial	213	-	11.3	-
3	Municipalities	472	-	70	-
4	SME Clusters ¹³	15.31	-	0	-
5	Domestic	1399	-	280	-
6	Industries	953	-	67	339
Total		3323.31		487 MU	339

5.3.3 Grid electricity consumption and GHG emission

Greenhouse gases (GHG) are emitted mainly from burning of fossil fuels. Generation of electricity which is mostly from conventional power plant results in considerable amount

of GHG emission. An assessment is being made to account the contribution of GHG from consumption of electrical energy in the state. Since the state is interconnected to NEWNE grid the specific emission factor of the grid is being considered for estimation.

Table 23: Sector Specific Electricity consumption & GHG emission from electricity consumption

Consumption Categories	In MU	In MWh	Emission Factor ¹⁴ (NEWNE Grid)	tCO ₂	%
Domestic	1,431.98	14,31,980	0.94	13,46,061	12.91
Non-Domestic/ Commercial	333.26	3,33,260	0.94	3,13,264.4	3.00
Industrial	849.53	8,49,530	0.94	7,98,558.2	7.66
Irrigation/Agriculture	140.67	1,40,670	0.94	1,32,229.8	1.27
Public Lighting	35.66	35,660	0.94	33,520.4	0.32
Public Waterworks	636.14	6,36,140	0.94	5,97,971.6	5.74
State Central Dept.	695.65	6,95,650	0.94	6,53,911	6.27
General Purpose Bulk supply	144.10	1,44,100	0.94	1,35,454	1.30
T & D losses-MU	6,824.26	68,24,260	0.94	64,14,804	61.53
Total GHG Emission (tCO ₂)				104,25,775	

¹³SME clusters electrical energy savings potential is already included in Industrial Sector.

¹⁴CO₂ Baseline Database for the Indian Power Sector, User Guide, Version 8.0_January 2013

GHG Emission in J & K in FY 2011-12

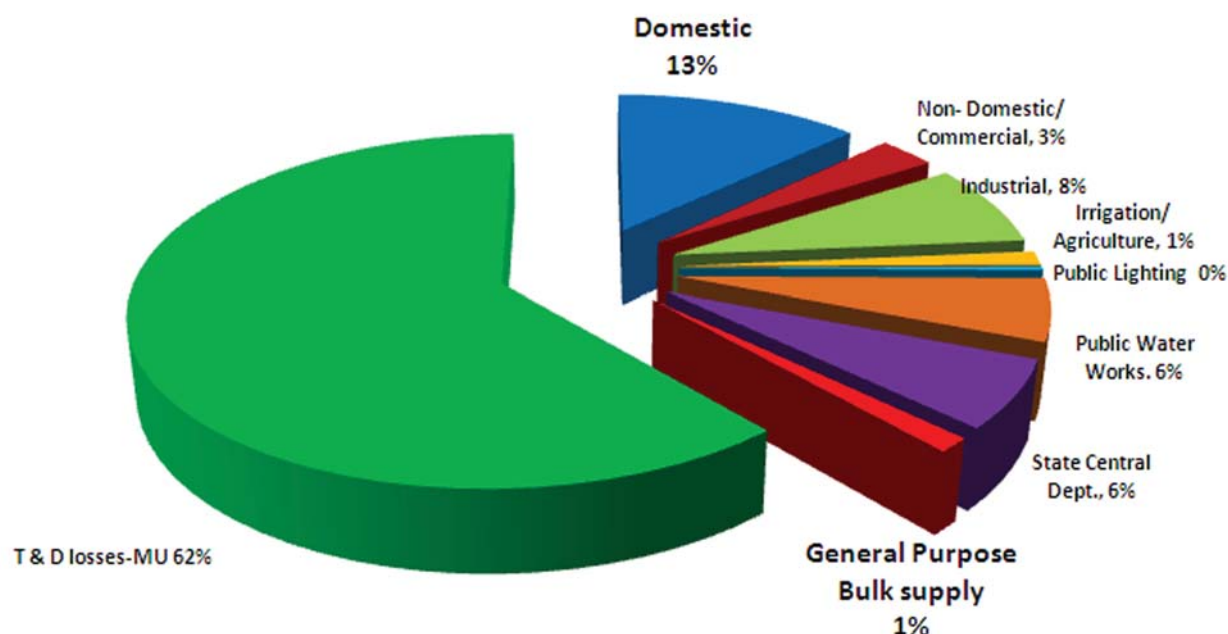


Figure 12: Sector specific GHG emission from electricity consumption

5.4 Programmes and Policies in the sector

5.4.1 National Mission on Enhanced Energy Efficiency

The National Mission for Enhanced Energy Efficiency is one of the eight missions of National Action Plan on Climate Change configured towards improvising cost effective strategies for end-use demand side management. The Mission, by 2014-15, is likely to achieve about 23 million tons oil-equivalent of fuel savings- in coal, gas, and petroleum products, along with an expected avoided capacity addition of over 19,000 MW. The carbon dioxide emission reduction is estimated to be 98.55 million tons annually.

In addition to the policies and programmes for energy efficiency implemented by BEE, NMEEE would enforce the following four initiatives. These initiatives are as follows:

- Perform Achieve and Trade (PAT), a market-based mechanism to make improvements in energy efficiency in energy-intensive large industries and facilities more cost-effective by certification of energy savings that could be traded.
- Market transformation for energy efficiency (MTEE) with a shift to energy-efficient appliances in designated sectors through innovative measures that make the products more affordable.

- Energy efficiency financing platform (EEFP), a mechanism to finance DSM programmes in all sectors by capturing future energy savings.
- Framework for energy efficient economic development (FEEED), or developing fiscal instruments to promote energy efficiency.

The NMEEE set out a comprehensive strategy, which comprise the following components;

- Create demand for energy-efficient products, goods, and services by creating awareness about the efficacy of these products and services, amending government policies and programmes to integrate energy efficiency, preparing bankable projects to stimulate the process, and offering right incentives to cost-effective improvements in energy efficiency in energy-intensive industries and facilities through certification of energy savings that could be traded.
- Ensure adequate supply of energy-efficient products, goods, and services. This is being done by creating a cadre of certified energy professionals; promoting energy service companies (ESCOs), standards, and labeling of end-use equipment and appliances; and preparing structured programmes to leverage international financing instruments including the Clean Development Mechanism (CDM) to reduce transaction costs to attract private investment, etc.
- Create and promote energy efficiency financing platform, set up partial risk

guarantee funds, and develop innovative financial derivatives of performance contracts and fiscal and tax incentives for investment in this sector.

- Create and adopt robust and credible monitoring and verification protocols to capture energy savings from all energy-efficient activities in a transparent manner.
- Take necessary steps to overcome market failures with appropriate regulatory and policy framework to support the measures mentioned above.

5.4.2 Restructured- Accelerated Power Development and Reforms Programme (R-APDRP)

The Govt. of India has proposed to continue R-APDRP as a Central Sector Scheme in the XI Plan with revised terms and conditions. The focus of the programme would be on actual, demonstrable performance in terms of sustained loss reduction. Establishment of reliable and automated systems for collection of accurate base line data, and adoption of Information Technology in the areas of energy accounting would be essential before taking up the regular distribution strengthening projects.

- R- APDRP is a scheme to bring down T&D losses to 15% within project period of 5 years with minimum target Of 3% p.a.
- The scheme is divided in two parts, PART-A and PART-B. Part-A is to cover

Ring Fencing, Feeder Metering, DT Metering, application of Information Technology and Supervisory Control and Data Acquisition / DMS in Distribution Sub Sector. Part-B is aimed to strengthen Distribution System in the identified project areas.

- The Power Finance Corporation (PFC) will be the Nodal Agency for the operationalization and implementation of the RAPDRP programme under the overall guidance of the Ministry of Power (MoP).
- GoI will provide 100% loan for part A of the R-APDRP schemes and 90% loan for part B of the R-APDRP.
- Under R-APDRP scheme 30 Towns have been identified as project areas in J&K State. Out of which 11 towns are in Jammu region including Jammu City and 19 towns in Kashmir region including Srinagar City. These towns are as under: Akhnoor, Anantnag, Budgam, Bandipore, Baramulla, Bhaderwah, Bijbehara, Doda, Duru-Verinag, Ganderbal, Handwara, Jammu, Kargil, Kathua, Kishtwar, Kulgam, Kupwara, Leh, Pattan, Pulwama, Punch, Rajouri, Ranbir-Singhpura, Samba, Shopian, Sopore, Srinagar, Sumbal, Tral and Udhampur. GoI has sanctioned Rs 191.25 Crore for part-A and Rs 1665.27 Crore for part-B of R-APDRP scheme. The DPR for SCADA / DMS for Jammu and Srinagar cities have been approved by GoI at an estimated cost of Rs. 52.89 Crore.

5.4.3 Power Sector Reforms

To support reform initiatives in power sector, three tripartite MOUs have been signed so far between J&KPDD, Planning Commission, GoI and Ministry of Finance GoI to avail Central assistance for the years 2006-07, 2007-08 and 2008-09. The State Government needs to prepare and submit a Blue Print for the turnaround of T&D power sector. Some of the key recommendations are given below;

- Projected reduction of T&D losses from 62.9% in 2009-10 to 28.6% in 2018-19.
- Legislative requirement i.e. enactment of J&K Electricity Act and J&K Conservation Act on the pattern of the Central Acts.
- Making SERC fully functional and Administrative Reforms for ensuring accountability.
- Establishment of base line data and undertaking T&D loss studies.
- Unbundling of T&D sectors.
- Tariff revision through SERC.
- Technical intervention like metering by 2012-13.
- Strengthening/upgrading computerized bills / collection system.
- Introduction of HVDS/ABC in power theft prone areas.
- Power factor correction - Installation of shunt capacitors.
- Energy saving incentive likes Time of Day tariff, Power factor incentives,

seasonal tariff and subsidy on energy savers etc.

- Strengthening training/capacity building.
- Augmentation of system network at various stages etc.

5.4.4 Hon'ble Prime Ministers Reconstruction Programme

The Hon'ble Prime Minister of India during his visit to J&K in November, 2004 announced a Re-construction Plan in infrastructure for the state. The objective of the scheme under the program is to develop infrastructure at 220KV and 132KV level by way of construction/ Augmentation of Grid Stations, new lines and thickening of conductors in existing lines. Originally the project was estimated to cost around Rs. 707.00 Crore based on cost data 2003. The project was scrutinized by CEA and the project cost was revised to Rs. 1006.51 Crore subsequently. Accordingly, the revised project amount of Rs. 1,351.00 Crore was furnished to CEA for approval and the Project Report was approved for 1,350.00 Crore. Out of this cost, an amount of Rs. 1,048.56 Crore was spent till March 2011.

5.4.5 Rural Electrification Programme

Under the rural electrification programme in J&K a total number of villages/ hamlets electrified rose to 14,229 by 31-03-2011. Under RGGVY schemes target was to electrify 283 un-electrified/ de-electrified villages, out of which 169 villages were successfully electrified by Oct-2012.

- RGGVY is a scheme for 100% Rural

Electrification.

- 14 Schemes covering whole of J&K were submitted to REC at a total estimated DPR(Detailed Project Report) cost of Rs. 1,051.73 Crore. All these schemes have been sanctioned by REC at a cost of Rs. 917.02 crore and Rs 741.36 Crores was released. An amount of Rs 568.57 Crore has been spent by end of March 2012.
- RGGVY Schemes aims at envisage electrifying 365 no. un-electrified villages and de-electrified villages, which are financially viable for electrification within the Grid System.
- 3,03,682 no. Rural Household connections are proposed under the RGGVY schemes.
- 1,10,337 no. BPL households are earmarked for free electrification and 5889 no. already electrified villages to be covered for intensive electrification under the scheme.
- As per guidelines of RGGVY, NHPC and J&KSPDC have been entrusted with the execution in each of the 7 districts, as under;
 - NHPC: Udhampur, Kathua, Jammu, Srinagar, Budgam, Leh and Kargil.
 - J&KSPDC: Kupwara, Anantnag, Doda, Baramullah, Pulwama, Rajouri and Poonch.

5.4.6 State Hydro Policy

State Hydro policy 2011, notified by the Govt. Of Jammu & Kashmir for any new hydro

projects of generation capacity from 2 to 100 MW is to be implemented by Jammu & Kashmir State Power Development Corporation Ltd. Projects upto 2 MW have to be implemented by Science and Technology Department. The main features are;

- Projects upto 10 MW are reserved for state subjects with relaxed technical criteria.
- 15% free power + 1% LADF has to be provided to State.
- Projects are to be allotted on BOOT basis to be returned to State after 35 years.
- Upfront Premiums ranging from 4.0 lacs to 8.0 lacs / MW as per the capacity of the projects.
- Tariff based bidding for project above 25 MWs.
- Terminal value for return of project should not be more than 10% of the cost of the project at the time of bidding.
- 30% power (excluding 15% free power + 1% LADF) has to be procured by the State with first right refusal for the remaining Allotment of project through transparent competitive bidding.
- Jammu & Kashmir State Power Development Corporation Ltd. should be the Nodal Agency and responsible for Land acquisition, IWT clearance, Forest clearance etc.
- Incentive should be procured by way of non imposition of water usage charges for 1st ten years, exemption from Entry Tax / Income Tax available to IPPs, besides incentive available from MNRE, GoI.
- Model Implementation Agreement and PPA are integral parts of the reserved Hydel Policy 2011. Timelines / mile stone have been fixed as part of the agreement.

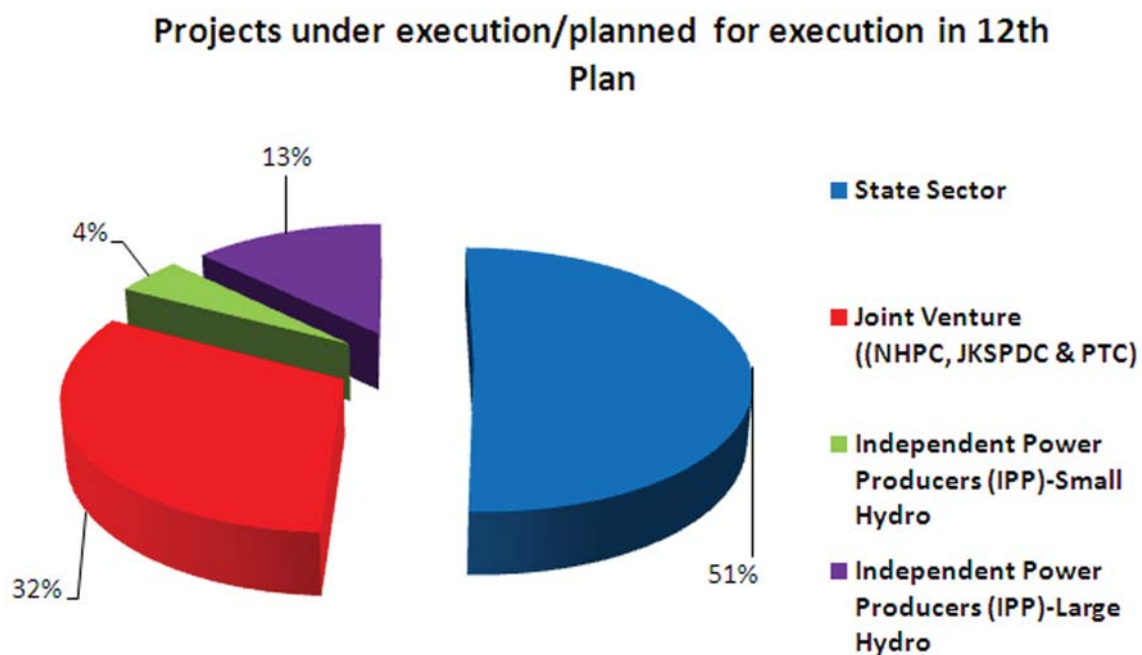


Figure 13: Projects under execution/planned for execution in 12th Plan

5.4.7 Jammu and Kashmir Energy Conservation Act, 2011

The Jammu & Kashmir Energy Conservation Act, 2011 as passed by the Jammu and Kashmir State Legislature was approved by the Governor on 23rd April, 2011. Salient features of the state EC act, 2011 are given below:

- Establishment and incorporation of Bureau of Energy Efficiency.
- The head office of the Bureau shall be at Jammu/ Srinagar.
- Management of Bureau: The general superintendence, direction and management of the affairs of the Bureau shall be under the Governing Council's purview. The Governor council would consist of ten to fifteen members appointed by the Government.
- The Bureau shall effectively co-ordinate with designated consumers, designated agencies and other agencies, recognize and utilize the existing resources and infrastructure, in performing the functions assigned to it under the Act.
- Bureau shall recommend amendments or modifications in the norms for processes and energy consumption standards to the Government to suit the regional and local climatic conditions of the State Bureau shall recommend on the energy conservation building codes to the Government with respect to use of energy in the buildings.
- Prescribe guidelines for energy

conservation building codes specified by the Government.

- The bureau shall support the Government for issuing energy savings certificates.
- Strengthening of consultancy services in the field of energy conservation.
- Promote research and development in the field of energy conservation.
- Develop testing and certification procedure and promote testing facilities for certification and energy consumption of equipment and appliances.
- Formulate and facilitate implementation of pilot projects and demonstration projects for promotion of efficient use of energy and its conservation.
- Promote use of energy efficient processes, equipment, devices and systems.
- Promote innovative financing of energy efficient projects.
- Support financial assistance to institutions for promoting efficient use of energy and its conservation.

5.5 Key Priorities

The following key priorities for the sector were identified after detailed deliberation in the working groups. The priorities are in line with the concerns raised due to the negative effect of the climate change and the state's response.

1. Establishment and incorporation of Bureau of Energy Efficiency, J & K EC act 2011

As per The Jammu and Kashmir Energy Conservation Act, 2011, a Bureau has to be formed. The Bureau shall be a body recognized by the name Bureau of Energy Efficiency. The bureau would have perpetual succession and a common seal, with powers subjected to the provisions of the Act. Under the Bureau, a statutory body/nodal agency need to be formed by the state to implement energy conservation measures at state level. The main objective of the scheme should be to build capacity of the nodal agency to enable them to discharge regulatory, facilitative and enforcement functions under the J & K EC Act 2011. The scheme would develop and implement Energy Conservation Action Plan (ECAP) based on a uniform template prepared to take necessary measures to build institutional and human capacity, enable the nodal agency to implement energy efficiency programmes and undertake evaluation and monitoring of the Energy conservation activities implemented in the state.

Newly formed statutory body/nodal agency have limited experience in energy efficiency / conservation activities. This has led to the need for capacity, enhancing the understanding and knowledge about energy efficiency, having a common action plan to implement measures to reduce energy intensity of the State etc. The proposed scheme of the Bureau of Energy Efficiency, J & K therefore enables statutory body/nodal agency to:

- a) Prepare a five year Energy Conservation Action Plan (ECAP)
- b) Implementation of ECAP with yearly deliverables
- c) Enhance capacity to undertake regulatory duties that they are required to perform under the Act.

The Bureau should provide financial support to the statutory body/nodal agency for strengthening their institutional capacity and carry out the following activities:

- Creation of IT infrastructure
- Creation of database for Energy Managers / Energy Auditors and Designated Consumers and other stakeholders.
- Organizing workshops / training programmes.
- Creating awareness through electronic media / print media.
- Preparation of Detailed Project Reports of Govt. Buildings under IGEA.
- Implementation of Energy Efficiency Demonstration Projects in the area of energy efficient street lighting, revamping of drinking water pumping system and energy efficiency in SMEs clusters.
- To replace the existing incandescent bulbs (ILBs) in the households and the street lights of one village with LEDs.

2. Promoting Energy Efficiency in Small and Medium Enterprises (SMEs)

Energy use & technology gap study and cluster specific manual on energy conservation opportunities need to be prepared for better understanding of EE in SME sector in J & K. SIDBI/ISTSL are already involved in preparation of DPRs on energy efficient technologies in SME clusters in different states. BEE also initiated the small group activities (SGA) and Total Energy Management (TEM) programme to introduce the energy efficiency practices in SMEs with support of experts from Japan.

3. Promoting Energy Conservation Building Code (ECBC)

The Energy Conservation Building Code (ECBC) was launched by the Government of India on 27th May, 2007. The ECBC sets minimum energy standards for new commercial buildings having a connected load of 100kW or contract demand of 120kVA in terms of Energy Conservation (Amendment) Act, 2010.

Labeling programme for 3 categories of buildings (day use office buildings/BPOs/ Shopping malls) have been developed and put in public domain. Till now 136 buildings have been found eligible for issue of label. The ESCO model has been promoted for carrying out energy efficiency measures

in existing buildings through performance contracting.

4. Renovation and Modernization of Hydro-power Projects

Hydro power stations are generally designed with a projected operational life of about 35 years, though equipment start wearing out from commencement of regular operation. As a result, the performance of the plants gradually declines over time. This necessitates Renovation & Modernization work of older power stations. Before the plants complete their productive lifecycle, they should be renovated and modernized to revamp their performance. The RM& U involves sizeable capital investment.

The main objectives of undertaking RM&U are as follows:

- (i) To protect against deterioration of performance.
- (ii) To improve availability, reliability, efficiency and safety of the equipment.
- (iii) To regain lost capacity.
- (iv) To extend the productive life.
- (v) To save investments on new equipment

Following projects are under RMU in the State:

Table 24: List of Hydro Power project (with capacity) under renovation and modernization

S. No.	Name of the Project	Capacity in MW
1.	LJHP	105
2.	Chenani-I	23.30
3.	USHP-I	22.60
4.	Karnah	12
5.	Mohra	9
6.	Satkna	4
7.	Iqbal	3.75
8.	Haftal	1
9.	Hunder	0.40
10.	Bazgo	0.30
11.	Sumoor	0.10
Total		181.45 MW

5. Development of comprehensive scheme for promotion of energy efficient pumps in agriculture sector

Jammu & Kashmir's Agricultural sector accounts for around 3.5% of the power requirement of the State. Promotion of energy efficient agricultural pump sets would reduce the overall power consumption, improving performance for ground water extraction and reducing the subsidy burden on the states without sacrificing the service obligation. It can also help in creating access to subsidies for beneficiary farmers for electricity conservation (estimated 30-40%) with replacement of inefficient pump sets with energy efficient ones. Implementation of the action plan is planned through following sub activities:

- (i) Preparation of scheme to provide with financial incentive to farmers on use of energy efficient agricultural pumps.

- (ii) Replacement of inefficient agricultural pumps with energy one.

6. Reducing Transmission and Distribution (T & D) Losses

Reduction of T&D losses is a prime focus in the energy sector as that would directly result in greenhouse gas emission reduction and ensure energy security. Implementation of the action plan is planned through following sub activities:

- (i) To assess the current T&D loss pattern and plan measures to reduce losses through increasing of energy efficiency and reducing pilferage.
- (ii) Renovation of existing transmission and distribution network.
- (iii) Replacement of Existing Transformer with star rated transformer.
- (iv) 100% consumer metering to reduce AT&C Losses - Involving Installation of SCADA system and metering

arrangement for on-line remote monitoring system right from grid sub-station till the consumer end having a connected load of 20 kW and above.

- (v) Implementation of feeder level metering at distribution level.
- (vi) Consumer level metering through a cell to formulate and carry out vigilance activities.
- (vii) To introduce franchise model for distribution to reduce commercial losses and better management of the distribution system;
 - a. Formulation of Project management unit to facilitate for franchisee for 3 years
 - b. Impart training
 - c. Training and providing support to prospective entrepreneurs.

7. Promoting Demand Side Management (DSM) and energy efficiency

DSM and energy efficiency would reduce the demand for energy and therefore reduce carbon emissions. Under this initiative, a comprehensive policy and plan to save energy in order to reduce the demand–supply gap and contribute towards abatement of climate change effects would be undertaken. This would include the following activities:

- (i) Implementation of utility level DSM measures,
- (ii) Awareness generation for Energy Conservation,
- (iii) Promotion and implementation of the National Bureau of

Energy Efficiency's ECBC norms in the state to limit the energy consumption in buildings and

- (iv) For proper energy monitoring, capacity building of energy auditors, strengthening the existing Energy Conservation Cell under the Energy Department supporting with manpower and infrastructure.

8. Decentralized power generation

The power sector of the state suffers from lack of proper transmission network. Transmission, Sub-Transmission and Distribution lines are very old and owing to the brittle conductors, energy loss during transmission is high. Promotion of evacuation of power from decentralized renewable energy power projects including micro hydro, run off the river, Solar (PV & Thermal both), IPP, CPP and high voltage transmission infrastructure for power evacuation and import of power from regional grid is planned through evacuation corridors. The fulfillment of the overall objective is planned through following sub activities:

Encouragement for Decentralized generation of power to reduce the T&D Losses by introducing schemes and policies

- a. Carrying capacity study and resource mapping of each of the districts for setting up possible power project.
- b. Preparation of techno economic feasibility report for the requirement of transmission network.
- c. Applying budgetary provision through international funding and private sector investment.

9. Harnessing the biomass potential

Promoting on-grid and off-grid biomass power projects would reduce the carbon emission through reduced use of grid power. Under this initiative, the following would be done to harness the biomass potential to its maximum: (i) Study the existing policy and develop investment friendly policy to promote additional biomass application (ii) Conducting a detailed feasibility study for scoping biomass-based project (iii) Developing a biomass supply chain involving agro, agro industrial and other biomass resources including dedicated energy plantation (iv) Promoting biomass based gasifier project in agro based industries (v) Raising awareness (vi) Creating a conducive scenario for investment and (vii) Implement demonstration/pilot projects if required.

10. Development of policy for mandatory use of efficient light particularly for commercial organizations in the state

Most of the energy is consumed for lighting purposes. The demand is on a rise with construction boom especially in state like Jammu & Kashmir. This means regulations in energy usage for lighting purposes can make a major contribution in containing the issue of climate change and energy efficiency. There are three key elements to achieve progress:

- (i) Use less energy
- (ii) Replacement of all inefficient lighting system with energy efficient systems.

- (iii) Use of renewable energy (e.g. Solar PV, lantern, etc).

State Govt. should employ a policy framework on using energy efficient lights in domestic & commercial buildings, Govt. & private offices, education institutes, hospitals, colonies etc.

11. Expansion Conversion of conventional street lights to solar LED/CFL street lights

Energy from the light of the sun (photovoltaic) can be used to produce electricity. Though the equipment is expensive, maintenance requirements minimal and there is no fuel cost. They are far less expensive than building new power plants and grid distribution systems for rural areas not covered by grid electricity. Energy from the heat of the sun (solar thermal) can be used economically for water and room heating purposes. Conventional street lights can be replaced by solar LED/CFL based street lighting system.

- (i) Preparation of pre-feasibility study.
- (ii) Fund identification.
- (iii) Implementation strategy

12. Maximizing solar power usage and Encouraging use of Solar Gadgets (especially in Industries):

The use of fossil fuel based power and therefore carbon emissions can be reduced with use of solar gadgets. Hence adequate incentives should be provided for the same. Under this initiative, the state would promote both solar photovoltaic as well as solar thermal. An increase in market penetration

of stand-alone solar systems for use by institutions, communities and individuals is being proposed. A range of activities –promotional policy initiatives, survey, feasibility reports, demonstration projects, awareness creation and capacity building, and strengthening the manufacturing base would be undertaken. All of these would lead to increased use of solar gadgets in the state. Industrial energy measures should strengthen the business case for investing in higher efficiency equipment and solar gadgets. Compliance of the action plan is planned through following sub activities:

- (i) State level policy framework for use of solar gadgets.
- (ii) Using solar PV for outdoor lighting.
- (iii) Using solar water heating system.
- (iv) Carrying out awareness campaign for more solar system installation and use of star rated devices.

13. State energy audit policy and Implementation of pilot Energy efficiency project and IGEA

The objective of the pilot initiative is to demonstrate the scope of energy efficiency through implementation of energy efficiency measures. Such action would enhance the capacity of the state nodal agency to undertake similar initiative in the future and also promote such activities amongst the sector. Compliance of the action plan is planned through following sub activities:

- (i) Conducting energy audit by the empanelled organization.

- (ii) Undertaking IGEA in Govt. Buildings, Hospital, Universities, Banks, Hotels etc.
- (iii) Implementation of Pilot energy Efficiency one of the identified pumping Station.
- (iv) Implementation of Pilot Energy Efficiency Street lighting Project in one of the ULB area.
- (v) Identifying and empanelling ESCO companies.
- (vi) Taking up joint pilot project with ESCO companies.

14. Promoting Energy Efficiency Practices in the State

The objective is to create awareness among the sectors (domestic, commercial and industrial) about the benefit and necessity of taking up energy efficiency measures and also the pathway of achieving the same including promotion of use of star rated gadget. Complying of the action plan is planned through following sub activities:

- (i) Identification of Agencies for the activities.
- (ii) Undertaking sectoral and scoping study of the possibility of energy efficiency in each sector and barriers in taking up the energy efficiency initiatives.
- (iii) Undertaking Training Need Analysis Study for the department, preparation of manual and carrying out pilot workshop.

- (iv) Carrying out awareness campaign towards adopting energy efficiency measures and use of star rated devices.

15. Awareness and implementation for use of CFL and replacing incandescent lamp under Bachat Lamp Yojna, and Umbrella program of BEE

Bachat Lamp Yojana (BLY) promotes energy efficient and high quality CFLs as replacements for incandescent bulbs in households at the price of an incandescent bulb, i.e. Rs 15. This would remove barrier of high CFL price (which is currently Rs 80 - Rs 100 per CFL) which is negating its penetration into households. It targets replacement of about 400 million incandescent bulbs in use in the country, leading to a possible reduction of 4,000 MW of electricity demand, and a reduction of about 24 million tonnes of CO₂ emissions every year. The price differential would be borne by project implementer through carbon credits earned which could be traded in the international market under Clean Development Mechanism (CDM) under the UNFCCC Kyoto protocol. BEE has developed the umbrella framework BLY-Programme of Activities (PoA) which has been registered under UNFCCC-EB on 29th April, 2010. The PoA would define key CDM requirements, including the project baseline, additionality, methodology, monitoring protocols through which CO₂ emission reductions would be assessed. The PoA approach reduces time and transaction costs for registering the projects since the key CDM requirements does not need to be addressed by area-

specific projects within the PoA.

16. Standards And Labeling Programme

Standards and labelling (S&L) programme has been identified as one of the key activities for energy efficiency improvement. A key objective of this scheme is to provide the consumer an informed choice about energy saving and thereby the cost saving potential of the relevant marketed product. The scheme was launched by the Hon'ble Minister of Power on 18th May 2006 and is currently invoked for 12 equipments/appliances, i.e. ACs, Tube lights, Frost Free Refrigerators, Distribution Transformers, Induction Motors, Direct Cool Refrigerator, Geysers, Ceiling fans, Colour TVs, Agricultural pump sets, LPG stoves and washing machines, of which the first 4 have been notified under mandatory labelling from 7th January, 2010. The other appliances are presently under voluntary labelling phase. The energy efficiency labelling programs under BEE are intended to reduce the energy consumption of appliance without compromising with the services it provides to consumers. The scheme aims at providing information on energy performance of appliances based on the standards issued by BEE so that the consumers can make judicious decisions while purchasing appliances. The STAR rating ranges from 1 to 5 in the increasing order of energy efficiency. In addition to the objective of informed choices to consumers, this program leads to huge energy savings and thereby cost savings. It also reduces capital investment in energy supply infrastructure, enhances the product quality, strengthens

the competitive markets, builds position for domestic industries to compete in such markets where norms for energy efficiency

are mandatory, removes indirect barriers to trade, reduces carbon emission and helps meet climate change goals.



5.6 List of Key Priority Action

Table 25: List of Key Priority Action Mission on Enhanced Energy Efficiency

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
1	Policy development for mandatory Energy Audit in Govt. Building: 1. Identification of Agencies for taking up activity. 2. Scoping study of the possibility of energy efficiency in Govt. buildings (e.g. Civil Secretariat, Hospitals, Irrigation pumping system, Universities, etc). 3. Barrier identification of taking up the energy efficiency initiatives. 4. Undertaking Training Need Analysis Study for the department, preparation of manual and carrying out pilot workshop.	PDD, PWD, Dept of Industry, Department of Irrigation, Health Department, Universities.	Nil	2.5	2.5	GOI
2	Development of comprehensive scheme for promotion of energy efficient pumps in agriculture sector. Replacement of Inefficient agricultural pumps by efficient pumps	PDD, Irrigation Dept., Agriculture Dept.	Nil	0.4	0.4	GOI, EFA
3	Conversion of conventional street lights to solar LED/CFL street lights. 1. Preparation of pre-feasibility study 2. Fund identification. 3. Implementation strategy	UDD, Municipality, PDD	1	100	101	Govt. of J &K, GOI
4	Encourage use of Solar Gadgets in Industries 1. Industrial Energy Efficiency Auditing and Benchmarking 2. State level policy framework for solar gadgets use 3. Using solar PV for outdoor lighting. 4. Using solar water heating system. 5. Carrying out awareness campaign towards taking up solar system installation and use of star rated devices.	Dept. of Industry, PDD, Science & Technology Department	Nil	2.5	2.5	GOI, EFA

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
5	T & D Loss reduction - - o To assess the current T&D loss pattern, plan & implement measures to reduce losses through increase of energy efficiency in DTs and reducing pilferages through pilfer-proof Boxes, Service Cables and ABC cables for LT lines. - o Renovation of Existing transmission and distribution network. - o Replacement of Existing Transformer with star rated transformer. - o 100% consumer metering to reduce AT&C Losses - Involving Installation of SCADA system and metering arrangement for on-line remote monitoring system right from grid sub-station up to the consumer end having a connected load of 20 kW and above (Bulk & other such categories). - o Implementation of feeder level metering at distribution level - o Consumer level metering a cell should be formulated and carry out Vigilance activity. - o To introduce franchise model in distribution to reduce commercial losses and better management of the distribution system. - o Formulation of Project management unit to facilitate for franchisee for 3 years - o Impart training • c. Training and imparting support to prospective entrepreneurs.	JKSERC & JKPDD	1	1000	1001	Govt. of J & K, GOI

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
6	Encouragement for Decentralized generation of power to reduce the T&D Losses by introducing schemes and policy - Pugah Geothermal Project (will replace Diesel generation in Ladakh) - Achoora small hydel Gurez (will replace Diesel generation in Gurez) - Preparation of DPRs/PFRs of Micro Mini & Small Hydel Projects. (100 Projects @ Rs 10 lakhs per project) - Incentive to Developers for Small Hydel Projects @ 5 % of estimated cost.	JKSERC, PDD, Science & Technology Department, IPPs,	Nil	3600	3600	GoI, APDRP, RGGVY Funding from R&M
7	Development of policy for mandatory use of efficient light particularly for commercial organizations in the state. 1. State Govt. policy framework on using energy efficient lights in domestic, commercial building, offices, education institute, hospitals, etc.	JKPDD, PWD, Dept. of Industry	Nil	2	2	GOI,
8	Awareness and implementation for use of CFL and replacing incandescent lamp under Bachat Lamp Yojana, and Umbrella program of BEE.	JKPDD, UDD, Municipality, PWD.	Nil	200	200	GOI
9	Promoting efficient brick manufacturing process leading to fossil fuel conservation, preventing denudation of top soil and GHG emission reduction by mandatory use of efficient brick heating/burning.	Dept. of Industries	Nil	2.5	2.5	GOI, EFA
10	Promotion of use of star rated domestic appliances may be encouraged. The shops selling TV, Refrigerators or Washing Machines, Fans etc. may be asked to keep only star rated products. 1. Incentivizing local equipment supplier (10% on MRP) for first 3 months from the launching of the scheme.	PDD, Dept of Industries	Nil	1	1	GOI, EFA

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
11	Development of policy to mandate ECBC adoption in state.	PDD, S&T Department, PWD, Industry, UDD	Nil	3	3	GOI
12	Formulation of DSM project for JKSERC 1. Implementation of utility level DSM measures 2. Awareness generation for energy conservation. 3. For proper monitoring, capacity building of energy auditors, strengthening of energy conservation cell under the Energy department supported manpower and infrastructure.	PDD, JKSERC, Dept. Of Science & Technology.	Nil	2.5	2.5	GOI/ EFA
13	Replacement of electric boiler with solar & grid connected hybrid boiler in industrial units	PDD, UDD, Department of Industries.	Nil	2	2	GOI
14	Energy Audit of Generation Stations & Renovation and modernization of the existing hydro power station	JKSPDD, JKSERC	1570	280	1850	Govt. of J & K, GOI
15	Transport 1. Promotion of low emission vehicles, electric vehicles 2. Introduction to concept of mass rapid transit	Department of Transport, Department of Urban Development Department	Nil	300	300	GOI
16	Maximize harnessing biomass potential in the state through cogeneration/ thermal/ power plant/gasification to feed in the grid as green power. Increase in application of CPP both in grid and stand alone mode	PDD, JKREDA, Dept. of Industries	Nil	50	50	CFA- MNRE/ IREDA / CDM revenue/ EFA
Total			1572	5548.4	7120.4	







6 Mission on Sustainable Habitat

6.1 Introduction

Urban sector faces significant impacts of climate change. These impacts potentially have serious consequences on human health, livelihood and socio-economic assets, especially for the urban poor, informal settlements and other vulnerable groups. Climate change impacts range from increase in extreme weather events, flooding to high temperatures leading to public health concerns. As the urban regions of the state are dynamic systems that face unique climate impacts, their adaptation must be location specific and tailored to local circumstances.

The importance of infrastructure and improvement of living standards of population is crucial to sustained economic development. A detailed analysis of the development process over last four decades shows that one of the major reasons for slow economic and social development is the unplanned population growth. The capital cities and other district towns of the state are challenged with problems of rapid urbanization, expansion of

informal settlements, substantial poverty, inadequate infrastructure and environmental degradation. These and other concerns plague cities' ability to grow and prosper. Many of these conditions also limit resilience to current climate variability.

The state government is thus planning to mainstream climate change adaptation in urban infrastructure development. Creating this new possibility of adaptive resilience would support a new future in the face of climate change. The concerned departments prepared the action plan prioritizing actions in line with National Climate Change Action Plan for the sustainable habitat sector.

6.2 Key trends in the sector

The state is situated at high altitude and has a mountainous terrain with unfavorable climatic conditions. Most of the urban population of the state is usually concentrated in towns and cities. More than 60% of the urban population of the state is concentrated in class I towns, also the net addition to the population of these towns has been the highest (SDR, J&K 2003). The population size,

growth, composition and quality play an important role in the development process.

Table 26: Urban Population¹⁵

Description	2001	2011
Total Population	10143700	12548926
Male	5360926	6665561
Female	4782774	5883365
Rural	7627062	9134820
Urban	2516638	3414106
Urban population as percentage of total population	24.81	27.21



Figure 14: District Map of J&K

The main developmental issues in the context of urban areas of the state are being taken care of by two main Municipal Corporations -Jammu Municipal Corporation and Srinagar Municipal Corporation. There are about 83

Municipality/ Municipal councils all over the state. The urban population is spread across all the districts of the state, with Srinagar, Jammu, Leh, Baramulla and Udhampur having high population concentration. All

these districts have major cities and divisional administrative headquarters. The capital cities of Srinagar and Jammu have always been the traditional growth centres. Most of the small and medium towns are extending due to uncontrolled population growth.

The existing urban infrastructure along with the core urban public services such as water supply, sanitation and sewerage, urban roads and solid waste management do not suffice for the growing urban population. The Sanitation facilities in urban areas lack sewerage systems in urban areas resulting in pollution of the nearby water bodies due to release of untreated effluents. Poverty and lack of livelihood amenities in rural areas has led to migration of population to urban areas. This has posed threat to the availability of proper human habitat. Lack of sewerage system, indiscriminate developmental activities have led to contamination of the water bodies, drain snagging and obstruction in the rainwater outlets.

Overcrowding and expansion of slums, increasing scarcity of water supply, inadequacies of public health facilities and sanitation system, and mismanagement of waste materials are the major problems dogging the urban areas. Moreover, rampant growth of private housing colonies, which in most cases are not well planned, leads to issues for urban local bodies to ensure basic amenities to these colonies. Most of these colonies have turned into slums adding to the problems in the Srinagar city.

6.2.1 Solid waste

Solid waste management is emerging as an essential urban and industrial issue. The Municipal Solid Waste (MSW) generated in Srinagar city in 2010 was around 394 MT/day of which around 220 MT/day was disposed as landfill at SyedporaAchan(SOE report 2012). Majority of slaughter houses in Srinagar city have the slaughtering areas within their premises. The waste is released into the surface and underground drainage system. In Jammu, the MSW generation is estimated to 350 – 400 MT/day, out of which 300 MT/day solid wastes are collected and sent to Dumping Site at Bhagwati Nagar.

6.2.2 Transport

The transport sector contributes majorly to the Green House Gas (GHG) emission in the state. It is estimated that the number of vehicles in the State would be multiplied two times every five years adding significantly to the quantum of emission of GHG. Roads are the primary mode of transport available in Jammu and Kashmir. Therefore, along with the road length, the number of vehicles has risen significantly. The State Government regulates road transport facilities under the provisions of Motor Vehicles Act, 1988. The number of registered vehicles has reached to 6.68 lakhs as on 31st March 2009 which accounts for 47,000 extra vehicles than the last year. This figure corresponds to an annual percentage growth of figure 7.63%. Details of public and private transport are given in Table blow:

Table 27: Types of vehicles registered in the State¹⁶

Sl. No.	Type of vehicle	Total Registration
1	Buses	10360
2	Mini buses	15411
3	Trucks/ tankers/ MGV/LGV	48190
4	Cars	192793
5	Jeeps	11791
6	Taxies	25308
7	Motor cycle/ Scooter	430844
8	Auto rickshaws	39753
9	Tractors	20296
10	Trailers	667
11	Other (Ambulance, Record rollers dozers etc)	15081
	Total	810494

6.3 Vulnerability of the sector

Cities and towns of Jammu & Kashmir have always been vulnerable to natural hazards such as earthquakes, cloudbursts, snow storms, floods, droughts, avalanches and landslides. In some situations, cities experienced increase in the frequency of existing climate-related hazards, such as flooding. Due to the peculiar topography, rugged terrain, extreme weather conditions and underdeveloped economy, the state has always been prone to natural disasters and it's after effects. Jammu and Kashmir has witnessed many cases of natural disaster in the 19th and early 20th centuries.

Such events not only disrupt economic activity but also lead to immense hardship for the affected population. The signs of changing climate has already manifested

through extreme weather conditions which have been further aggravated by many ill-practices resulting in deforestation and environmental degradation.

Within the cities of Jammu & Kashmir, the poor are typically the most vulnerable. They tend to live in slum/informal settlements, often located in areas most exposed to the effects of global climate change, steep slopes and ravines. Little or no infrastructure exists to provide protection from storm events or to ensure mobility. Reflecting the low and unstable incomes of the residents, but exacerbated by poor land tenure characteristics, little or no building regulation, and lack of housing finance, low housing quality results poor resistance to natural disasters. The inherent vulnerability of these settlements within the urban areas of the state is amplified as the effects of climate change become more pronounced.

¹⁶J&K Transport Department, 2009-10

The signs of climate change at present are mainly visible through rise in temperature or increase or decrease in rainfall. Also frequent rainfall makes urban living highly vulnerable to floods and landslides. Indian Meteorological Department (IMD) monitoring reveals that temperature is increasing in both the regions of Jammu & Kashmir with significant increase in Maximum temperature by 0.05° C per year in Kashmir valley and minimum temperature in Jammu region by 0.08° C per year (*Jaiswal&Rao, IMD Pune, 2007*).

6.3.1 Flashfloods and Landslides

Cloudbursts and flash floods are common features in different parts of the state which are usually responsible for loss of life and property. Flash floods are mainly caused due to glacial melting as a result of glacial melting. Suru basin and Kolhai are the prominent examples. They have lost about 16% and 18% of the total volume respectively. It is observed that flash floods cause tremendous loss in catchment of Jhelum, Chenab and Tawi rivers. Low-lying areas of the Kashmir Valley, especially Sonawari, Awantipora, Srinagar, along with parts of Jammu are prone to flood. Upper catchments of all tributaries of Jhelum, Indus, Chenab and Tawi rivers are also prone to flash floods.

Landslides are common phenomenon across the state. Large and small landslide occurs every year in all three region of the State. These are complex disaster phenomenon caused by heavy rainfall, snowfall and earthquakes. Also, downhill creep causes landslips and landslides. The area between Batote and Banihal on National Highway is

prone to landslides.

6.4 Key issues in the sector

The issues related to the people's habitat are considered the vital indicators for assessment of the living condition in an identified geographical area.

Solid Waste generated from urban household is normally collected and dumped in an unscientific manner to nearby dumping grounds. Absence of proper segregation system and scientific treatment and disposal procedure leads to unhygienic condition near the urban areas across the state. Different wastes and waste management activities have varied impacts on energy consumption, methane emissions and carbon storage. Recycling reduces Green House Gas (GHG) emissions by reducing methane emissions from landfills. If energy is generated from the waste the consumption of energy from fossil sources would reduce. Inadequate collection and improper disposal system currently lead to spillage and contamination of soil and surface as well as groundwater streams. Integrated Solid waste Management facility is still not implemented within the state. Moreover, the solid and liquid waste generated from the slaughter house has a detrimental effect on the nearby water bodies and also the aesthetic environment.

Sanitation possesses major problems in the absence of any sewerage system in urban areas resulting in release of domestic effluent into nearby rivers and streams leading to contamination of water sources. Indiscriminate developmental activities also

add to the problem by jamming drains and encroaching on rainwater flow paths.

The urban transport sector has been largely neglected in the State, characterized by heavy traffic congestion due to narrow roads, rapid growth in number of vehicles along with highly topographic and concentric development. Often there are days when some areas remain inaccessible due to

landslides, snowfalls or other damages caused by heavy rains. Public transport is limited due to inadequate road network, poor infrastructure and scattered demand.

In order to combat these odds through a sustainable strategy for climate resilience, the state has envisaged some key priorities in the urban sector.

Table 28: Adaptation Pathways with respect to climate change in State

Issues	Impact	Pathways
Warm and Humid summer and cold winters	Increased demand for cooling	Create awareness to retrofit building with green design; policy incentive for usage star rated HVAC products
Energy Usage	Higher concentration and higher use	Utility DSM measures in street lighting, solar water heating
Increase in natural hazard like earthquake, flashflood, Cloudbursts, landslides etc.	Increase loss of human settlement and loss of lives	Policy formulation on hazard resistant settlements within populated urban areas and awareness for hazard preparedness to minimize loss of property and lives.
Heavy and aberrant precipitation	Increased storm-water runoff	Development of storm water management plan and investment in sewerage; re-assessment of master plans/land use plans of urban agglomerations, policy incentive use of permeable surfaces and incorporation in the PWD codes
Enhanced waste generation due to urban agglomeration by population influx	Health hazards, soil contamination through leaching, odour pollution	Awareness for waste segregation and policies for land filling of waste
Decline in the forest cover	Decrease in bio-sequestration of atmospheric carbon dioxide, incur significant adverse soil erosion and frequently degrade into wasteland.	Planting heat tolerant trees, city wide programmes for tree watering and maintenance, roadside plantation programme, development of parks
Transport system congestion and ageing	Congestion and higher GHG emission	Phase out of old vehicles, integrated traffic study, congestion reduction plan and implementation of intelligent transport system

6.5 Programme and Policies in the sector

The following key policies are in place that would be able to address the impacts of climate change and also reduce the sectoral contribution to climate change. The Housing and Urban Department is responsible for implementation of these schemes.

1. Jawaharlal Nehru National Urban Renewal Mission (JNNURM)

The State government has adopted several urban development schemes such as the Jawaharlal Nehru National Urban Renewal Mission (JNNURM). JNNURM was launched by Government of India on 3rd December, 2005 and it started its implementation in the State in 2007. It is a reform driven fast track programme to ensure planned development of identified cities. It focuses on efficiency in urban infrastructure/Service delivery mechanism, community participation and on enhancing accountability of the Urban Local Bodies. The main thrust areas of JNNURM are Water Supply Including Sanitation, Sewerage, Solid Waste Management, Road Network, Urban Transport, Integrated Development Of Slums, Re-Development Of Inner (Old) City.

2. Capital City Development Programme (CCDP)

The scheme is being implemented in two capital cities of Srinagar and Jammu to provide basic civic amenities primarily in Construction of lanes/ drains/ culverts, construction of bath rooms/ lavatory blocks, development of Parks/ play grounds/

graveyards, development of roads etc.

3. Integrated Development of Medium Towns

The scheme is implemented by the Urban Local Bodies in all towns other than Srinagar and Jammu Cities for providing basic civic amenities. The scheme is implemented through the Municipal Councils/Committees and the funds are routed through the Directors of Urban Local Bodies Jammu and Kashmir. The other scheme with same objectives known as Urban Development Scheme (UDS) is a District Sector scheme and administrated by the District Development Commissioners.

4. Urban Transport

The funds under this scheme were provided to Srinagar Development and Jammu Development Authorities for construction and up-gradation of Bus Terminals and other Mini Bus stands to resolve the transport related problems of Srinagar and Jammu Cities till 2009-10. During the previous financial year the scheme was extended for other towns of the State and the funds were released to two Directors of Urban Local Bodies for Urban Transport.

5. Drainage Sector

The Department contemplates to provide proper drainage network in the State. Accordingly various schemes/works under this sector have been taken up to be executed through Srinagar/Jammu Municipal Corporations and Directorate of Urban Local Bodies, Kashmir/ Jammu. An allocation of

2000.00 lacs INR were provided during the year 2011-12 and have been proposed to retain the same for the current financial year under this sector.

6. Integrated Low Cost Sanitation Scheme

This is a centrally sponsored scheme aiming to convert /construct low cost sanitation units through sanitary two pit pour flush latrines and new latrines for EWS households without latrines. . About 56 projects for conversion/ construction of 5901 units have been sanctioned by Government of India (GoI).

7. Dal Development

The Dal Development Programme has two components - Conservation/ Management Plan of the Lake and Rehabilitation / Re-Settlement of Dal Dwellers & Lateral Sewer lines/ house connectivity. The Conservation Management Plan of the Lake is funded

100% by GOI and the Rehabilitation has to be borne by the State Government. The rehabilitation plan has been worked out and land at Rakh-i-Arth, measuring about 3000 canals is being developed in the first phase. There are other components for establishing the Housing colony at Rakh-I-Arth to be funded through State Plan besides funds for construction of secondary sewers and laterals of the Sewerage System.

8. Sewerage Sector

Three sewerage schemes viz Sewerage Project TalabTiloo at Jammu, Sewerage Project in BrariNumbalat Srinagar and Sewerage Project in Khushalsarat Srinagar are under execution under this sector. The pilot Project TalabTillo has been commissioned. The amount shall be utilized to provide house connections through chambers in major portion of the area covered under the project. The STP of 10 MLD capacity is ready to be commissioned provided the house connections are made.



6.6 Key Priorities

Jammu & Kashmir is characterized by an increasing altitude as one moves towards north, thereby decreasing temperature. Thus, Jammu is 366 m above the mean sea level with annual mean temperature of 24.5°C and sub-tropical climate. Kashmir is at an altitude of 1585 m above mean sea level with a mean annual temperature of 13.3°C and comes under temperate climate, while Ladakh is at a height of 3,505 m above the mean sea level and experience sub-arctic climate. The state has a hilly topography and one of the most vulnerable states to the effects of climate change such as warmer temperatures, unusual rainfall, landslides. At the same time, better urban planning and policies can reduce energy use and Green House Gas (GHG) emissions and improve the resilience of urban infrastructure to climate change, thereby shaping future trends. Lack of suitable livelihood and employment facilities added to uncontrolled exploitation of natural resources. Moreover, the state is situated adjacent to three other countries which are equally vulnerable to natural disasters. This necessitates special attention for initiatives on mitigation and adaptation under sustainable habitat mission.

The state has identified key priorities with high importance and out of that five actions are adaptation activities and eight are mitigation activities. The high priorities identified are based on the basis of cost effectiveness, cost-benefit, feasibility, ease of implementation and overall sustainability. Within the Sustainable Habitat sector, six key priorities

are put under the transport infrastructure sector and seven priority actions are under sustainable Urban development sector. The total budget proposed for the thirteen key priorities along with sub activities is 96885 Lacs INR or 968.85 Crore INR.

Under a policy scenario with implementation of national emission reduction strategies, aggregated mitigation costs can be reduced if economy-wise environmental policies are complemented by urban policies, such as congestion charges or increasing spatial density. This is due to complementarities with other policy objectives, such as lower local pollution and health benefits, and enhancement of city attractiveness and competitiveness through lower local pollution levels. The identified key priorities consist of multiple opportunities for the mitigation activities to explore the carbon markets with several mechanisms like **Clean Development Mechanism (CDM), Voluntary Carbon Standards, Nationally Appropriate Mitigation Action (NAMA) etc.** The revenue generated from the mitigation activities can be used for their effective operation. Involvement of State Government through **“climate sensitive”** urban planning and management can help achieve national climate goals and minimise tradeoffs between environmental and economic priorities at local levels. While local authorities can help to achieve national climate goals through urban policies to reduce energy demand and improve resilience to climate change, national governments can help to create a sound institutional foundation and knowledge base to support local decision makers engage with

stakeholders to identify and carry out cost-effective actions.

State governments are acting on climate change issues, even without national policies through local regulations, urban services, and programme administration. The two

capital cities can stimulate green jobs by raising consumer awareness, raising the eco-efficiency of local business, facilitating clean-tech start-ups and supporting training programmes. The following action points have emerged out of several rounds of discussions among the working group members.

Key Priorities: Sustainable Habitat

1. Promotion of battery operated (charged from solar power) transport systems in tourist places, battery driven motor boats instead of diesel engine boats (battery charged from solar module mounted on the boat) and use of alternative fuel instead of diesel to drive Barges and other diesel driven equipment used for maintenance purpose in the lake
2. Capacity building of all levels of stakeholders to make them understand their role on their regards and maintain proper environment management system
3. GHG accounting study on emission by transport vehicles
4. Establishment of intelligent transport system
5. Policy framework in transport sector
6. Promotion of public transport and mass transport & Promotion of water transport
7. Establishment of Rail based transport system
8. Reallocation of the hotels or application of STP near Dal lake & DPR preparation on STP
9. Establishment of end-to-end solid waste management system
10. Base line study on solid waste and industrial waste generation and DPR preparation & four pilot project implementation
11. Capacity Building/ Awareness Generation Program on segregation of solid waste at household level
12. Establishment of segregation mechanism of bio-medical waste in urban hospitals and in rural PHCs
13. Application of methane capture technology and preparation of fertilizer from weeds generated in Dal lake

1. Promotion of battery operated (charged from solar power) transport vehicles for tourist zones, battery driven motor boats instead of diesel engine boats (battery charged from solar module mounted on the boat) and use of alternative fuel instead of diesel to drive Barges and other diesel driven equipment used for cleaning the lake

State Government has planned to promote research, development and demonstration projects in the field of Battery Operated Vehicles (BOVs) under the Alternative Fuel for Water Transportation primarily in tourist places. The objective of the programme is to promote non-polluting Battery Operated Vehicles (BOVs) in the state, which would help in conserving diesel oil and curb environmental pollutions.

Dachigam National Park is kept out of bounds for the daily tourists by the Forest Department, only few tourists are given entry with special permission from the Tourism Department This is to prevent the sound and noise pollution caused by the tourist Vehicles that enter the sanctuary boundaries. Green or Eco – Friendly Vehicles such as Battery or Fuel cell operated vehicles including solar/ battery operated vehicles with low sound pollution could be the effective. Similarly, for water transportation which is a primary transportation system throughout the Dal lake, Battery Operated Vehicles (charged from solar power) should be introduced. These

measures would reduce the Green House Gas (GHG) emission from the diesel operated motor boats as well as the consumption of fossil fuel.

2. Capacity building of all levels of stakeholders for better understanding of their role and maintain proper environment management system

The state government has emphasized on the need for capacity building of the officials and other stakeholders on climate change related issues, implications and possible adaptive and mitigating measures. This would enable them to include climatic considerations and proper environmental management strategies in their departmental planning as well as day to day operation and monitoring activities. The process should begin with a training needs assessment for all relevant departments and agencies. This might be followed with training modules especially on overall climate change issues, environment management, solid waste management, water management and urban management would be conducted and subsequent training would be imparted. Capacity building would also be extended to awareness generation on good practices such as source segregation of waste and energy efficiency.

The identified key priorities are based on the cost effectiveness, feasible options, sustainability and ease of implementing with respect to the present condition. New or reformed institutions are needed to enable state governments to facilitate capacity building and decision-making on climate

change at the local levels. A comprehensive capacity building programme on climate change is necessary which would help to generate awareness and increase the knowledge base of responsible stakeholders, officials, policy makers for better climate change adaptability. The action is necessary before implementation of any climate change mitigation initiatives for a comprehensive knowledge base for better understanding and better implementation of the initiatives. Department of Urban Development would be primarily responsible for this key priority action.

3. GHG accounting study on emission by transport vehicles

Measures to reduce GHG emissions from all sectors must be immediately taken to minimize climate impacts. While the role of GHGs in global climate change is well established, there are disagreements about what might constitute unacceptable damage and a range of projected temperature changes and resulting impacts. Transportation sector contributes largely to overall GHG emissions throughout the state and is rapidly growing. The state government would undertake GHG emission estimation from the transport sector (both surface and water transport sector) so that proper mitigation action could be planned. Based on the outcome of the result state government would also formulate new transport policies throughout the state.

4. Establishment of intelligent transport system

Road traffic congestion is a regular problem within the state of Jammu & Kashmir. Due to its unique geographical characteristics and adverse climatic conditions road networks are affected severely. The problem is mainly felt in the state capitals as well as district headquarters. This is primarily because infrastructure growth is slow compared to growth in number of vehicles, space and cost constraints. Better traffic management would limit the CO₂ emission, reducing stop-go traffic and discouraging excessive speed. Intelligent Transport System (ITS) techniques have to undergo adaptation and innovation to suit the traffic characteristics of the hilly terrain and narrow roadways in J&K. Apart from adopting traditional sensors to chaotic traffic conditions, efforts can be made to design new sensing solutions for chaotic traffic. There is scope for evaluating existing ideas for the challenging traffic scenarios, innovate new solutions and empirically evaluate ideas in collaboration with public and private sectors.

State Government is planning to adopt the ITS for better management of increasing traffic congestion leading to higher GHG emission. Adopting ITS can reduce the GHG emission and more efficient transport system which would also contribute to the overall economic growth of the state.

5. Policy framework in transport sector

Climate Change poses two fundamental challenges to the transport sector; the sector would have to significantly reduce Green House Gas (GHG) emissions which

needs investment for adaptation to impacts of climate change. The scale and scope of emission reductions sought by policy makers is a challenging task but there is much that can still be achieved within the sector at low cost while the cost of energy prices are increasing rapidly.

The state government is planning to formulate cost effective mitigation policies that can minimize the GHG emissions from the transport sector. Regional circumstances within the state would play an important role in determining the allocation of effort. The new transport policy would focus to curb GHG emissions resulting from the transport sector throughout the state.

6. Promotion of public transport and mass transport & Promotion of water transport

Transport is the fastest growing source of Green House Gas (GHG). A shift towards sustainable transport and urban development would reduce environmental costs and reliance on fossil fuels. Vehicular emissions have two negative impacts on environment: one is global climate change through increase in GHG emissions and the other is air pollution and its negative health repercussions. The total number of registered motor vehicles has increased in the two wheeler segment comprising of motorcycles, scooters and mopeds. The number of two wheelers has grown rapidly amongst personalized modes of transportation emitting the maximum Carbon Dioxide (CO₂) and Carbon Monoxide (CO) than the other vehicles. Out of all the pollutants, CO₂ is a potent GHG and CO is

considered as the major air pollutant.

General public transportation falls into the category of Mass Rapid Transit (MRT), or modes of urban transportation that carry large volumes of passengers quickly. State Government is planning and prioritizing their budgets to allocate more funds in order to build new public transportation systems and expand or improve upon old ones. The improved public transportation systems would encourage use of public transport in place of personal vehicles. This would require employers to subsidize mass transit fees for their employees. Water transport is a major transportation system within the state and contributes significantly to the economy of J&K. Water transport system can also be encouraged for mass transit using cleaner fuels instead of diesel. This would also significantly reduce GHG emissions throughout the state. The state government is encouraging the usage of water transportation system through introduction of Battery Operated Vehicles (BOVs) and motor boats run by solar charged batteries instead of diesel oil to reduce GHG emission.

7. Establishment of Rail based transport system

Energy use for freight transportation is expected to increase significantly in the next 25 years throughout the state. GHG emissions are largely based on energy use. Therefore, GHG emission would also increase significantly. But large percentage reduction is possible if intermodal shift, (for example from truck to rail) is encouraged. For example, reduction of the order of 85% is possible if

long-haul truck transport is replaced with a combination of rail and truck transport. This estimate takes into account a portion of the total freight shipment using truck transport at the start and end of the trip.

Works are on to connect the Kashmir Railway to the rest of the country. Officially termed the Jammu Udhampur Srinagar Baramulla Railway Link, the railway tracks would be laid over 345 km (214 mile) from Jammu and to Baramulla on the northwestern edge of the Kashmir Valley. The route crosses major earthquake zones, and is subjected to extreme temperatures and un-favorable terrain, making it an extremely challenging engineering project. The state government is also planning to expand the network of railways to reduce the dependency over road transportation.

8. Reallocation of the hotels or application of STP near Dal lake & DPR preparation on STP

Dal Lake is located on the eastern part of Srinagar city on the right bank of Jhelum. 15 major drains of the city open into the Dal Lake, polluting its waters on a daily basis. The untreated sewerage brings with it nitrogen and phosphorus which has changed physical and chemical properties of Dal waters. The sewage generated from entire Srinagar city discharged into Dal lake pollutes the water body significantly. Moreover, increasing sewage discharged from the nearby hotels and resorts around Dal lake result in a synergistic effect.

Thus state government is planning to

implement several sewage treatment plants which would treat the sewage wastes produced from the city and hotels situated near Dal lake before discharging it into the water body. This would reduce the pollution load as well as methane generation resulting from anaerobic degradation of the high organic load of the city sewage. Also the potent GHG emission would be reduced along with restoration of the Dal lake.

9. Establishment of end-to-end solid waste management system

The MSW management decisions taken by mayors, county executives, and city and county councils and boards can influence the rate of greenhouse gas (GHG) emissions. Emissions of CH₄ results from the decomposition of biodegradable components in the waste stream such as paper, food scraps, and yard trimmings. The potential of GHGs to cause global climate change is being debated both nationally and internationally. Solid waste management subprojects include construction and upgrading of landfill sites, transfer station, storage and parking facilities for the collection vehicles and procurement of collection and disposal equipments, as eligible under the subproject selection criteria for the Investment Program. The activity is proposed to establish an Integrated Waste Management Plan for cities including measures to improve efficiency of existing solid waste and sewerage management systems, and incorporate a plan for management of Construction and Demolition (C&D) waste, biomedical waste, and domestic hazardous waste.

Collection and dumping of the solid waste in urban areas is one of the basic responsibilities of the Department. However, the Department does not have sufficient machineries and equipment's for collection and disposal of Municipal Solid Waste. In order to deal with the issue the Department has introduced a scheme in the Annual Plan 2010-11 with proposed 6 SWM projects for the district of Reasi (Katra), Udhampur, Kothmohar (Jammu) Budgam, Pulwama and Kulgam. The SWM is an ever increasing challenge due to constant migration of people towards towns/cities. There are several issues which should be addressed before venturing into a solution for handling huge amount of garbage.

The Urban Development Department might draw comprehensive plan for power generation from Municipal Solid Waste (MSW) and manure production from Municipal Solid Waste. The project could be implemented either through public/private partnership or fund made available through JNNURM sources. MSW projects are also highly suitable to attract CDM benefits. Measures to reduce Green House Gas emission and adapt to expected climate change impacts would put additional pressure on city budgets and increase the need for additional public resources. These mitigation activities would reduce significant amount of GHG emissions and the revenue flow from the sale of emission reductions would help in sustaining the project. The problem of littering can be well managed with development of suitable linkage or setting up of appropriate infrastructure for recycling of plastics, glass, paper etc. With

active participation of people, this can be a sustainable source of livelihood for a larger section of urban population.

10. Base line study on solid waste and industrial waste generation and DPR preparation & four pilot project implementation

There is an urgent need for detailed analysis of the sources of solid wastes from urban and industrial sectors for proper management plan to be formulated. The urban department of J&K is thus planning for a detailed baseline study and preparation of Detailed Project Report (DPR) for the solid waste management activities. The Department is exploring the possibility of engaging consultants for a typical cost effective design accommodating the latest technologies in the area. Subsequent allocations would be made. Thus the study shall contribute to a scientific product such as the development of a methodological approach which can be used by municipalities or research institutions or consultants with similar tasks.

11. Capacity Building/ Awareness Generation Program on segregation of solid waste at household level

The Municipal Solid Wastes (Management and Handling) Rules 2000 are not the only policy documents targeting the issue. The National Environment Policy 2006, lays down the action plan for soil pollution comprising actions for strengthening the capacities of ULBs for segregation, recycling, and reuse of municipal solid wastes and setting up and operating sanitary landfills, in particular

through competitive outsourcing of SWM services. The garbage collection services are often not available to many households. They often fail to dump their garbage on time when the collection vans arrive at their doorstep. Also many have to walk to the municipal containers. The surroundings of the container collection points are often filthy due to littering by residents, animal activities, or lack of site maintenance by the responsible authorities. Indiscriminate dumping into open drains, down banks on the road side and down into gullies, is a major problem within the city area. This other than leading to a visually filthy environment leads to increased health risk by physical contact, vector proliferation and infrastructure damage by flooding.

The Urban development department of the state is planning to provide mass awareness programme with capacity building at household level for proper waste sequestration at source. The wastes can be categorized into biodegradable (wet) and non-biodegradable (dry) wastes. The non-biodegradable (dry) waste would thereafter be segregated into recyclables, non-recyclables and domestic hazardous waste. Each household would be provided two buckets in different colors free of charge for the wet and the dry waste respectively. This would reduce the segregation cost of the SWM projects and enhance the efficiency of environmental friendly disposal of the solid waste.

12. Establishment of segregation mechanism of bio-medical waste in urban hospitals and in rural PHCs

As per a survey report of J&K Pollution Control Board, the lack of coordination between various government departments and the departments within the health care institutions is causing issues for implementation of bio-medical waste management rules. The report says that the management of bio-medical waste should be a priority while allocation funds for health care institution. Tons of bio-medical waste has accumulated in associated hospital of the Government Medical College Srinagar in absence of proper collection mechanism. The state hospitals are littered with used syringes, needles, plastic items, bandages and empty medicine and saline bottles.

State Urban department needs to establish an effective mechanism for proper segregation of biomedical wastes in urban hospitals and rural public health centres. This would reduce the environmental degradation as well as harmful effects to public health within the state. The mechanism would significantly help in establishment of proper disposal of the generated biomedical waste from the hospitals of urban area as well as rural PHCs.

13. Application of methane capture technology and preparation of fertilizer from weeds generated in Dal lake

Weed control methods with use of chemicals or mechanical mowers address the problem less harmfully than using enzymes and dispersal equipment. The excess nutrients have the same harmful effects as those that come from the surrounding watershed. Fertilizers, animal droppings, septic tank

seepage and nutrients recycled from the sediment reduce the dissolved oxygen level in a lake thereby increasing the organic load. This leads to the habitat destruction and eutrophication in the Dal Lake and destruction of the major water source of Srinagar city. Government is planning to implement five Sewage Treatment Plants (STPs) with the combined capacity of 36 Million Litres per Day for treatment of sewage generated by households, commercial establishments and the hotels.

However the weeds harvested from the lake can be utilized for the extraction of methane through controlled anaerobic digestion system which can be utilized for power generation or any other energy sources. The digested sludge will be rich in nutrients and can be utilized as organic manure. Thus the process would restore the oxygenation characteristics of the Dal Lake, promoting healthy aquatic habitat while reducing undesirable weed growth, snails and blood suckers like leech etc along with the associated scum and algae population.





6.7 List of Key Priority Action

Table 29: List of Key Priority Action Mission on Sustainable Habitat

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
1	Promotion of battery operated (charged from solar power) transport systems in tourist places, battery driven motor boats instead of diesel engine boats (battery charged from solar module mounted on the boat) and use of alternative fuel instead of diesel to drive Barges and other diesel driven equipment's used for maintenance purpose in the lake	Housing & Urban Development (H&UD) Department	0	1.0	1.0	GoI, EFA
2	Capacity building of all levels of stakeholders to make them understand their role on their regards and maintain proper environment management system	H&UD Dept.	0	2.5	2.5	GoI, EFA
3	GHG accounting study on emission by transport vehicles	H&UD Dept.	0	2.5	2.5	GoI, EFA
4	Establishment of intelligent transport system	H&UD Dept.	0	50.0	50.0	GoI, EFA
5	Policy framework in transport sector	H&UD Dept.	0	2.5	2.5	GoI, EFA
6	Promotion of public transport and mass transport & Promotion of water transport	H&UD Dept.	0	800.0	800.0	GoI, EFA
7	Establishment of Rail based transport system	H&UD Dept.	0	320000	320000	GoI, EFA
8	Reallocation of the hotels or application of STP near Dal lake & DPR preparation on STP	H&UD Dept.	0	400.0	400.0	GoI, EFA
9	Establishment of end-to-end solid waste management system	H&UD Dept.	0	400.0	400.0	GoI, EFA
10	Base line study on solid waste and industrial waste generation and DPR preparation & four pilot project implementation	H&UD Dept.	0	50.0	50.0	GoI, EFA

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
11	Capacity Building/ Awareness Generation Program on segregation of solid waste at household level	H&UD Dept.	0	10.0	10.0	Gol, EFA
12	Establishment of segregation mechanism of bio-medical waste in urban hospitals and in rural PHCs	H&UD Dept.	0	10.0	10.0	Gol, EFA
13	Application of methane capture technology and preparation of fertilizer from weeds generated in Dal lake	H&UD Dept.	0	60.0	60.0	Gol, EFA
	Total Budget (INR in Million)		0	321788.5	321788.5	
N.B: The cost estimates are indicative and may change after preparation of Detailed Project Reports (DPRs)						





7 Green India Mission

7.1 Introduction

Forests and climate change are inherently linked. When kept healthy, forests play a key role in our national climate change strategy. Over geologic time, changes in disturbance regimes are a natural part of all ecosystems. However, loss of forest's ability to mitigate the effects of climate change will mean losing many important ecosystem good and services. Development, climate change, invasive species, unsustainable forestry practices, diseases and insects, all of these factors and more, endanger the forests that we rely on for so much. The variety and severity of the threats to our forests means that simply planting trees for sheer numbers is not enough. We have to develop detailed restoration plans that take every factor into account - from the needs of local wildlife to the changing climate in the region. Even if changes cannot always be predicted, it is important to consider ways in which impacts to forest systems can be mitigated under likely changes in disturbance regimes. The task for

the next decade is to understand better how climate affects disturbances and how forests respond to them. Improved monitoring programs and analytic tools are needed to develop this understanding. Ultimately, this knowledge should lead to better ways to predict and cope with disturbance-induced changes in forest.

7.1.1 Current Status of Forest and Forest Cover Change

According to the interpretation of satellite data of October-December 2008, the forest cover of Jammu and Kashmir is spread over 22,539 km² (including area outside the LoC) accounting for 10.14% of the State's geographical area (Source: India State of Forest Report, 2011). In terms of forest density parameters fixed by Forest Survey of India, a total of 4,140 km² area is presently covered under very dense forest, 8,760 km² under moderately dense forest and 9,639 km² under open forest; distribution of which is depicted in the figure below.

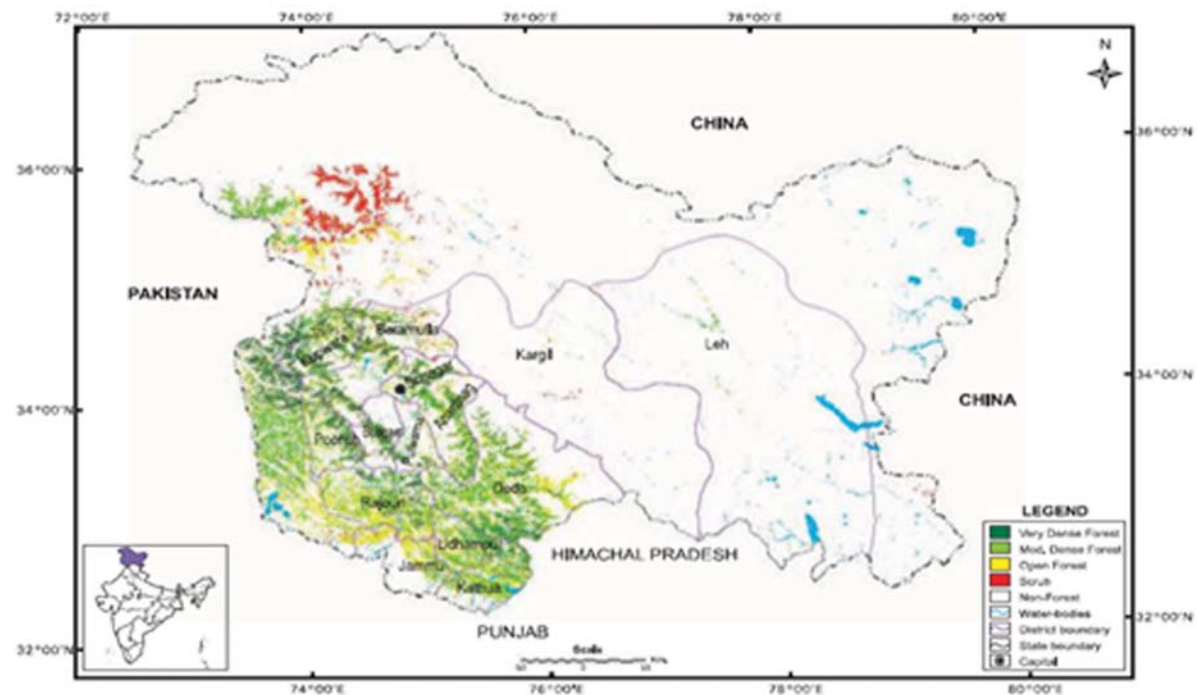


Figure 15: Forest cover in the state¹⁷

Jammu and Kashmir has 4 National Parks, 15 Wildlife Sanctuaries and 34 Conservation Reserves, spread across an area of 11,774.50 km². The Dachigam National Park is famous for Hangul. The Wular Lake in Baramulla

district, having a spread of 8,900 hectares, has been designated as a Ramsar site. The district-wise forest cover in different canopy density classes is given in the following table:

¹⁷India State of Forest Report 2011, Forest Survey of India, Dehradun



Table 30: Assessment of variation in forest cover (Area in km²)¹⁸

District	Geo. Area (GA)	2011 Assessment				Percent of GA	Change* over 2009 assessment	Scrub
		Very Dense forest	Moderately Dense Forest	Open Forest	Total			
Anantag	3984	196	664	578	1438	36.09	0	23
Baramulla	4588	425	286	446	1157	25.22	0	86
Budgam	1371	99	69	52	220	16.05	0	8
Doda	11691	619	1689	1659	3967	33.93	0	5
Jammu	3097	0	210	672	882	28.48	0	43
Kargil	14037	0	3	21	24	0.17	0	19
Kathua	2651	112	673	614	1399	52.77	1	12
Kupwara	2379	472	366	322	1160	48.76	0	3
Leh	45110	0	48	57	105	0.23	0	6
Outside LOC	120848	1326	2471	2686	6483	5.36	1	1810
Poonch	1674	187	300	242	729	43.55	0	9
Pulwama	1398	110	106	78	294	21.03	0	10
Rajouri	2630	49	439	752	1240	47.15	0	8
Srinagar	2228	196	307	249	752	33.75	0	16
Udhampur	4550	349	1129	1211	2689	59.10	0	47
Grand total	222236	4140	8760	9639	22539	10.14	2	2105

According to the 2011 assessment of FSI, it is observed that the area under forest cover has increased by 2 sq. kms in Jammu and Kashmir State. Actually, however, there is a net decrease of 147 km² from

the reported area in the ISFR 2009. This is due to interpretational changes on account of refinement of methodology as well as availability of improved satellite data in appropriate seasons (Source: ISFR 2011).

Table 31: Forest Cover Change Matrix (Area in km²)¹⁸

	2011 Assessment					2009 assessment
	VDF	MDF	OF	Scrub	NF	
Very Dense Forest	4140	0	0	0	0	4140
Moderately Dense Forest		8760	0	0	0	8760
Open Forest	0	0	9637	0	0	9637
Scrub	0	0	0	2101	0	2101
Non-Forest			2	4	197592	197598
Total 2011	4140	8760	9639	2105	197592	222236
Net Charge	0	0	2	4	-6	

¹⁸India State of Forest Report 2011, Forest Survey of India, Dehradun

¹⁹ISFR 2011

7.1.2 Forest cover under different forest type

Forest Survey on India has undertaken forest type mapping using satellite data with reference to Champion & Seth Classification. The assessment has identified 38 forest types belonging to eight forest type groups, viz. Tropical Dry Deciduous, Subtropical Pine, Subtropical Dry Evergreen, Himalayan Moist Temperate, Himalayan Dry Temperate and Sub Alpine Scrub. Percentage-wise distribution of forest cover in various forest types is given in the chart below.

actions for enhanced ecosystem (like carbon sequestration and storage in forest & other ecosystems), hydrological and biodiversity conservation along with provisioning services like fuel, fodder, timber and NTFPs. The “greening” in the context of climate change aims at a combination of adaptation and mitigation measures, which helps in:

- Enhancing carbon sinks in sustainably managed forests and other ecosystems.
- Adaptation of vulnerable species/ ecosystems to the changing climate.

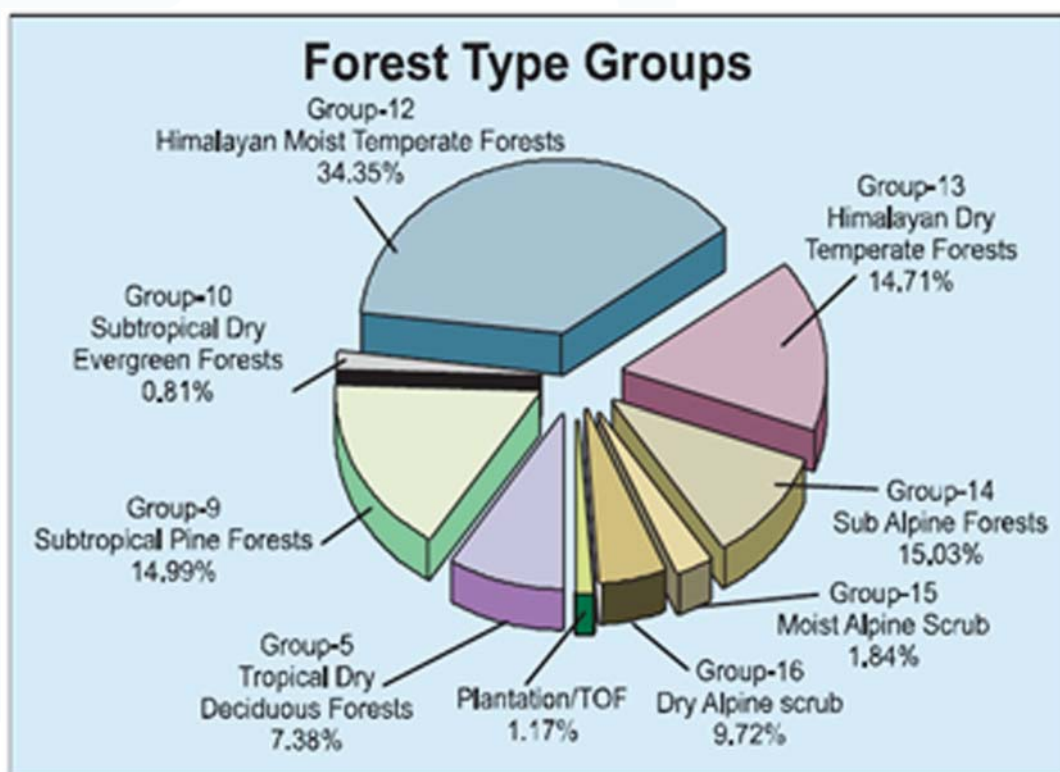


Figure 16: Different Forest Types in States²⁰

The Greening of India Mission (GIM) under the National Action Plan on Climate Change (NAPCC) aims at adaptation and mitigation

- Inculcating resilience in the existing forests by various silvi-cultural and forest management interventions

²⁰India State of Forest Report 2011, Forest Survey of India, Dehradun

- Adaptation of forest-dependent communities

The objective of the State Mission on Forest & Biodiversity sector is to achieve the national goal of GIM under NAPCC & to respond to the possible climate change impacts in the state.

Climate change impacts can be reduced by way of conservation and protection of forests and biodiversity resources. Revised policies for the protection and management of the forest should be integrated with conservation of the catchment areas in the hills.

The Green Mission of Jammu and Kashmir is prepared in compliance to the guidelines of National Climate Change Action Plan-Green Mission with a series of discussions with the

Officers of the Forest Department and the Prl. Chief Conservator of Forests.

7.1.3 Role of Forest in state

The State of Jammu and Kashmir has rich diverse forest resources and plays an important role to preserve the fragile ecosystem of the state and it also serves as catchments for Himalayan Rivers. In Jammu and Kashmir, the perennial water supply including groundwater recharge, and health of soils, is primarily depending upon the area and quality of forest cover. A healthy forest cover is essential for long term operation of hydroelectric projects. In the State forest also provide some environmental services like carbon sequestration, pollution abatement, amelioration of climate, in-situ conservation of biodiversity and maintenance of ecological



balance. Since there is unique aesthetic value of forest landscape and huge potential for tourism in the State, the forests play a pivotal role to make the State an ideal tourist destination.

Forests are also essential for soil conservation, water security, and for meeting the daily necessities of local population for timber, firewood, fodder and other forest produce. As per the State specific demarcation rules the Forests of the State were constituted along with the initial revenue settlement. As the largest land based resource in Jammu and Kashmir, forests have huge potential for supporting livelihoods and poverty alleviation. Forests also have some economic value. As per the expert estimation (Down to Earth, issue of July 2005) the net present environmental value of forests of Jammu and Kashmir was Rs.1,92,000 crores.

7.2 Key trends in the sector

The varied and distinctive geo-climatic conditions across the three regions of the state, viz, Jammu, Kashmir and Ladakh is responsible for the diversified forest cover ranging from sub-tropical to temperate and alpine. The three regions have witnessed exponential growth in human and livestock population, rapid industrialization and spurt in developmental activities, especially in past few decades. These developmental processes have resulted in depletion of forest cover accompanied by an overall degradation and forest soil. Earlier, forests mainly served the purpose of timber production. Although, commercial felling of green trees has been discontinued, but drying of trees due to

natural reasons is unavoidable. Further, due to continuous and unrestricted grazing, most of the forests in the State are deficient in regeneration. Other factors like forest fires, invasive weeds, unregulated tourist movement and lack of timely silvicultural operations prohibit the regeneration. Similarly, a marked reduction in population of other productive species especially medicinal plants, has been observed. Over exploitation of forests has not only degraded them but also impaired their ability to provide environmental benefits. The effects of forests degradation are visible through drying up of perennial water sources, accelerated soil erosion, flash floods, silting of reservoirs loss of biodiversity and reduced forest productivity.

7.3 Vulnerability of the sector

7.3.1 Climate Variability in J&K

Climate variability refers to shorter term like daily, seasonal, annual, inter-annual, several years' variations in climate, including the fluctuations associated with El Niño (dry) or La Niña (wet) events. The overall effects of climate variability are puzzling in Jammu and Kashmir. Based on Percent of Normal, majority of the districts in the northwest part of the country consisting of Rajasthan, Gujarat, Jammu & Kashmir, Punjab, and Haryana have drought probabilities of $\geq 20\%$. Severe drought will produce massive tree mortality, but there is also evidence of periodic tree staffing during extreme rainy years. As per INCCA Report 2010 the whole Himalayan region is exhibiting an increase in the precipitation in the 2030s scenario.

The increase varies between 5% and 20% in most areas, with some areas of Jammu and Kashmir showing an increase of up to 50%. As per INCCA Report 2010 due to climate variability in the State, food production deficit is growing in recent times. Due to low rainfall, the rain-fed agriculture will suffer the most. Horticultural crops like apple are also showing decline in production and areal coverage particularly due to decline in snowfall.

7.3.2 Impact of climate change on forest ecosystems in Jammu and Kashmir

Indian Institute of Science has conducted an assessment of the impact of climate change on forest ecosystems in India. A dynamic vegetation model IBIS (Integrated Biosphere Simulator) was used to assess the impacts of climate change on forests in India. The study indicates that about 39% and 34% of the forested grids are likely to undergo shifts

in vegetation type under A2 and B2 climate scenarios, respectively with a trend towards increased occurrence of the wetter forest types.

Approximately 47% and 42% of tropical dry deciduous grids are projected to undergo shifts under A2 and B2 scenarios respectively, as opposed to less than 16% grids comprising of tropical wet evergreen forests. Similarly, the tropical thorny scrub forest is projected to undergo shifts in majority of forested grids under A2 (more than 80%) as well as B2 scenarios (50% of grids). According to another study projected impacts of climate change, using a moderate A1B scenario and IBIS vegetation model for the period of 2030s and 2080s, are projected in figure 3.3 and 3.4, which show the impacts on current forested locations in India. The figure shows that Jammu and Kashmir is projected to undergo vegetation change in both A1B (2035 and 2085) scenarios.



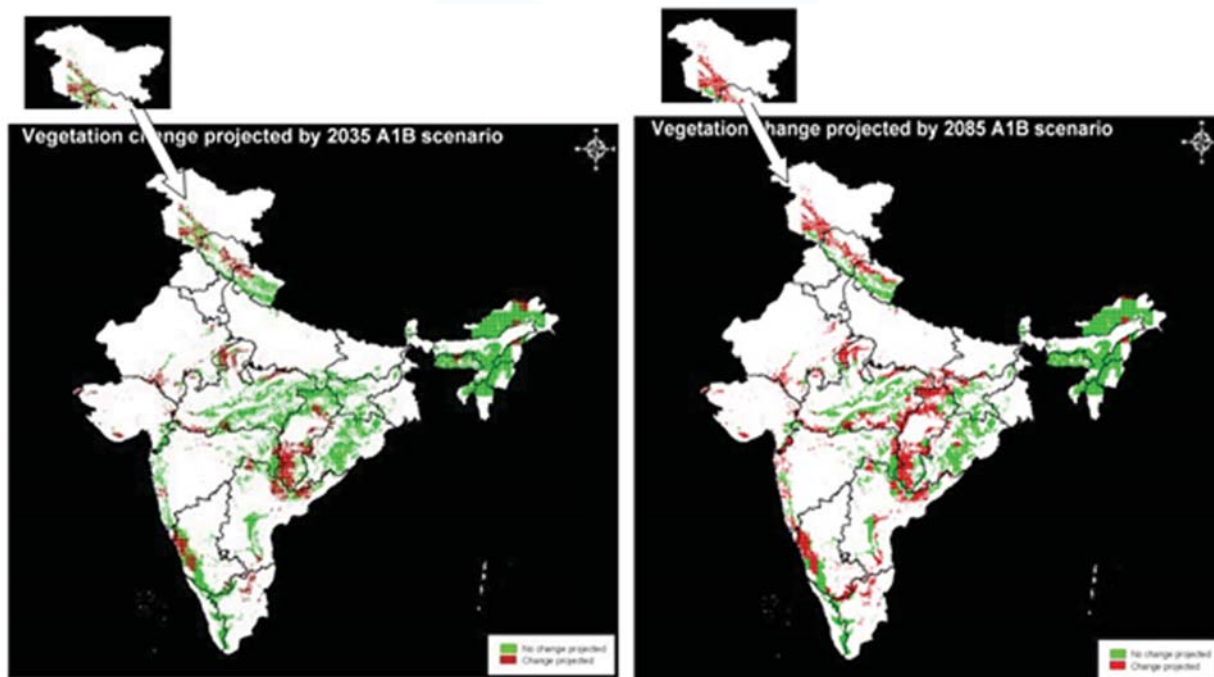


Figure 17: Forest Grids in India

All forested grids in India are shown in color (red or green): red indicates that a change in vegetation is projected at that grid in the time-period of 2021-2050, and green indicates that no change in vegetation is projected by that period.

All forested grids in India are shown in color (red or green): red indicates that a change in vegetation is projected at that grid in the time-period of 2021-2050, and green indicates that no change in vegetation is projected by that period. The timeframe is 2071-2100

7.3.3 Vulnerability to the sector

Number of pressures is responsible for deforestation and degradation in Jammu and Kashmir. Due to high dependency on forest for timber and fuel wood deforestation is an ongoing process. On the other hand the livestock population in the State is 9.90 million resulting in huge demand for fodder. In addition most of this livestock graze in

forests. Grazing and trampling of saplings by livestock is the biggest threat to regeneration of vegetation in all forested areas of the State. Low tree density, low bio-diversity status as well as higher levels of fragmentation contribute to the vulnerability of these forests. As per Chatuvedi, (2010) , most of the mountainous forests (sub-alpine and alpine forest, the Himalayan dry temperate forest and the Himalayan moist temperate forests) are susceptible to the adverse effects of climate change. This is because climate change is predicted to be larger for regions that have higher elevations. There is a need to explore win-win adaptation practices in such regions such as anticipatory plantations, sanitary harvest, and pest and fire management.

It is also seen that a significant part of the Himalayan bio-diversity hotspot that stretches along the northwestern part of India along the states of Punjab, Jammu and Kashmir and Himachal Pradesh is projected

to be highly vulnerable. This may be mostly attributed to the higher elevation of these regions. The study has shown that these regions will experience higher levels of warming.

According to the estimated vulnerability index suggests that upper Himalayas, northern and central parts of Western Ghats and parts of central India are most vulnerable to projected impacts of climate change. The forests in the central part of India, especially the north-western part of India are highly vulnerable.

Due to changes in precipitation and temperature in Jammu and Kashmir the probable vulnerability of the forestry sector is as follows:

- Climate change would increase many existing stress on forest ecosystems through increase in invasive species, insect pests and pathogens, and disturbance regimes. Climate change would result in ecosystem shifts and conversions.
- Many tree species would have insufficient adaptation rates to keep pace with climate change.
- Forests would not be able to provide a consistent supply of forest products.
- Climate change impacts on forests would impair the ability of many forested watersheds to produce reliable supplies of clean water.
- Climate change would result in a widespread decline in carbon storage

in forest ecosystems across the region.

- Many contemporary and iconic forms of recreation dependent on forest ecosystems would change in extent and timing due to climate change.
- Climate change would alter many traditional and modern cultural connections to forest ecosystems.

7.4 Key issues in the sector

Loss of forest cover and habitat destruction:

Many species are threatened due to habitat destruction, forest fragmentation and loss of forest cover. Developmental activities (road construction, transmission lines, power projects, etc.), encroachments, illicit felling of trees, uprooting of trees due to natural calamities like heavy snowfall, and wind storms are some of the reasons responsible for loss of forest cover.

Spread of Alien species: In some of the sub-tropical forests areas of the state, spread of alien species like Lantana camara and Parthenium hysterphorus, has posed an increasing threat on forest productivity and native biodiversity. Few species of fish like mirror carp have caused decline in the population of native fish species like the Schizothorax. Other species of trout in several hill streams have adapted well but their impacts on local biota unknown.

Unregulated grazing by domestic livestock:

Rapid increase in the livestock population of the State has resulted in a huge gap between demand and supply of fodder.

Consequently, the forests and pastures of the State are subjected to heavy grazing, much beyond their carrying capacity. This problem has remained largely unaddressed, resulting in severe degradation of forests and pasturelands which has further reduced the fodder availability. The palatable grass and shrub species have been replaced by un-palatable species. To meet the fodder demand, more and more areas are subjected to overgrazing, further compounding the problem. This is a vicious cycle which often results in many conflicts between different stakeholders for use of scarce grazing resources. Over-grazing of forest areas by livestock has also seriously impaired the ability of natural forests to regenerate.

Encroachments: Human encroachments into biologically rich forest areas have increased alarmingly resulting in loss of forest cover and honeycombing of forests. Encroachments into water bodies like lakes are common and have reduced the expanse of lake bed.

Anthropogenic pressure on forest areas: Both protected and unprotected forest areas and grazing lands are under tremendous biotic pressure for timber and firewood extraction/ collection. Indiscriminate collection of many native species having commercial value or medicinal importance continues despite protection accorded under different laws (Wildlife Act, Forest Act, and Biodiversity Act).

Jammu and Kashmir is endowed with rich biodiversity and gene pool resources, such as dense lush green forests, alpine pastures, snow bound peaks, glaciers, rivers, streams

and fascinating wetlands like Wullar, Dal, Hokersar, Hygametc offering conducive breeding grounds for the migratory birds. In the sub-tropical forests of the state, Chir (*Pinus roxburghii*), which occurs between 600 m to 1500 m, is the predominant species. The common tree associates of this species are *Quercus leucotrichophora*, *Quercus ilex*, *Oleacuspidata*, *Pistacia integerrima*, *Lyonia ovalifolia*, *Pyrus pashia*, *Alnus nepalensis*, *Acacia modesta*, *Celtis australis*, *Rhus* spp and *Corylus colurna*. Shrubs commonly found in this zone are *Myrsine africana*, *Dabregeasia hypoleuca*, *Berberis* spp, *Nerium indicum*, *Plectranthus rugosus*, *Daphne cannabina*, *Zizyphus* spp, *Spiraea sorbifolia*, *Prinsepia utilis*, *Rubus ellipticus*, *Woodfordia floribunda*, *Rosa moschata* and *Punica granatum*. These shrubs constitute the under growth in Chir forests. The intensity of under growth in Chir forests varies from light to heavy. The ground flora generally comprises *Fragaria vesca*, *Viola canescens*, *Polygonum* spp, *Girardinia heterophylla*, *Ainsliaea aptra*, *Impatiens glandulifera*, and a variety of grasses and ferns. The common climbers of this zone are *Clematis* spp, *Jasminum humile*, *Vitis* spp, *Rosa moschata*.

The temperate forests occur at altitudes between 1600 m and 3200 m. These temperate forests are dominated by Blue pine (*Pinus wallichiana*), Spruce (*Picea smithiana*), Deodar (*Cedrus deodara*) and Silver fir (*Abies pindrow*). The common broad leaved tree associates of Deodar, Kail and Fir are *Aesculus indica*, *Alnus nepalensis*, *Ulmus wallichiana*, *Prunus padus*,

Quercus dilatata, *Quercus semecarpifolia*, *Q. leucotrichophora*, *Fraxinus excelsior*, *Betula utilis*, *Rhododendron* spp, *Celtis australis*, *Pyrus pashia*, *Corylus colurna*, *Toonaciliata*, *Toonaserrata*, *Populus* spp. The incidence of the under growth in these forests varies in intensity from scant to moderate and the commonly found species are *Viburnum foetens*, *Viburnum nervosum*, *Skimmialaureola*, *Podophyllum hexandrum*, *Parrotia Jacquemontiana*, *Desmodium* spp, *Sarcococcaligna*, *Indigofera pulchella*, *Prinsepia utilis*, *Rubus* spp, *Rosa* spp, *Berberis* spp, *Daphne cannabina*, *Acer pictum*, *Cotoneaster* spp, etc.

The herbaceous ground flora includes *Viola canescens*, *Fragaria vesca*, *Rumex hastatus*, *Anemone* spp, *Polygonum* spp, *Girardinia heterophylla*, *Impatiens glandulifera*, *Atropa belladonna*, *Podophyllum hexandrum*, and a variety of miscellaneous grasses and ferns.

Alpine pastures immediately follow the temperate tree line from 3200 to 3600 m, the alpine vegetation is scarce between 3600-4100 m and largely comprises of trees like junipers, *Rhododendrons*, *Betula* etc sprinkled with multiple herbs and shrubs.

The fauna of Jammu and Kashmir includes wild goats like Markhor and Ibex, antelopes like Chiru, deer's like Hangul or Kashmir Stag, Musk Deer, Spotted Deer, Shapoo, Black Bears, Leopards, Wolf, Fox, Monkeys and birds like Western Tragopan, Himalayan Monal Pheasant etc. Though the man vs wild conflict has always existed in the past, it has certainly escalated in the past two decades.

Asiatic black bear (*Ursus thibetanus*) and common leopard (*Panthera pardus fusca*) frequently attack humans and livestock resulting in death/ serious injuries, besides causing loss to property. The obvious reasons are the loss of wild habitat as a result of human activities such as deforestation causing loss of forest cover, change in land use pattern, human interference, stone quarrying and mine blasting near the forest areas, forest fires, and natural factors like increase in wild life populations, decline in the natural prey base, change in the predator behavior due to easy availability of prey like stray dogs, etc.

7.5 Programme and Policies in the sector

The following are the major schemes and Programmes implemented in the forestry sector of the State under State/District/ Centrally Sponsored schemes:

- Consolidation and Demarcation
- Development of Minor Forest Produce including Medicinal Plants
- Eco Task Force
- Urban Forestry
- Pasture and Fodder Development
- CM's participatory Afforestation Scheme
- Forest Protection
- Participatory Grazing Land Development Programme
- Eco-restoration of Degraded catchments
- Integrated Forest Protection Scheme
- CAMPA

It is seen that majority of the above schemes are related to afforestation and reforestation. Afforestation on the degraded forests has yielded good results. The initiative has been recognized and awarded by the government of India for the year 2009 and 2010. To maintain ecological balance, the felling of trees has also been limited. As a result, the outturn of forest produce has declined. The outturn of timber which was 102.48 thousand cubic meters in 2000-01, declined to 69.70 thousand cubic meters in 2009-10, and has further come down to 37.14 thousand cubic meters during the current financial year 2012-13 (ending Nov)²¹. Therefore, the continuous ecological restoration activity in Jammu and Kashmir would certainly have climate change mitigation effects besides addressing biodiversity issues. Forests have to be conserved and managed in a sustainable manner with active participation of local communities so as to ensure a continuous flow of benefits to all stakeholders.

At the same time, in the Government has adopted the J&K State Forest Policy in January 2011 for effective conservation and management of forest ecosystems. The State Forest Policy addresses need for conservation of forest resources, meeting needs of people, and poverty alleviation through developmental activities. Strategies for protection, conservation and management of forests are fully defined and documented in the State Forest Policy which has been formulated through series of consultation with different stakeholders.

Salient features of Forest Policy 2010:

Being the largest land based resource in the State; forests have immense potential for supporting livelihoods and alleviating poverty. State forest policy was launched in the year 2010.

The basic objectives of the Jammu and Kashmir State Forest Policy are as follows:

- Conservation of biodiversity and natural habitat through preservation of natural forests with the vast variety of flora and fauna.
- Rehabilitation of degraded forests so as to optimize their productivity and restore their potential to provide ecosystem goods and services on sustainable basis.
- Poverty alleviation by meeting livelihood needs of forest dependent communities through sustainable supply of forest produce by improving productivity of existing forests, and through forestry activities, schemes and programmes.
- Extending tree cover outside forests to reduce pressure on natural forest for supply of forest produce.

7.6 Key Priority

Trees utilize carbon dioxide in the atmosphere for biomass production, and provide mitigation and adaptation services. In order to tap the potential of forests for climate change mitigation and adaptation, following strategy will be adopted:

²¹http://www.jandkplanning.com/images/Economic_Survey/19-env.forests.pdf

- Conservation, sustainable management of forests and enhancement of forest carbon stocks will be encouraged to offset emissions from core sectors of energy, industry and transport.
- Forest carbon stocks will be conserved and enhanced by reforestation of regeneration deficient and open forest areas.
- Unutilized areas suitable for tree growth will be afforested by planting fast growing species to meet the energy requirements of local population in order to offset use of fossil fuels.
- Technologies for efficient conversion of energy from biomass will also be developed and popularized.

In order to benefit from international incentives towards conservation, maintenance and enhancement of forest carbon stocks, institutional framework will be established to synergize actions of individuals, organizations and Government Departments. Cent percent incentives will flow to the local community wherever such incentives accrue because of involvement of the community in protection, afforestation and reforestation activities.

The key priority actions for climate change adaptation and mitigation adopted by the state would aim at maintaining a healthy forest resource, reduce the continuous depletion and degradation of forest and, and also recognize the close association between climate change and forests.

1. Implementation of J&K State Forest Policy :

The Jammu and Kashmir State Forest Policy-2011, provides a comprehensive roadmap for conservation of existing forests and restoration of the forests that have degraded. Implementation of strategies provided in the State Forest Policy will address the climate change issues related to forestry sector in a significant manner. The policy provides that the Forest Department will prepare an implementation schedule for realizing the objectives of this Policy, and, a high powered institutional mechanism will be put in place to monitor and review the implementation of the Policy. Therefore, the first and foremost priority for the Forest Department is to implement the State Forest Policy in letter and spirit.

2. Capacity building and awareness for all levels of stakeholders

Capacity building of forest department personnel would be undertaken to improve their skills and professional competence to enable them to recognize the impacts of climate change on forest resources. They should be trained on the legal aspects of forest and environmental issues. Infrastructural requirement and professional needs are to be identified for proper guidance on climate change issues to meet the requirements at all levels of Forest officers & staff.

3. Gene bank development for climate adaptable species

Forests have to cope with new environmental

stresses, both biotic and abiotic. These include habitat shifting and alteration, fragmentation, drought, temperature extremes, flooding, wildfire, and novel insects and diseases. To maintain the healthy ecosystems in the State vulnerable species and habitats should be identified. Therefore, genetic conservation is the key to preserving vulnerable species. At the same time, species having wider adaptability over larger geographical and climatic regimes also need to be identified and conserved as a part of adaptation strategy for climate change.

4. Eco-restoration through afforestation and Climate oriented eco-restoration plan

Ecological restoration helps in recovery of degraded, damaged, or destroyed ecosystem. The state government wants to take Ecological restoration activity particularly in terms of afforestation to respond to climate change in the State. These activities would help influence the State's carbon budget in a positive way. The state would prepare an eco-restoration plan for climate change mitigation.

5. Study on identification of tree phenology

Climate change may cause deviation in timing, duration and synchronization of phonological events in forest species. Changes in rainfall, precipitation and temperature would alter tree phenology since they differ widely with respect to adaptations to seasonal drought and indications for bud break of vegetative and flower buds. Erratic precipitation and

increasing temperature are likely to alter the span of the growing season of trees. The impact of climate change can be better assessed at the level of functional types based on the duration of deciduousness and timing of onset of the reproductive phase (first-visible-flower). The State government would take a study to see the significant variation (advanced or delayed) in onset dates of flowering and fruiting responses in tree species as a result of climatic change.

6. Flora and Fauna vulnerability study

Fauna and flora though can gradually adapt to changes in temperatures but they fail to adapt to sudden weather variability. Reduced snowfall in the mountains of the State may cause habitat shift for mountainous plant and bird communities to higher elevations for survival. Species having physiological limitations to adapt to sudden weather changes might become extinct. Since horticulture is one of the major sources of livelihood and revenue, the State Government wants to do a study to see the floral and faunal changes in the State.

7. Carbon sequestration study for Jammu and Kashmir forest

The goal of carbon sequestration study is to identify potential approaches for enhancing terrestrial carbon sequestration as a partial mitigation of CO₂ induced climate change. The primary objective of the State Government is to provide data to determine carbon stocks and sequestration rates spanning the entire life cycle of the major forest species. The

study would be designed to provide improved estimates based on micrometeorological and inventory methods, litter, deadwood and soil carbon stocks and sequestration rates for different species.

8. Study on per capita firewood consumption and alternative livelihood

Firewood derived from forests is an important source of domestic energy in rural and urban areas of Jammu and Kashmir. Fuelwood is collected directly from forests and fields in the Himalayan region. Unrestricted grazing in forest areas has reduced the tree regeneration. This would affect future supply of fuelwood and carbon sinks. The per capita fuel wood consumption in the State needs to be quantified so that future plans can be formulated accordingly. This study seeks to document annual per capita fuelwood consumption, availability and production through case study research in all the districts of Jammu and Kashmir and associated fuelwood harvesting areas. This study would help in improvement of livelihood of the forest dependent communities and also encourage forest conservation which will ultimately result in GHG sequestration.

9. Study on soil organic carbon of forest area

The soil represents the largest depository for fixing carbon. The forest soil is one of the major carbon sink due to high organic carbon content. A study would be conducted in Jammu and Kashmir to estimate the SOC pool under different forest covers. Due

to wide variations in weather, soil types, topography and altitude the forest species vary from broad leaved to conifers and from deciduous to evergreen. SOC pool also varies as per the forest species. Findings of this study would serve as benchmark for the future investigations to ascertain the changes in SOC pool over a period of time and to take climate change mitigation measures if required.

10. Climate impact study in undisturbed/ protected forest areas

Most of the forests are subjected to disturbances on account of human activities. The forest degradation in these areas is primarily due to anthropogenic causes and it would be difficult to segregate the effects of climate change on forests in these areas. Therefore, climate impact studies should also be carried out in undisturbed/ protected forest areas where the climate change effects, if any, are not confounded with other disturbance effects.

11. E-green portal with geo-reference

The State Government would do the field Survey with modern instruments/ Standalone GPS with reasonable accuracy for collection of associated Master Data inclusive of Legacy Data as per Standard Operating Procedure (SOP) prescribed by FSI / Department. The E-green portal would provide accessibility to a variety of geospatial layers available including thematic information, topographic maps and satellite images. It would also provide

access to important documents relevant to the geospatial layers shown on the map. The e-portal would help to develop adaptation and mitigation strategy for forestry and other sectors in the Jammu and Kashmir.

12. Nursery development for climate adaptable species

The State Government wants to run a programme for producing high quality climate adaptable seedlings by using advanced technology, which would be initiated under State Forest Department. Under this programme some root trainer and clonal nurseries would be established in different forest divisions. The seedlings produced are utilized for plantations in the land under forest Department and the surplus seedlings would be sold to the public for better survival.

13. REDD+ feasibility study for carbon sequestration

Forests are not only a major carbon sink, they are also a renewable source for imperishable forest products and a replacement for fossil fuels. This scoping study aims at exploring the feasibility for a new developing mechanism under the UN Framework Convention on Climate Change (UNFCCC) to Reduce Emission from Deforestation and forest Degradation (REDD). The State has become a huge forestry resource and is especially interested in the opportunities of the most recent development REDD+, which would include financial award for conservation, sustainable management of forests and enhancement of carbon stock in the State.

14. Study of climate change impact on wild life specie

Wildlife lifecycle (breeding, nesting, growing or feeding) depends on the climatic conditions of the area. Some species sensitive to climate change need to move to suitable climate conditions. Other species which are able to adapt to the new climate conditions would become more dominant. Biodiversity management strategies are essential for maintenance of biodiversity resources. The most important strategy would be to leverage our current system of nature conservation reserves by creating connectivity among the protected areas. To help animals to survive in the weather variability or climate change, the State government wants to do a study on climate change impact on wildlife species. A team of officers should be involved for wildlife conservation.

15. Climate change impacts on undisturbed forest area like national parks

Climate change would affect the habitats and wildlife in National Parks though few habitats are capable of being effective carbon sinks. Rates of glacier melting would be noted and the impact on these areas needs to be assessed. Although maximum National Parks can be affected by impact of climate change like flooding, droughts, changes in plant and animal feeding, reproductive and migration patterns. Since tourism is one of the major sources of revenue generation of the State, a detailed study on climate change impact on National Parks of the State is the need of the hour.

7.7 List of Key Priority Action

Table 32: List of Key Priority Action Green India Mission

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
1	Implementation of J&K State Forest Policy	Department of Forest	0	150.0	150.0	Gol, Govt. of J&K, EFA
2	Capacity building and awareness for all levels of stakeholders	Department of Forest and CAMPA Division	0	200.0	200.0	Gol, Govt. of J&K, EFA
3	Gene bank development for climate adaptable species	Department of Forest	0	50.0	50.0	Gol, Govt. of J&K, EFA
4	Eco-restoration through afforestation and Climate oriented eco-restoration plan	Department of Forest and CAMPA Division	0	450.0	450.0	Gol, Govt. of J&K, EFA
5	Study on identification of tree phenology	Department of Forest	0	7.5	7.5	Gol, Govt. of J&K, EFA
6	Flora and Fauna vulnerability study	Department of Forest	0	6.0	6.0	Gol, Govt. of J&K, EFA
7	Carbon sequestration study for Jammu and Kashmir forest	Department of Forest	0	10.0	10.0	Gol, Govt. of J&K, EFA
8	Study on per capita fuel wood consumption and alternative livelihood	Department of Forest	0	10.0	10.0	Gol, Govt. of J&K, EFA
9	Study on soil organic carbon of forest area	Department of Forest	0	5.0	5.0	Gol, Govt. of J&K, EFA
10	Climate impact study in undisturbed/protected forest areas.	Department of Forest	0	10.0	10.0	Gol, Govt. of J&K, EFA
11	E-green portal with geo-reference	Department of Forest	0	100.0	100.0	Gol, Govt. of J&K, EFA
12	Nursery development for climate adaptable species	Department of Forest	0	150.0	150.0	Gol, Govt. of J&K, EFA

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
13	REDD+ feasibility study for carbon sequestration.	Department of Forest	0	5.0	5.0	Gol, Govt. of J&K, EFA
14	Study of climate change impact on wild life species	Department of Forest and wildlife Division	0	10.0	10.0	Gol, Govt. of J&K, EFA
15	Climate change impacts on undisturbed forest area like national parks	Department of Forest and wildlife Division	0	10.0	10.0	Gol, Govt. of J&K, EFA
			0	1173.5	1173.5	





8 Water Mission

8.1 Introduction

Climate change is projected to influence the hydrological cycle of water bodies, water supply systems and exacerbate requirement of water in different regions. The disturbance of the hydrological cycle as projected due to the variability in the climatic conditions are likely to enhance the current stress on water resources from surging population, growing economic development and land use change including urbanization. Increase in ambient temperature and changes in rainfall pattern might influence the hydrological cycle to a large extent. Other consequences like increase in rates of evapotranspiration decrease in fresh water availability, mass losses from glaciers and reductions in snow cover might severely impact the hydrological cycle. Change in the hydrological cycle may affect the spatial and temporal distribution of runoff, soil moisture, ground-water reserves and may increase the frequency of droughts and floods. Event of increased frequency of rainfall and variation in rainfall pattern including extreme events like flood risk poses challenge to the society, physical

infrastructure and quality.

Economy of the State relies heavily on water intensive sectors like agriculture, horticulture, energy generation, tourism, etc. Since most of the water-bodies in the state are supplied by melt-water from mountain range therefore loss of glacier might reduce the water availability impacting hydro power potential and lower availability of water for irrigation impacting the agrarian economy in the state. Regions in the state that are already subjected to water stress are projected to be adversely affected with the potential of impact on multiple sectors e.g. agriculture, domestic water supply and health.

It is therefore imperative that effective plan be strategized towards increasing the efficiency of water use, explore options to augment water supply in critical areas and ensure more effective management of water resources. Integrated water resources planning and basin management are strategized to cope up with the projected variability in rainfall and river flow. The Broad objectives of the mission addressed are:

- Promoting research studies on all aspect related to impact of Climate Change on Water Resources including quality aspects of water resources so as to address the growing concern of water and food security issue.
- Expeditionary implementation of water resources projects- irrigation, Hydro power, domestic water supply, water transport etc.
- Promotion of traditional and modern system of water conservation and preservation of water bodies –lakes.
- Planning on principles of integrated water resources development and management
- Ensuring convergence among various water resources programmes.
- Intensive capacity building and awareness programme including those for Panchayati Raj Institutions, Urban local bodies, Water User Associations and youths towards water use efficiency, water conservation and managing water related disaster like flood and drought.

8.2 Key trends in the sector

The State is endowed with ample amount of surface and ground water resources mostly of Himalayan origin. The three distinct physiographic division of the State are drained by three distinct river basins namely Chenab, Jhelum and Indus River. Apart from the river the state is also gifted with numbers of water bodies including tributaries, lakes

and wetlands. The following section provides a brief of the existing water bodies in the state, the availability of water from various resources and its demand across different sector in the state.

8.2.1 River System

The water flowing in the Himalayan rivers is combined flow from rain, snow and glaciers. Flow from snow and glaciers make these rivers perennial where heavy monsoon rainfall contribute to the bulk of water in these basins which are used for irrigation and Hydro-power generation.

1. Indus River

The Indus river originates at the confluence of the Sind river and the Gar river which is Mansarovar lake in Tibet at an elevation of 518 meters, and enters India through South-eastern corner of J&K State. The Indus River is fed by massive Tibetan glacial waters to become a mighty river with further feeds from other glacial catchment areas in Karakoram and Zaskar ranges. The Indus river flows in between the Ladakh range and Zaskar range before crossing the international boundaries into Pakistan. The Indus traverses a distance of 1800 miles (2900 km) through Tibet, India, Pakistan occupied Kashmir (PoK), and Pakistan before draining into the Arabian Sea south of Karachi. Out of the total catchment area of the Indus River in India a very small portion of about 500 sq km lies in the State of Himachal Pradesh in Lahaul district. The main catchment area of the river is in the Kargil and Ladakh districts of J&K (about 59,146 sq km). The catchment area up to Indo-Pak



border is about 168355 sq km out of which 30755 Sq km is in Tibet and 137000 Sq Km is in the international boundaries of India (out of which 59646 Sq Km within the line of control.

Indus has several tributaries principle among them ate Gilgit River, Gumal River and Kabul

River. The western tributaries of the Indus that include the Swat, Kurram, Gomal, Kohat, Zoab and Kabul are not discussed herein. The river has been variously known as the Sengge or Lion River by the Tibetans, Abbasseen or Father of Rivers by the Pathans of present NWFP Pakistan, and MithoDariyo or Sweet River by the denizens of the arid Sindh.



Figure 18: Indus River and its tributaries with in Jammu and Kashmir



2. Jhelum River

Jhelum originates from Verinag spring in Anantnag and after passing through Srinagar it flows into Wularlake and then passing through Baramula and Uri it flows to Pakistan. The total geographical area of Jhelum basin up to Indo-Pakistan border is about 34755 km². The total catchment area upto the ceasefire line of about 17622Km², spreads in seven districts namely Anantnag, Pulwama, Srinagar, Badgam, Baramulla, Kupwara, and Poonch. The Lidder River is the biggest tributary of Jhelum and is fed by a large number of glaciers. Jhelum is joined by many small streams like Vishow, Rambiar, Wanbran, Aripal, Romeshi etc.

3. Chenab River

The merging of the two Himalayan River ; Chandra River and Bhaga River at Tandi located in upper Himalayas forms the Chenab. It enters the plains of Jammu and then enters into Pakistan. The Salal, Dulhasiti, Sawalkot and Baglihar Hydel projects have been constructed over it.

4. Kishan Ganga River

Kishan Ganga is in Drass (Kargil) inner Himalayas. After passing through the mountain tracks of Gurez, Keran and Karnah it joins Jhelum at Doemel.

5. Tawi River

Originating from Kailashkund spring at Soej hills in BhaderwathTawi passes through Jammu and joins Chenab in Ranbir Singh Pura and then enter Pakistan.

Pakistan. The river drains a total catchment area of 14442 km² in India after flowing for a length of 720km.

8.2.2 Lake

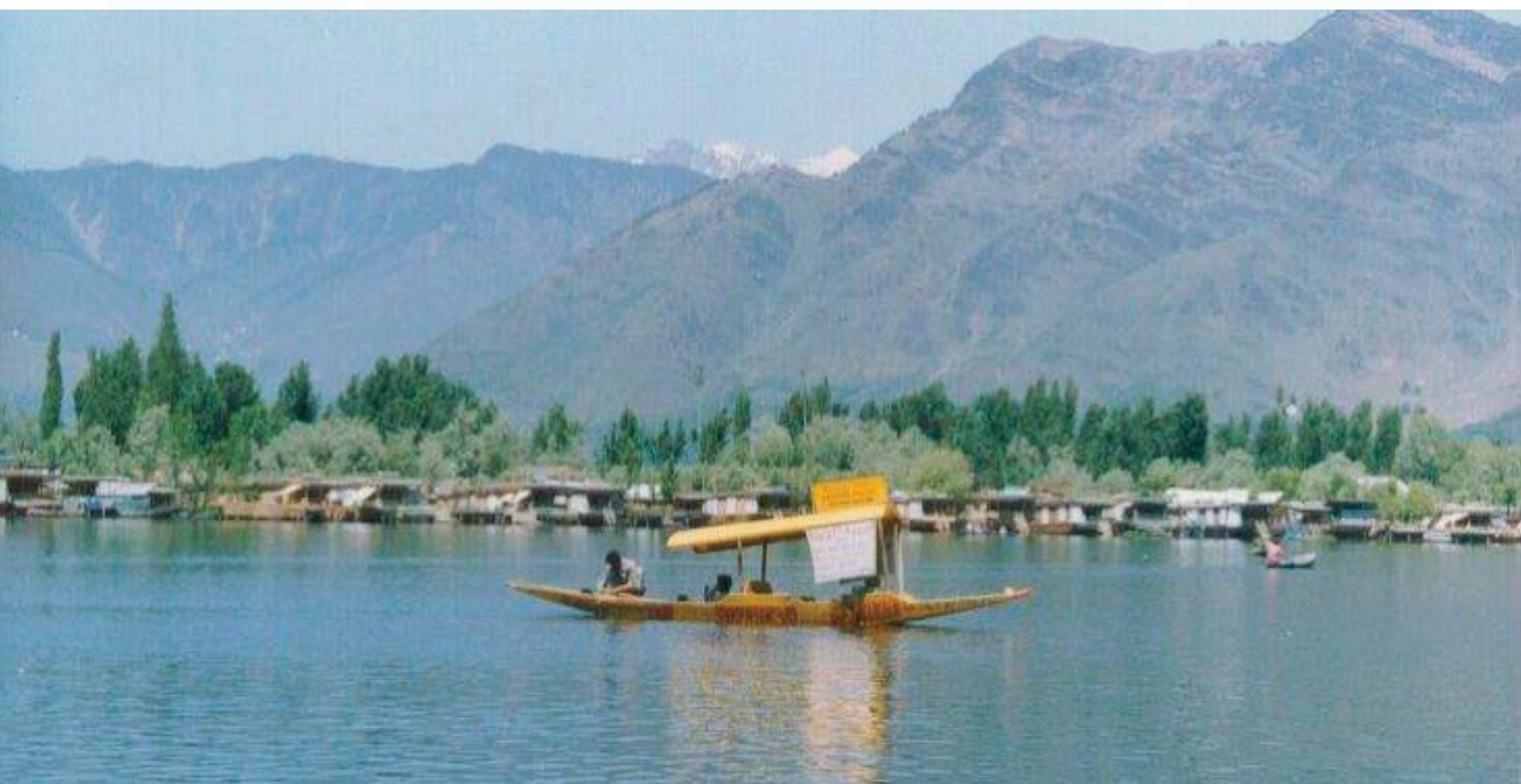
6. Ravi River

Ravi is a trans-boundary river flowing through North-western India and North Eastern

Number of lakes and water bodies in the state that acts as a source of surface water are represented as follows:

Table 33: District wise number of lake and water bodies

Sl. No	District	Number of Lakes and Water bodies	
		SOI Toposheet of 1967	LISS III Datasheet of 2012
1	Anantnag	88	98
2	Baramulla	119	124
3	Budgam	25	33
4	Doda	13	88
5	Jammu	15	15
6	Poonch&Rajouri	46	27
7	Pulwana	61	20
8	Kathua	1	1
9	Kupwara	111	66
10	Ladakh	637	665
11	Srinagar	52	74
12	Udhampur	36	19
	Total	1248	1230



8.2.3 Major supply of water in the state

1. Precipitation

Precipitation is the major source of water resource in the state over and above surface water and replenishable groundwater. The precipitation is also one of the important

constituents of climate. The presence of the water vapors in the climate, which depends upon the feed of the moisture from evaporation, both from the seas and the land, and the movement of this moist water, its condensation and the precipitation are part of the climatic process. The regime of the precipitation in the hydrologic cycle, thus heavily depends on the climatic factors.

Table 34: Availability of Water from precipitation across the administrative divisions of Jammu

Sl. No.	Station	2004	2005	2006	2007	2008
		Total Rainfall in mm	Total Rainfall in mm	Total Rainfall in mm	Total Rainfall in mm	Total Rainfall in mm
1.	Jammu	1098.9	1112.4	1235.8	1294.1	1339.4
2.	Banihal	982.8	1553.2	1865.4	1038.4	1180.7
3.	Batote	1151.2	1703.5	1912.5	1038.4	1265.9
4.	Bhaderwah	1193.5	1597.4	1407.9	918.8	1075.5
5.	Katra	1819.7	1879.2	2546.9	2283.2	2375.5
6.	Kathua	1005.4	1301.8	1162.9	1896.4	1360.4





An analysis of availability of precipitation between 1980 and 2008 reveals an un-uniform pattern. Analysis of the rainfall trends indicate that since 1980, the rainfall has shown declining trend in Mansar, Udhampur, Kathua, Kishanpur, Mandi, Basoli, Rajouri, Akhnoor, Khorh, Poonch, Bilawar, Ramnagar & Barmeen stations while the number of rainy days being increased in Kishanpur, Mandi, Rajouri & Akhnoor. Rainfall pattern started increasing from 2000 in Jammu, Banihal, Batote, Bhaderwah, Katra & Kathua along with increase in number of rainy days in Jammu, Banihal, Batore & Bhaderwah.

Analysis of pattern of precipitation in Kashmir division between 1990 to 2008 reveals that though rainfall in Srinagar, Pahalgam, Kukernag, Qazegund, Gulmarg & Kupwara showed significant variation, the number of days having rainfall has increased in the areas.

2. Glacial

The “frozen water” in the Himalayas is crucial for the people inhabiting the mountain areas as well as those inhabiting the downstream regions. The Indus basin has 7,997 glaciers

Table 35: Availability of Water from precipitation across the administrative divisions of Kashmir

Sl. No.	Station	2004	2005	2006	2007	2008
		Total Rainfall in mm	Total Rainfall in mm	Total Rainfall in mm	Total Rainfall in mm	Total Rainfall in mm
1.	Srinagar (IMD)	635.2	680.8	887.6	501.8	649.7
2.	Pahalgam	976.6	1300.7	1514.7	921.2	1038.2
3.	Kukernag	671.8	1107.2	967.1	713	1215.3
4.	Qazigund	1006.1	1275.7	1407.1	879.8	1075.3
5.	Gulmarg	1057.8	1663	1664.3	937.8	1222.9
6.	Kupwara	868.8	970	1196.8	654.2	989.5

with a total glacier cover of 33,679 km² and total ice volume of 363.10 km³. The contribution of snow to the runoff of major rivers in the eastern Himalayas is about 10% but more than 60% in the western Himalayas.

Table 36: Major glaciers in the Indus basin of India

Basin	No. of glaciers	Glacier covered Area (km ²)	Ice volume (km ³)
Ravi	172	193	8.04
Chenab	1,278	3,059	206.30
Jhelum	133	94	3.30
Beas	277	579	36.93
Satluj	926	635	34.95
Upper Indus	1,796	8,370	73.58
Shyok	2,454	10,810	NA
Nubra	204	1,536	NA
Gilgit	535	8,240	NA
Kishenganga	222	163	NA
Total	7,997	33,679	363.10

Table 37: The characteristic of glacial coverage in State

Glaciers (no.)	Area (km ²)	Average Size (km ²)	Glacier%
526	29163	10.24	61.8



3. Ground-water

The major areas of the state are occupied by high hills. Occurrence of the ground water is primarily confined to alluvial regions which have been classified into piedmont deposits of outer plains of Jammu, Dune belt in outer Himalayas, isolated valley fills in lesser Himalayas, fluvio-Lacustrine deposits in Kashmir valley and moraines and fluvioglacial deposits in Laddakh. Dug well in Kashmir has limited yield. Ground water occurs in perched condition and give rise to springs in phreatic zone water table. The Annual Replenishable Ground water Resource of the state has been assessed as 2.70 bcm and the net ground water availability is 2.43 bcm. The annual ground water draft of the state is only 0.33 bcm. Ground Water is being used through dug wells, bore wells and tube wells for drinking and irrigation purposes.

Two main aquifer systems lies in this area, the hard rock or fissured aquifer and soft sedimentary or porous aquifer. Occurrence and movement of ground water is mainly controlled by primary inter- granular porosity in the soft sedimentary quaternary alluvium and Karewas formations. The unconsolidated sedimentary deposit form multi-layer aquifer system in Jammu, outer plain and in Kashmir, valley areas. In hard rock or fissured formation the thick weathered zone and the fractured secondary porosity govern the occurrence and movement of ground water. There exists large scope for further ground water development, particularly in the valley area as well as Kandi areas by constructing suitable ground water structures.

The groundwater resources have two components, viz. static and dynamic.

Table 38: Basin-wise Replenishable Groundwater Resources (in bcm)

Annual Replenishable Ground water Resources	Monsoon Season	Recharge from Rainfall	0.61
		Recharge from other sources	0.77
	Non-Monsoon Season	Recharge from Rainfall	1.00
		Recharge from other sources	0.32
	Total	2.70	
Natural Discharge during non-monsoon season			0.27
Net Annual Ground water Availability			2.43
Annual Ground water draft	Irrigation		0.10
	Domestic and industrial uses		0.24
	Total		0.33
Projected Demand for Domestic and Industrial uses upto 2025			0.42
Ground water availability for future irrigation			1.92
Stage of Ground water development			14

Table 39: Basin-wise Replenishable Groundwater Resources (in bcm)

Parameters			Anan- tnag	Bad- gan	Bara- mula	Jammu	Kathua	Kupw- ara	Pulw- ama	Srin- agar
Annual Replenishable Ground water Resources	Monsoon Season	Recharge from Rainfall	3038	1969	4689	31424	13748	1921	3544	1158
		Recharge from other sources	12237	8229	9885	21921	8747	4074	7877	4175
	Non-Monsoon Season	Recharge from Rainfall	13851	6805	29245	15700	6869	7002	16160	4028
		Recharge from other sources	3465	1050	1260	16032	6223	105	3045	840
	Total		32591	18053	45079	85077	35587	13102	30626	10201
Natural Discharge during non-monsoon season			3259	1805	4508	8508	3559	1310	3063	1020
Net Annual Ground water Availability			29332	16248	40571	76569	32028	11792	27563	9181
Annual Ground water draft	Irrigation		40	36	236	6636	2193	14	53	295
	Domestic and industrial uses		1377	2285	3488	6854	1786	1981	2256	3903
	Total		1417	2321	3724	13490	3979	1995	2309	4198
Projected Demand for Domestic and Industrial uses upto 2025			2534	4017	6257	11721	2676	3948	3691	6694
Ground water availability for future irrigation			26758	12195	34078	58212	27159	7830	23819	2192
Stage of Ground water development			5	14	9	18	12	17	8	46

8.2.4 Major Usage of Water and its sourcing in the State

1. Irrigation

The total cultivable area of the state is 25.96 lakh²² acres (10.51 lakh ha). The agriculture and its allied sector including horticulture and sericulture are highly dependent upon

the availability of water and as such the irrigation is the crucial input for development of agriculture and horticulture sector in the State. The irrigation potential of the State is 2.50 lakhs hectares through major/medium surface water scheme and about 4.00 lakhs ha through surface water minor schemes. The ground water irrigation potential is

²²1 lakh = 0.1 million

estimated to be 7.08 lakhs ha. Two major and 16 medium and a large number of minor irrigation completed schemes provides irrigation facilities to about 2.877 lakhs

hectare area in the State through Indus, Jhelum, Chenab and Ravi rivers. Currently, 14 medium and 13 renovation projects are under construction in the State.

Table 40: Total area of irrigated land from different sources²³

Year	Net Area Irrigated by (000 ha)				
	Canals	Tanks	Well	Other	Total
1950-51	244.00	3.00	3.00	11.00	261.00
1955-56	277.00	1.00	3.00	9.00	290.00
1960-61	256.00	-	5.00	13.00	274.00
1965-66	270.00	-	1.00	7.00	278.00
1968-69	252.00	-	1.00	11.00	264.00
1974-75	279.00	-	3.00	13.00	295.00
1980-81	285.00	2.00	4.00	13.00	304.00
1985-86	288.69	2.67	4.12	14.13	309.61
1990-91	278.58	1.98	1.33	16.20	298.09
1995-96	284.86	2.57	1.42	17.73	306.58
1998-99	283.81	2.60	1.32	21.42	309.15
1999-00	278.35	2.57	1.37	20.80	303.09
2000-01	284.15	2.71	1.53	22.48	310.87
2001-02	284.42	2.79	1.61	21.35	310.17
2002-03	274.50	2.66	1.57	20.49	299.67
2003-04	282.41	3.87	1.06	19.19	306.53
2004-05	266.28	3.93	1.08	19.60	310.89
2005-06	289.28	4.21	1.05	17.57	312.11
2006-07	286.64	4.24	1.04	17.52	309.44
2007-08	285.78	4.22	0.99	17.05	308.04
2008-09	287.77	4.84	3.80	17.32	313.73
2009-10	287.80	5.11	4.33	20.03	317.27
2010-11	288.48	6.22	11.65	14.28	320.63

²³Digest of Statistics 2010-11

The above statistics clearly reveals that majority of irrigation in the state is canal fed comprising about 90% of the total.

Irrigation Development in Indus Basin

Due to sub-zero temperatures during the winter the Ladakh terrain remains under snow cover. There is only one crop season available during summer time which is highly dependent on irrigation due to very scanty rainfall in the area and arid climate condition. The irrigated area as on effective date of treaty was 42179 acres (17069 ha) which is now increased to 49338 acres (19967 ha). Therefore, another 62837 acre (25430 ha) area can be brought under irrigation under the provision of treaty. However, the total cultivable area in the Ladakh region is only 71540 acre (28953 ha). Therefore, almost all the cultivable area of Ladakh region can be brought under irrigation under the treaty, but the provisions cannot be fully utilized

as the cultivable area is limited. In the Indus basin there is only one crop season available during the summer time, the main crops in the two districts (Ladakh and Kargil) are wheat, gram and small millets. Major sources of irrigation are gravity canal or kulhs drawn from rivers or streams. Due to topography and availability of agriculture land in patches there is limited scope of major and medium irrigation projects. Minor irrigation schemes are well suited for the region; about 476 minor irrigation schemes are providing irrigation to about 16,000 ha area in the Ladakh region. One medium project namely IgoPhey canal project provides irrigation to about 4000 ha of land. One ParkachikKhowas irrigation canal project for irrigation of 2235 ha land in the Kargil is under construction. It is estimated that another 8000 ha area can brought under irrigation by minor irrigation schemes in the districts, leaving a scope of about 15,000 ha area without irrigation under the treaty.



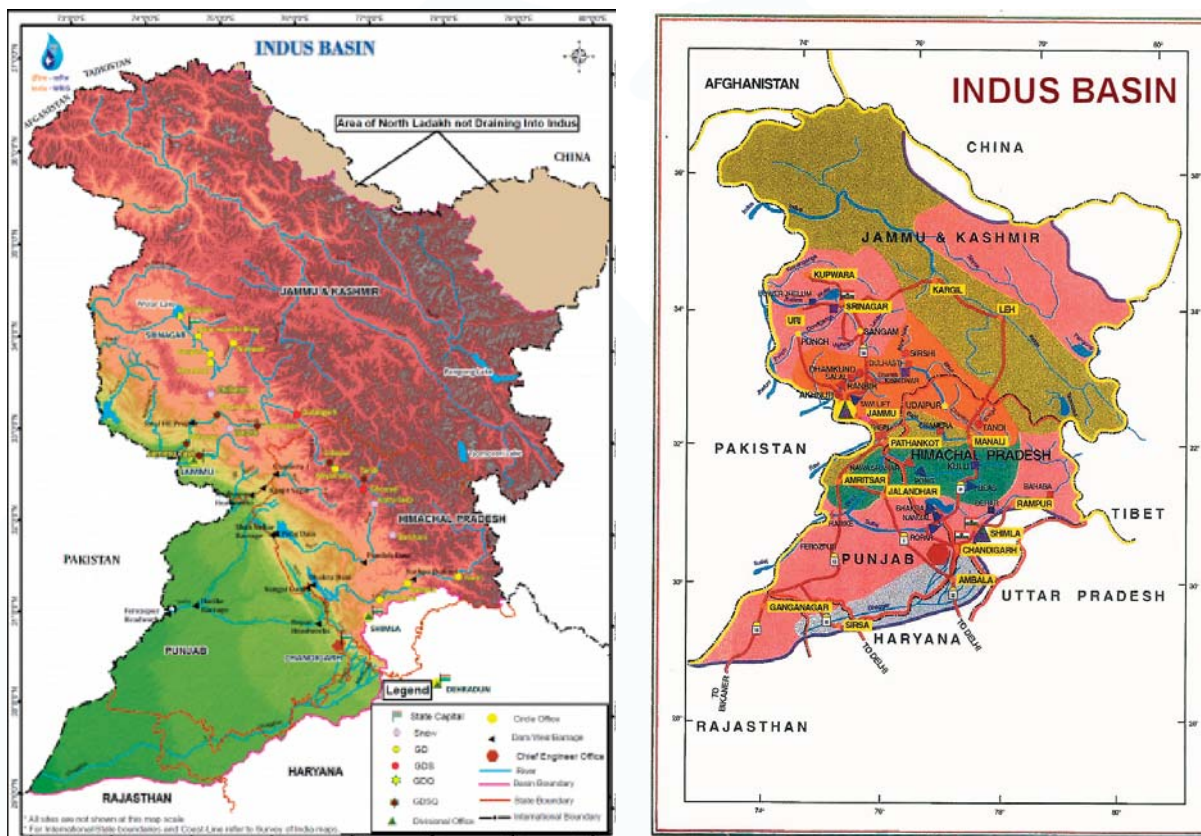


Figure 19: Indus Basin

Irrigation Development in Jhelum Basin

Due to hilly terrain Jhelum basin has got limited scope of major and medium irrigation projects. Large numbers of Zamindari canals (kuhl) were constructed by people to supply irrigation water to the fields. A large number of government canals schemes like lift schemes, diversion schemes and storage schemes have been providing irrigation facilities to the large track of cultivated and horticulture area. As on effective date of Indus treaty, irrigated area in Jhelum was 5,17,909 acres (2,09,594 ha) and additional area permissible without any storage in the treaty, works out to be 6,67,909 acres (2,70,300 ha), out of which 65,150 acre

irrigation area (2,63,658 ha) has already been achieved. There is large number of surface flow minor irrigation schemes (3,540 no's) and surface water lift irrigation schemes (411) as per minor irrigation census (2000-2001). Dug wells, shallow tube wells as well as deep tube wells have also been developed to a limited extent. An area of about 15,000 acre (6,000 ha) is irrigated through ground water. During the survey and studies conducted by Central Water Commission, it was found that there were not many feasible storage schemes in the tributaries of Jhelum basin. Only feasible storage option found out was Drang site. The Irrigation and Flood Control Department, J&K has proposed 14 storage sites where storage are feasible and an area of about 80,000 acre

can be brought under irrigation from these storages. However, these sites have to be further surveyed and investigated as a small height of dam of about 20 M height, could relate to serious problems of submergence of forest, land and villages.

Irrigation Development in Chenab Basin

About 16% of the basin area is sown with mostly wheat, maize, rice, onion, fruits, vegetables and beans. The area in Chenab basin in Jammu region is mostly irrigated by gravity canals. Besides, irrigation from other sources like tube wells, tanks, well, springs etc are also carried in much smaller magnitude. In Doda district, two medium irrigation schemes namely Shiva canal and Kashtigarh canal scheme have been constructed during Fifth Five Year Plan. In Udhampur district, two canals namely Udhampur and Duddar canal were constructed in Second and Fifth plan respectively. Similarly, in Rajouri District also two medium irrigation projects namely Ans and Sajal have been functioning. In Jammu district, land is generally flat and two major canal systems, namely Ranbir canal and Pratap canal were constructed way back during 1903-04 to irrigate vast track of land on either side of river. As these are restrictions under the Indus Treaty for expanding the irrigation area, without creating storages,

there is urgent need to build storages as to exploit all the additional ICA permissible under with storage.

Ground Water is being used through dug wells, bore wells and tube wells for drinking and irrigation purposes. Two main aquifer systems underlies in the area, the hard rock or fissured aquifer and soft sedimentary or porous aquifer. Occurrence and movement of ground water is mainly controlled by primary inter- granular porosity in the soft sedimentary quaternary alluvium and Karewas formations. The unconsolidated sedimentary deposit form multi-layer aquifer system in Jammu, outer plain and in Kashmir, valley areas. In hard rock or fissured formation the thick weathered zone and the fractured secondary porosity govern the occurrence and movement of ground water. There exists large scope for further ground water development, particularly in the valley area as well as Kandi areas by constructing suitable ground water structures.

Development of additional Irrigated Cropped Area within the provision of Indus Water Treaty

As per provisions of Indus Water Treaty, additional Irrigated Cropped area (ICA) can be taken by India and is as follows:



Table 41: River basin wise irrigated crop area permitted under Indus Water Treaty

River Basin	ICA as on Effective Date (1-4-1960)	ICA permissible as per Treaty		Total ICA permissible		Total ICA 2010-11 (Actual)
		Without storage	With storage & releases (Addl)	Without storage	With storage & releases	
1	2	3	4	5	6	7
Indus	42179	70000	-	112179	112179	51175
Jhelum	517909	400000	150000	667909	917909	512575
Chenab	82389	231000	50000	132389	313389	48952
	-	-	150000			
Total	642477	701000	350000	912477	1343477	612702
# includes 6,000 acres outside drainage basin of Chenab						

2. Domestic Water

Jhelum basin

The population of Jhelum basin in 2001 Census was about 40 lakhs. The total domestic water demand is estimated to be 460 MLD. The areas falling in the basin are hilly and mountainous with inter mountain valleys. The valley areas are flat and mildly undulating topography. In such condition small surface water streams/ rivers and ground water is most suitable for domestic water supply. The ground water utilization in the valley has not much developed but 95% of development of ground water has been utilized for domestic and industrial use. As per State statistics, Jhelum valley has 1485 tube wells and 5216 dug well for the domestic water supply purposes. Given the relatively low stage of ground water development in each of these districts, it seems that increase in the water demand in the future decades can be taken care of through further development of ground water resources. Similarly, small industrial

requirement of water can be taken care of by exploiting ground water.

Chenab basin

Total population in the Chenab basin district in J&K is 34.28 lakhs as per 2001 census. The domestic water demand is roughly estimated to be 240 mgd. The stage of ground water development is quite low (Jammu 16% & Kathua 12%). These are about 600 tube wells and 331 dug wells in the Chenab districts, mostly in Jammu and Kathua districts. For meeting the domestic water requirement at the remote places not covered with the irrigation projects commands, ground water potential has to be tapped.

In the whole of the State of J&K 555700 water taps have been installed out of which 152400 are public stand post and rests are private pipe connections.

3. Drinking Water

Drinking water is widely available in the State. However, some villages/ habitations

once covered for drinking water supply faces problematic scenario due to quantitative or qualitative deficiencies leading to water scarcity. Issues related to less service level, sources depletion, outlived design period of water supply schemes needs to be addressed. The Govt. of J&K through its administrative department is facilitating availability of portable water to the extent of 135 ltr per person per day in urban areas to 70 ltr per person per day in rural areas. It is envisaged to provide deep tube wells/ production wells to address the drinking water problem in extremely water scarcity zones and areas where water sources have completely depleted due to drought.

8.3 Vulnerability of the sector

8.3.1 Projected variation in the precipitation pattern

Overall Precipitation pattern: The PRECIS run for 2030's indicate that annual rainfall in the Himalayan region is likely to increase in 2030's with respect to 1970's range from 5% to 13% with some areas of Jammu and Kashmir showing an increase of up to 50%. All seasons in the Himalayan regions indicate an increase in rainfall, with monsoon months of June, July, August and September showing the maximum increase in the rainfall. The winter rain in the month of January and February are also projected to increase whereas minimum increase is indicated in October, November and December.

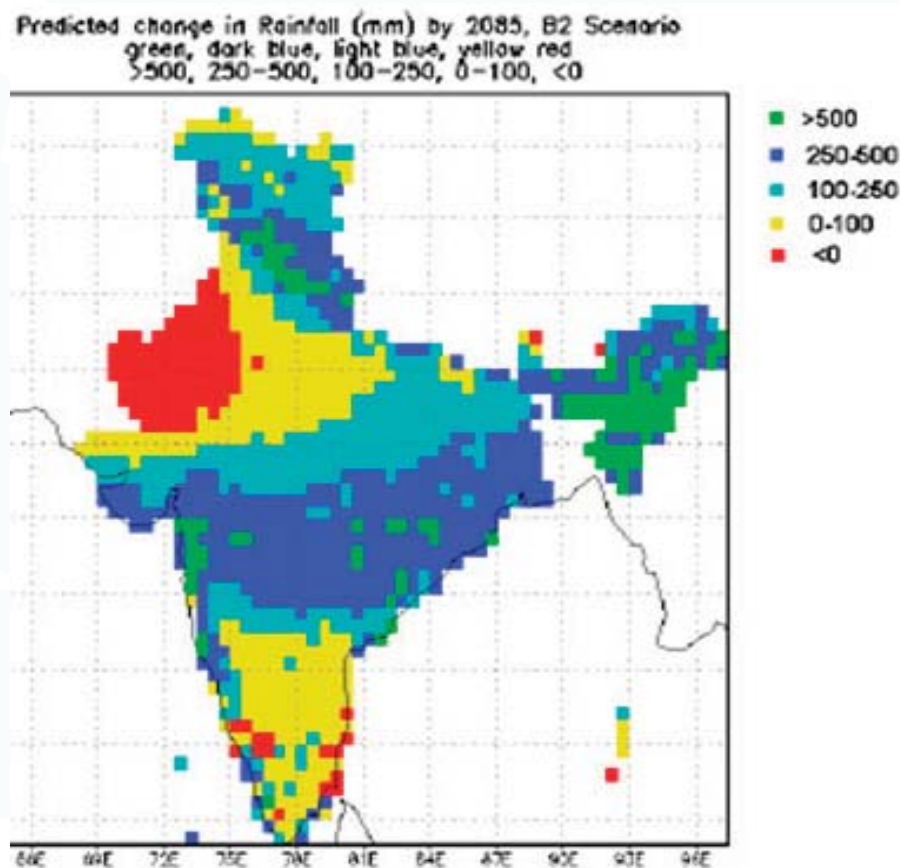


Figure 20: Predicted change in rainfall (mm) by 2085, (Source: Current Science)

Extreme precipitation: The number of rainy days in the Himalayan region in 2030s may increase by 5-10 days on an average, with an increase by more than 15 days in the eastern part of the Jammu and Kashmir region. The intensity of rain fall is likely to increase by 1-2 mm/ day.

Associated Vulnerability: Increase in the rainfall is likely to enhance the incidence of flood and might impact livelihood sectors like agriculture and tourism to a great extent. The general impact of increase in the precipitation is reflected in the increase in sediment yield. The other major factors that dictate sediment yield are the intensity of rainfall, land use and the soil type of the area. The increase in the sediment yield in the Himalayan region is up to 25%, which can be detrimental for the existing water resources projects and has the potential to cause considerable damage

to the environment.

8.3.2 Projected variation of the glacial reserve

The snowline and glacier boundary are sensitive to changes in climate conditions. Trend analysis of various glaciers from the past data gives some ideas of future projections. Glaciers are not well represented by PRECIS modeling. Warming is likely to rapidly increase the rate at which glaciers are melting, leading to greater ablation than accumulation. According to SASE report, Himalayas have grown warmer by 2.2°C in last two decades. Almost all glaciers reportedly show negative mass balance.

The rate of recession of glaciers is reportedly varying which is being attributed to winter precipitation climate warming and anthropogenic elements of landscape.

Table 42: Recent retreat pattern of selected glaciers in Jammu and Kashmir

Name of the Glacier	Retreat of snout (m)	Observation Period	Trend	Avg. retreat rate (m/yr)
Drang-Drung	NIL	2004-2007	No Change	-
Kangriz	NIL	1913-2007	No Change	-
Siachin	NA	1862-1909	Advancing	15.42
	NA	1909-1929	Retreating	2.5
	NA	1929-1958	Retreating	14
	NIL	1958-1985	No Change	NIL
	NA	1985-2004	Retreating	3
	NA	2004-2005	No change	NIL

Associated Vulnerability: Glacial melt is expected to increase under change in climatic conditions, which would lead to increased summer flows in some rivers systems leading to enhanced flood scenario for few decades, followed by reduction in the flow as glaciers disappear resulting into drought conditions.

8.4 Key issues in the sector

8.4.1 Flood risk from water bodies in the State

Destructive floods both large and small magnitudes, have been taking heavy toll of life and property in the State as seen from the historical records. The problem in Kashmir valley is basically of overflowing of banks of river Jhelum and its tributaries, land erosion in Jammu region due to Chenab and Tawi.

Flood Problem in Jhelum Basin

The flood problem in the valley is due to inadequate carrying capacity of the river Jhelum in its length from Sangam to Khandanyar. Just upstream of Srinagar at Padshahibagh, a flood spill channel with capacity of $481 \text{ m}^3/\text{s}$ (17,000 cusecs) takes off to by-pass the city. However, in spite of flood spill channel, floods can be caused by Jhelum in the Srinagar town, if the discharge of river through the town exceeds $991 \text{ m}^3/\text{s}$ (35,000 cusecs).

The outfall channel below Wular Lake is the only outlet for drainage of whole valley. The off take channel is only about 20,000 cusecs ($570 \text{ m}^3/\text{s}$). During high discharges in the river, the flood water gets accumulated in the Wular Lake, which acts as an absorption

basin. When the Wular lake level rises, the water heads up in the river reach above, almost up to Srinagar. Many times the rising water breaches the embankment at a number of places, mostly on the left side, submerges tracts of rich cultivated land, causing heavy damages to the crops. The problem is aggravated on account of the silt brought down by Pohru River entering into outfall channel and causing blockages.

The major risk is projected as the town is thickly populated along its bank and any widening of bank or raising embankments is not feasible.

Historical instances of flood are

- During the year 1928, Jhelum caused heavy flood in the valley when the river carried a discharge of $2265 \text{ m}^3/\text{s}$ (about 80,000 cusecs) against the capacity of $906 \text{ m}^3/\text{s}$ (32,000 cusecs).
- In 1950, flood flow was encountered due to discharge of about $1000 \text{ m}^3/\text{s}$ (35,000 causes).

Flood Problem in the Chenab basin

The Chenab in the hilly area up to Akhnoor does not pose any flood problem as it passes in the deep rocky steep gorges. In the plains, the slope of river is flat and it spreads in a width of 1 Km to 8 Km with many channels. The river channels erode the cultivable lands and at higher discharges, the river inundates vast area causing devastation.

8.4.2 Retreat of Glacial System

1. A study of snow cover monitoring by Negi et al, (2009) in the PirPanjal and Shamshawari ranges in Kashmir valley indicated spatial and temporal variability in the snow cover. Advance melting was observed in Banihal and Naugam/ Tangdhar regions than Gurej and Machhal regions.
2. Shekher et al (2010) studied the snowfall patterns in the Western Himalayan range which indicated a decrease in total seasonal snowfall of 280 cm over the entire Western Himalaya between 1988 to 2008. The snowfall decreased by 280 cm over the PirPanjal, 80 cm over Shamshawari and 440 cm over the greater Himalaya ranges. Whereas over the Karakoram range, only 40 cm decreasing trend was noticed.
3. Study by A V Kulkarni (2006) indicated significant glacial retreat. The study investigated 466 glaciers in Chenab, Parbati and Baspa basin since 1962. The investigation has shown an overall reduction in glaciers area from 2077 km² to 1628 km² since 1962, an overall deglaciation of 21%. However, numbers of glaciers have increased due to fragmentation. MoEF report on Climate Change and India (A 4x4 Assessment report) in Nov 2010, gives an overall status as summarized below.
4. Studies conducted during last three decades by the National Institute of Hydrology (NIH) reveal that glaciers in Ladakh, Zaskar, and the greater Himalayan ranges of Jammu and Kashmir are generally receding and the glaciers volume changes range between 3.6% and 97%, with the majority of glaciers showing degradation of 17% - 25%. The studies however, do not reveal any significant retreat during 2004-07. Most of the glaciers in Western Himalayan are receding (except a few in Jammu & Kashmir which are not changing or rather are advancing).
5. The Nubra valley of Jammu and Kashmir has 114 small-sized glaciers varying between less than 5km and 10km in length. The glaciers of the valley do not show much change in their length and area during the period 1989–2001. However, variable decline in the glacial area of the Siachin glacier has been observed. The area has reduced from 994.99 km² in 1969 to 932.90 km² in 1989. However, small change in the area (932.90 km² to 930 km²) has been noticed during the following decade (1989–2001). Recession patterns of 466 glaciers in the Chenab, Parbati and Baspa basins of the western Himalayas have been studied for the period 1962–2008. Here, a reduction in the glacial area from 2,077 km² to 1,628 km² and an overall deglaciation of 21% has been observed.
6. Most of the glaciers in western Himalayas are receding (except a few in Jammu and Kashmir, which do not show any change or are advancing). The processes controlling the rate of retreat of glaciers are complex and vary with location and topography of

the area. However, the impact of rising temperature and reducing snowfall on glacier mass balance is reflected in these findings, which may require a sound long-term database for precise climate change assessment.

8.4.3 Deterioration of the Water bodies

The lakes are the worst sufferers of the human interference and rapid urbanization. Some of these water bodies have disappeared due to natural cause like glacial action, low precipitation or are on the verge of extinction. Lakes that are within the urban areas have deteriorated over the period of time of which many are non-existent. The reason for such shrinkage/deterioration is an outcome of massive habitations around the lakes and encroachment/choking of drainage system. Even the source/catchment areas for water changing of this lakes have been built up due to construction of the buildings. Also massive erosion in the catchment area is resulting in siltation of these lakes thereby converting the water area into landmass. Dal lake which exists as water body system with its adjoining lakes i.e. Gilsar, khushalsar and Aanchar are almost nonexistent. The drainage systems of the Dal lake which feed into these small water bodies have got silted up/landmasses created which would normally flush the Dal lake into the river Jhelum resulting into natural decaying. The deterioration of water bodies is largely due to:

1. Indiscriminate cutting of surrounding vegetation thereby increasing silt and nutrient load in the lake bodies.

2. Disposal of Sewage and domestic waste.
3. Open Defecation.
4. Excessive use of pesticide and insecticide in the catchment.
5. Soil erosion in the catchment and siltation
6. Prolific plantation in Wular
7. Unchecked housing
8. Weed intensification and eutrophication

8.5 Programme and Policies in the sector

8.5.1 Indus water Treaty

Water utilization and hydro-power generation in the State is largely governed by the Indus Water Treaty. The Indus system comprises three eastern rivers namely the Sutlej, the Beas, the Ravi and three western rivers namely the Indus, Jhelum and the Chenab. Under the Indus Water Treaty (IWT) of 1960, the waters of eastern rivers stand allocated to India and those of western rivers largely to Pakistan except the specified uses by India mostly in the State of J&K.

As on effective date of agreement on 1.4.1960, 6.42 lakhs acres were being irrigated by India from the Western Rivers and is entitled to irrigate additional Irrigated Cropped Area (ICA) of 7.01 lakhs acres (total of 13.43 lakhs acres) as per details below.

Table 43: Maximum area irrigated

Name of the river	The Indus	The Jhelum	The Chenab	Total
ICA in acres	70,000	4,00,000	2,31,000 ²⁴	7,01,000

As per Annex – E of the Indus Water Treaty, the aggregate storage capacity of all single purpose and Multi-purpose reservoirs which may be constructed by India after Effective Date shall not exceed the following table:

Table 44: Aggregate Storage Capacity

SI No.	River System	Conservation storage MAF		Flood Storage MAF
		General Storage	Power Storage	
(a)	The Indus	0.25	0.15	Nil
(b)	The Jhelum (excluding the Jhelum Main)	0.50	0.25	0.75
(c)	The Jhelum Main	Nil	Nil	As provided in Paragraph 9 of Annexure to the Treaty
(d)	The Chenab (excluding the Chenab Main) ²⁵	0.50	0.60	Nil
(e)	The Chenab Main ²⁶	Nil	0.60	Nil

8.5.2 Jammu and Kashmir Water Resources (Regulation and Management) Act, 2010

The State Act of 2010 provides the law relating to use of water, the measurement, construction, control and management of works with respect to water storage, conservation and protection, the irrigation water supply, drainage, flood control and prevention, the improvement in the flow of

water, the protection and improvement in the physical integrity of water courses, lakes and springs, the safety and surveillance of dams. The Act also provides for the establishment of State Water Resources Regulatory Authority for regulating water resources, ensuring judicious, equitable and sustainable management, allocation and utilization of water resources, fixing the rates for use of water and matters connected therewith and incidental thereto.

²⁴Includes 6000 Acres outside drainage basin of Chenab

²⁵India shall have the option to increase the Power Storage Capacity specified against item (d) above by making a reduction by an equal amount in the Power Storage Capacity specified against items (b) or (e) above.

²⁶Storage works to provide the power storage capacity on the Chenab main specified against item (e) above shall not be constructed at a point below Naunut (Latitude 33°19' N and Longitude 75°59' E)

The tariff fixation and revenue collection from different water usages particularly from Hydro power will provide huge revenue realization to the State which can be further reinvested in construction of Hydro and Irrigation Schemes in the State and proper operation and maintenance of the water system including water bodies. The Act covers all the aspects of proper regulation of surface and ground water in the State.

In the Jammu and Kashmir Water Resources (Regulation and Management) Act, 2010, prioritization of water allocation has been given in the following order, but subject to modification if so warranted by area/situation specific consideration:-

- Drinking water including washing and bathing
- Irrigation
- Generation of electricity
- Ecology and
- Agro industries and non-agricultural industries.

The act has adopted policy for establishment of a network of water quality laboratories at district level to ensure the water quality and regular monitoring in terms of parameters laid down by Government of India so as to ensure that the health of people is not affected adversely. The Act provides for prevention of encroachment on existing water bodies and deterioration of water quality in such water bodies due to pollution. The Act also provides for treatment of effluents, solid/gaseous wastes which are discharged into

the natural streams, water bodies and have the potentials of contaminating the ground water through seepage, treating to acceptable standards before these are released from the industries, institutions, residential and commercial areas.

8.6 Key Priorities

1. Generate baseline data with terrain analysis (contour/slope/ gradient/aspect, etc.), including comprehensive land classification, catchment demarcation, river networks, lakes and water bodies, forest coverage, etc.

Assessment of the water resources is the first step in understanding the state of water resources. This state is essential for all water resource related activities such as efficient management of the resources, planning of the water resource for future use, re-adjustment of water use plans considering the changed supplies and demands etc. Baseline assessment consists of

- A hydrologic study to establish the likely natural water availability of the basin/region/area;
- Monitoring of Ground water level and quality
- Estimation of surface runoff and Evapotranspiration
- Spatial Distribution and Seasonal Variability of precipitation and number of dry days

- Monitoring of steam level and flow
- An assessment of the resultant flows that would continue to be available in future
- Estimate available catchment areas and basin studies to evaluate the actual scenario of water availability
- Undertaking study on Glaciology, Snow Hydrology, River hydrology and sedimentation
- Deciding on the acceptability of the resultant flows from the environmental and cultural aspects, and changing the water use plans to obtain an acceptable situation.

Such data information will help in ground water, surface water vulnerability mapping and framing plan for integrated water management.

2. Promoting and implementing water use efficiency measures

Implementing water use efficiency has been accorded with the highest priority under national water mission. Following are the measures proposed to be undertaken as a part of the priority action:

- Mandating water audit across industrial and commercial sector to benchmark unit wise water usage and enforcing adaptation of water conservation measures.
- Introduction of water metering across domestic, industrial and commercial sector to ensure judicious usage of

water.

- Creating awareness amongst the farmer group towards judicious use of water and promoting sprinkler/deep irrigation system, lowering evapotranspiration. Farmer should also be trained over computing water requirements of crops and build capacity towards applying as much water as needed and not irrigate randomly rather follow a proper schedule. Promoting System of crop intensification and achieving water use efficiency across agriculture sector.
- Water recycling (including recycling of tail end water for irrigation)/reuse
- Canals should be ensuring canals are free from seepages.

3. Develop hydrological models for different seasons and calibration with the existing data

Planning adaption or DRM measures requires consistent monitoring, archiving (including past weather data information) and analysis of the weather related parameters so as to assess the variation/trends in the meteorological cycle and project future climate scenario using Global Circulation Models (GCMs). On regional basis GCMs are down scale to Regional Climate Models called RCMs. The study should also include data collection programme about the hydraulic parameters in the three sub-basins in the State. The Soil and Water Assessment Tool (SWAT) model usually used to predict the hydrological response of the basin catchment to the natural inputs (meteorology) as well

as manmade interventions (irrigation etc) can also be used for the study.

The weather and basin level hydrological data parameters will be used to develop the hydrological model under short; medium and long term scenario. Hydrological models give the hydrological forecasting of stream flows. These models are also used for flood forecasting of stream flows. These models are also used for flood forecasting and lean period discharges. The projections of variability in the hydrological cycle will be used to develop the hazard risk vulnerability and plan adaptation measures. The hydrological model developed will be calibrated on time to time basis based on the continual monitored data parameters so to facilitate projection with lowest error.

4. Comprehensive water budgeting in lieu of the current and future demand from domestic supply, irrigation, industry, tourism and for other sectors

The total water balance of the state, its basins, sub-basins, areas etc. depicting the quantified hydraulic cycle is the main tool for understanding the water situation. Water budgeting should be based on this hydraulic cycle and inter-action between rain water, surface water, ground water, evaporation etc. It is therefore essential that water budgeting be carried out to chalk out efficient planning for

5. Monitoring river flows and mapping of flood and drought zones

Using the river geometry and the flood area maps, it is possible to judge the discharges at any point of the stream network. For this, the digital elevation model (DEM) would have to be linked to the flood flow model. After the model gets validated in the process, flood prone areas for a given flood frequency could be mapped from the model, for the changed hydraulics. This would be an aid to planning or modifying the flood protection works. Again, the proposed flood management plan can be studied on the model. If, an unsteady flow model is linked to the topography, effects of these, on the upstream and downstream, can be studied.

6. Formulate comprehensive rejuvenation plan for Dal Lake, Mansar Lake, Manasbal Lake, Pangong Lake and all other major lakes

Lakes are the essential source of surface water which suffice the domestic and drinking water requirement in the state. Moreover these lakes are also main areas of tourist attraction. Shrinkage and deterioration of the lakes and water qualities along with encroachment and choking of drainage system might force

7. Assessment and Implementation of rain water harvesting and artificial recharging of ground water

Artificial Recharging of Ground water

The artificial recharge to ground water aims at augmentation of ground water reservoir by modifying the natural movement of surface

water utilizing suitable civil construction techniques. Artificial recharging facilitate to enhance the sustainable yield in areas where over-development has depleted the

Aquifer, Conservation and storage of excess surface water for future requirements, improve the quality of existing ground water through dilution, remove bacteriological and other impurities from sewage and waste water so that water is suitable for re-use. To facilitate ground water recharging it is essential to identify

- Availability of non-committed surplus monsoon run off in space and time.
- Identification of suitable hydrogeological environment and sites for creating subsurface reservoir through cost effective artificial recharge techniques

As a first step towards identification of suitable sites following studies are proposed:

(i) Identification of Areas : identification of areas should be carried out in

- Areas where ground water levels are declining on regular basis.
- Areas where substantial amount of aquifer has already been desaturated.
- Areas where availability of ground water is inadequate in lean months.
- Areas where salinity ingress is taking place.

(ii) Identifying site suitability through following studies

- Hydrometeorological Studies to decipher the rainfall pattern, evaporation losses and climatological features
- Hydrological Studies to ascertain the availability of source water for the purpose of recharging the ground water reservoir. Four types of source water may be available for artificial recharge viz. In situ precipitation on the watershed, Surface (canal) supplies from large reservoirs located within basin, Surface supplies through trans basin water transfer, Treated municipal and industrial wastewaters.
- Soil infiltration studies
- Hydrogeological study of the area relating to rock units, their ground water potential and general pattern of ground water flow and chemical quality of water in different aquifers including aquifer geometry
- Geophysical studies
- Chemical quality of source water
- Assessment of Sub-Surface Potential For Ground Water Recharge

Based on the identification of the area that are water stressed and possessing lower ground water level ten locations will be selected based on the assessment of quality of the soil, hydrogeology and hydrology and the level of vulnerability and artificial recharge be carried out.

Rainwater Harvesting

Rain water harvesting is the technique of collection and storage of rain water at surface or in sub-surface aquifers, before it is lost as surface run-off.

Following are the measure proposed as part of the action plan

- (i) Developing policy and incentives framework towards promoting rainwater harvesting in urban areas and new construction.
- (ii) Training of architect, urban local bodies and builders over the technology of rain water harvesting.
- (iii) Implementation of pilot and demonstration project in urban and rural areas
- (iv) Undertaking study of the water stressed area with declining ground water level and substantial amount of aquifer being de-saturated and demarcating areas for undertaking rain water harvesting project. Following are the important aspects to be looked into for rain water harvesting:

- Hydrogeology of the area including nature and extent of aquifer, soil cover, topography, depth to water level and chemical quality of ground water.
- The availability of source water, one of the prime requisite for ground water recharge, basically assessed in terms of non-committed surplus monsoon runoff.
- Area contributing run off like

area available, land use pattern, industrial, residential, green belt, paved areas, roof top area etc.

- Hydrometeorological characters like rainfall duration, general pattern and intensity of rainfall.

Following are the technologies proposed under the action plan in urban and rural areas subjected to suitable satisfaction of the ground condition:

Urban Areas

- Roof Top Rain Water /Storm run off harvesting through
 - o Recharge Pit
 - o Recharge Trench
 - o Tubewell
 - o Recharge Well

Rural Areas

- Rain Water Harvesting through
 - o Gully Plug
 - o Contour Bund
 - o Gabion Structure
 - o Percolation tank
 - o Check Dam/ Cement Plug/ Nala Bund
 - o Recharge shaft
 - o Dugwell Recharge
 - o Ground Water Dams/Subsurface Dyke

Emphasize the awareness and capacity building for conservation, concern and co-management of water

Creating awareness over the necessity of water conservation and propagating benefits in terms of reducing water purification and sewer cost, less electrical energy use for pumping, environmental and social benefits and preservation of ground water reserves is the essential steps towards planning water management and ensuring water use efficiency. In this context it is important to encourage schools to help develop and promote a water conservation ethics among children and also engage civil society organizations in imparting awareness.

8. Dealing with flood and Drought in the changing climate scenarios

Flood or Drought can derail the development trajectory currently in place in the state. Prolonged drought conditions can be harmful to the agriculture yield. State has to manage such conditions by development of irrigation system, ground water development, water harvesting, changing cropping pattern, adopting integrated farming systems. Studies in various regions with regard to climate change scenarios have indicated over increased frequency or probability of floods. Flood causes extreme damage to the human life and infrastructure and is therefore highly essential that the extreme event be judiciously managed so as to lower the impact on livelihood.

To control or reduce the impact of extreme hydrological event like drought or flood it

is essential that adaptation measures be planned to identify and recue the impact of climate variability.

1. Developing hydrological forecasting models for floods and lean period flows and fitting the models and calibration of the model with the existing data.
2. Linking digital elevation models for low lying areas with hydrological models to understand flood situations under different Scenarios

Drought Related Measures

3. Completion of on-going Irrigation projects (major, medium minor schemes) with the help of control funding / assistance central.
4. Renovation of existing irrigation Schemes
 - Major and medium schemes with Central Assistance
 - Minor irrigation schemes within State funding
5. Development of ground water/ irrigation and Drinking Water to cover all the area which is not reachable by surface water.
6. Implementation of rehabilitation of water bodies by desalting and raising the capacity of their reservoirs by gated arrangements,. The stored water to be useful for drought management in the cropped area.
7. Developing crop varieties considering the rise in air temperatures and

intermittent drought conditions.

Flood Related Measures

8. Preparation/ revision of Master Plans for flood protection with changed flood scenarios due to climate change. Some efforts for preparation of master plans for floods control and management in the Jhelum and Chenab basins have been carried out by the State government with the help of Central Water Commission. Considering the projections of enhanced magnitudes of floods due to Climate Change efforts, these masters plan may be revised and revised.
9. Urban storm water drainage improvements- guidelines to be framed to facilitate Implementation of various schemes under JNURM central assistance. Increased frequency of floods, urban drainage congestion may pose bigger problems. Studies should be carried out with the help of CWC for improved drainage designs for urban areas. Strict measures for cleaning of drainage and waterways and removal of encroachments and acquisition of land/ properties for effecting proper drainage measures are enforced through legal and procedural changes.
10. Designing and planning of Implementation of non – structural measure for flood management, and storage of water in numerous

depressions and lakes.

11. Development of flash flood forecasting system based on the alerts from Automatic Rain gauges.
12. Dam Break or Embankment break studies to be done routinely with the help of CWC, NIH and CWPRS.

9. Reducing impact of increased erosion and sedimentation

Whereas losses consequent to floods generally occur only in crop season, the losses due to sedimentation of fertile lands, bank erosion and changing course of river lead to loss of productive lands. The land erosion and deposition by sedimentation is likely to be accentuated in the climate change scenario because increasing flood frequencies will cause land erosions, and rainfall intensities may erode soil for the upper catchments. In the flood control & management master plan, river training and erosion control measures should also be studied and implemented. A master plan for soil conservation measures and catchment area treatment should be prepared for time bound implementation. It is also recommended to develop a universal Soil loss model depicting erosion and sediment transport and actuate the above model for changed rainfall regime due to Climate Change. It is also recommended to develop a master Plan for soil conservation measure and catchment area treatment and implementation.

8.7 List of Key Priority Action

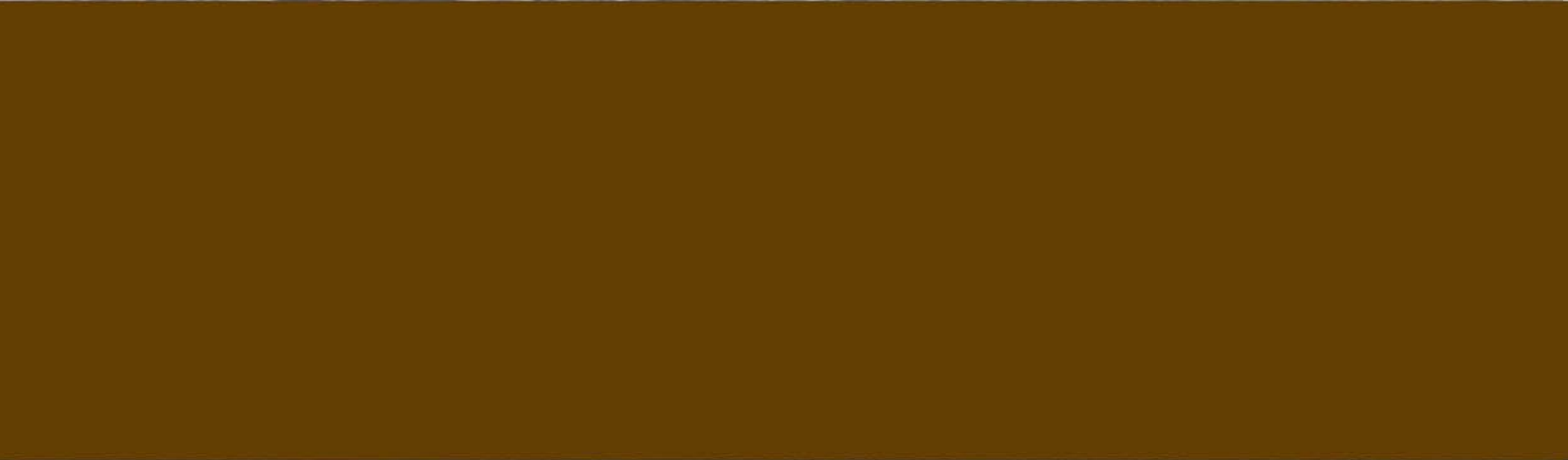
Table 45: Key Priority Actions State Water Mission

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
1	Generate baseline data with terrain analysis (contour/slope/ gradient/aspect, etc.), with comprehensive land classification with catchment demarcation, river networks, lakes and water bodies, forest coverage, etc	PHE, Irrigation and Flood control department	0.00	8.00	8.00	MOWR, CWC, CGWB, GoI, EFA
2	Promoting and implementing water use water use Efficiency measures	PHE, Irrigation and Flood control department , Housing and Urban Development Department, MOWR, CWC, CGWB , Agriculture	0.00	12.00	12.00	MOWR, CWC, CGWB, GoI, EFA
3	Develop hydrological models for different seasons and calibration with the existing data	PHE, Irrigation and Flood control department , MOWR, CWC, CGWB	0.00	8.00	8.00	MOWR, CWC, CGWB, GoI, EFA
4	Comprehensive water budgeting in lieu of the current and future demand from domestic supply, irrigation, industry, tourism and for other sectors	PHE, Irrigation and Flood control dept, Industry and Commerce dept	0.00	12.00	12.00	MOWR, CWC, CGWB, GoI, EFA
5	Monitoring river flows and mapping of flood and drought zones	PHE, Irrigation and Flood control dept	0.00	12.00	12.00	MOWR, CWC, CGWB, GoI, EFA

Sl. No.	Title	Organizations	Budget (INR in Million)			Source of Funding
			Existing	Additional	Total	
6	Formulate comprehensive rejuvenation plan for Dal Lake, Mansar Lake, Manasbal Lake, Pangong Lake and all other major lakes	Lake Development authority, Housing and Urban Development Department	0.00	56.00	56.00	MOWR, CWC, Govt of India, and other external agencies
7	Assessment and Implementation of rain water harvesting and artificial recharging of ground water	PHE, Housing and Urban Development Department, Rural dev. Dept.	0.00	208.00	208.00	MOWR, CWC, CGWB, GoI, EFA
8	Emphasize the awareness and capacity building for conservation, concern and co-management of water	PHE, Housing and Urban Dev. Dept	0.00	10.00	10.00	MOWR, CWC, CGWB, GoI, EFA
9	Dealing with flood in the changing climate scenarios.	PHE, Irrigation and Flood control dept, MOWR, CWC, CGWB, Revenue and Relief department	0.00	300.00	300.00	MOWR, CWC, GoI, EFA
10	Reducing impact of increased erosion and sedimentation	Irrigation and Flood control department , NIH, Agriculture Department	0.00	50.00	50.00	MOWR, MoAgri, GoI, EFA
11	Creation of Storage facilities on Distributaries of River Chenab and Jhelum As per Indus Water Treaty	Irrigation and Flood Control Department MOWR, NIH	0.00	45000.00	45000.00	Govt of India MOWR
Total			0.00	45676.00	45676.00	







**Department of Ecology Environment
& Remote Sensing
Government of Jammu & Kashmir**